Chapter 13

Clinical Technique for Amalgam Restorations

"Endurance is not just the ability to bear a hard thing, but to turn it into glory."

—William Barclay

Amalgam has been the primary direct restorative material for more than 150 years. It has been the subject of intense research and has been found to be safe and beneficial as a direct restorative material.[1–3] The cost-effective nature of amalgam restorations has benefited many people. According to the U.S. Public Health Service, 'hundreds of millions of teeth have been retained that otherwise would have been sacrificed because restorative alternatives would have been too expensive for many people'.[4]

In addition to being *cost effective*, amalgam has the unique property of being *self-sealing*. Self-sealing occurs when percolation of oral fluids (i.e. micro-leakage) between the amalgam restoration and the prepared cavity walls results in corrosion of the amalgam and a subsequent accumulation of corrosion products in the microscopic space. Micro-leakage between the restoration and the adjacent tooth structure is reduced as corrosion products fill the space. The self-sealing process is self-limiting and requires several months. Amalgam is the only restorative material with an interfacial seal that improves over time.[5–7]

Important Amalgam Properties

I. The linear coefficient of the thermal expansion of amalgam is 2.5 times greater than that of tooth structure, but it is closer to that of tooth structure than the linear coefficient of thermal expansion of composite.[8–10]
II. Although the compressive strength of high-copper amalgam is similar to tooth structure, the tensile strength is lower, making amalgam restorations prone to fracture during flexure.[11–13]
III. Usually, high-copper amalgam fracture is a bulk fracture, not a marginal fracture. All amalgams are brittle and have low edge strength. As such, the amalgam restoration must have sufficient bulk (usually 1.5–2 mm in any occlusally loaded area, depending on the position within the tooth) and a 90° marginal configuration.
IV. Creep and flow relate to the deformation of a material under load over time. High-copper amalgams exhibit no clinically relevant creep or flow.[14,15]
V. As amalgam is metallic in structure, it also is a good thermal conductor. At minimum, a dentin desensitizer should be used immediately prior to amalgam placement to limit sensitivity secondary to rapid fluid movement in the dentinal tubules caused by thermal changes. A liner or base should be placed in areas of deep caries removal prior to amalgam placement to limit thermal sensitivity.

General Considerations for Amalgam Restorations

Restorative Factors

Amalgam is effective as a direct restorative material because of its easy insertion into a tooth preparation and, when hardened, its ability to restore the tooth to proper form and function (Fig. 13.1). The tooth preparation not only must remove the fault in the tooth and remove weakened tooth structure, but its form must also allow the amalgam material to function properly. The required tooth preparation form must allow the amalgam to

I. Possess a *uniform specified minimum thickness* for strength (usually 1.5–2 mm in any occlusally loaded area, depending on the position within the tooth) so that it will not flex and fracture under load,
II. Produce a *90° amalgam angle* (butt-joint form for maximum edge thickness) at the margin
III. Be *mechanically retained* in the tooth (Fig. 13.2A and B).

Amalgam restorations initially leak and therefore require steps to protect from pulpal sensitivity until self-sealing is able to occur (Fig. 13.3A).[16] After desensitizing the prepared tooth structure, mixing, inserting, carving and finishing the



Figure 13.1 Clinical example of an amalgam restoration. (From Hatrick CD, Eakle WS, Bird WF: *Dental materials: clinical applications for dental assistants and dental hygienists*, ed 2, St. Louis, 2011, Saunders.)

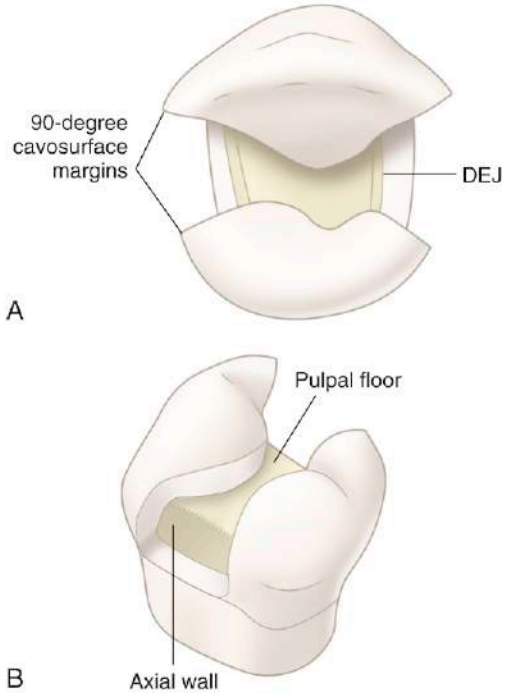


Figure 13.2 (A and B) Diagrams of Class II amalgam tooth preparations illustrating uniform pulpal and axial wall depths, 90° cavosurface margins and convergence of walls or prepared retention form or both.

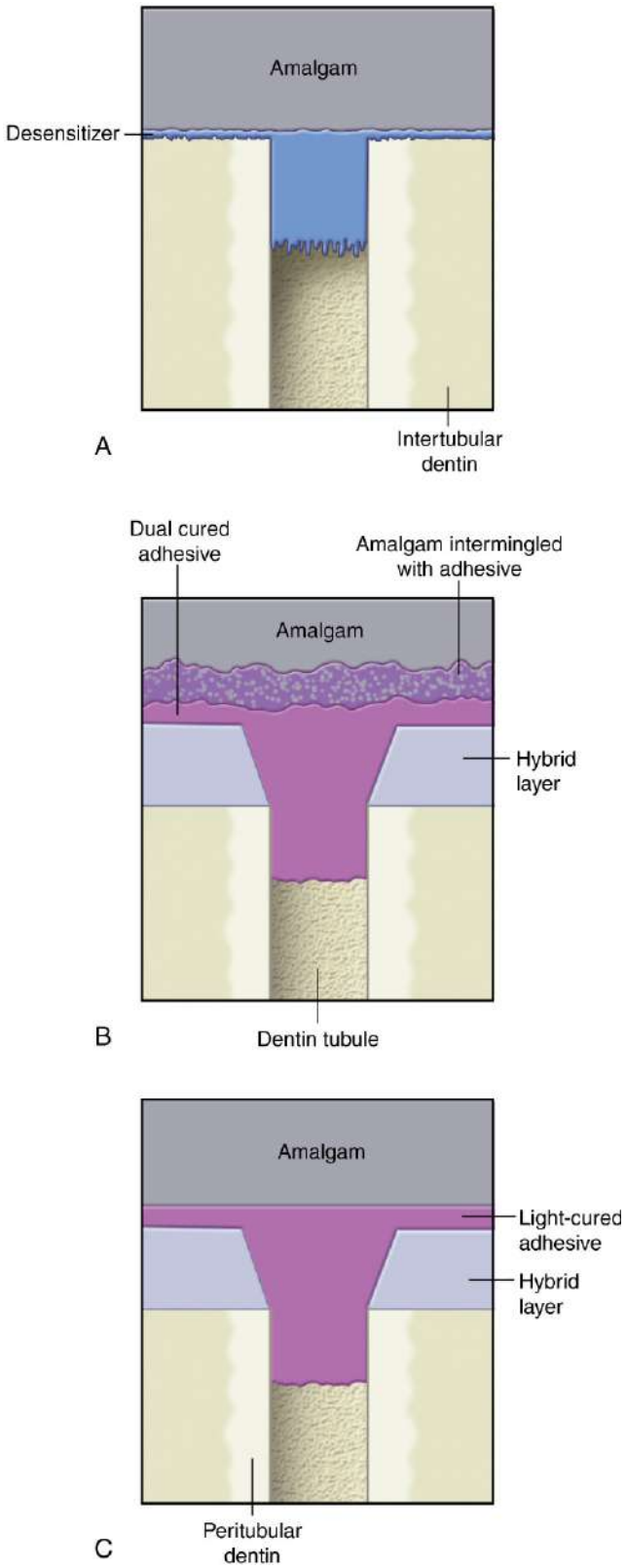


Figure 13.3 Methods to limit initial pulpal sensitivity after amalgam placement. (A) Dentin desensitization with 5% glutaraldehyde + 35% 2-hydroxyethyl methacrylate (HEMA); (B) amalgam bonding (amalgam intermingled with adhesive resin); (C) dentin sealing (adhesive resin placed and cured before amalgam placement).

amalgam are relatively fast and easy. The placing and contouring of amalgam restorations are generally easier than those for composite restorations.[17] For these reasons, it is considered a *user-friendly material* that is less technique (i.e. operator) sensitive as compared with composite.

Some practitioners have continued to use bonded amalgam restorations in their practice (Fig. 13.3B). As noted in Chapter 12, *this book does not promote the use of bonded amalgams*.[18–21] Another amalgam technique uses light-cured adhesive to seal the dentin under the amalgam material (Fig. 13.3C). This procedure, as is true of all procedures that use adhesive technology, requires proper isolation. The prepared tooth structure is etched (i.e. demineralized), primed and sealed with adhesive. The adhesive is polymerized before insertion of the amalgam. This technique seals the dentinal tubules effectively.[22,23]

Occlusal Factors

Amalgam generally has greater wear resistance than does composite in patients that have heavy occlusal function.[2,24,25] Amalgam also may be more appropriate than composite resin when there is little to no natural occlusal tooth structure remaining that requires the restoration to restore most or all of the occlusal contacts.[25]

Isolation Factors

Isolation of the operating field is important for moisture control, access and visibility, to protect the patient from aspirating or ingesting foreign objects, protecting the pulp in the event of pulpal exposure and protecting the operator medicolegally. However, minor moisture contamination of amalgam during the insertion procedure may not have as adverse an effect on the final restoration as the same contamination would for a composite restoration.

Operator Ability and Commitment Factors

The tooth preparation for an amalgam restoration is very exacting. It requires a specific design with depths that allow appropriate amalgam thickness and a precise marginal form. The failure of amalgam restorations is often related to inappropriate tooth preparation. However, the insertion and finishing procedures for amalgam are much easier than for composite.[17]

Indications of Amalgam

Amalgam is used for the

I. Restoration of carious or fractured posterior teeth as Class I and Class II restorations;
II. Replacement of failed posterior restorations;
III. As a foundation restoration;
IV. As a caries control restoration (Figs. 13.4 and 13.5) and
V. Due to its strength and ease of use, amalgam may be used (*not commonly though*) for restoring large defects

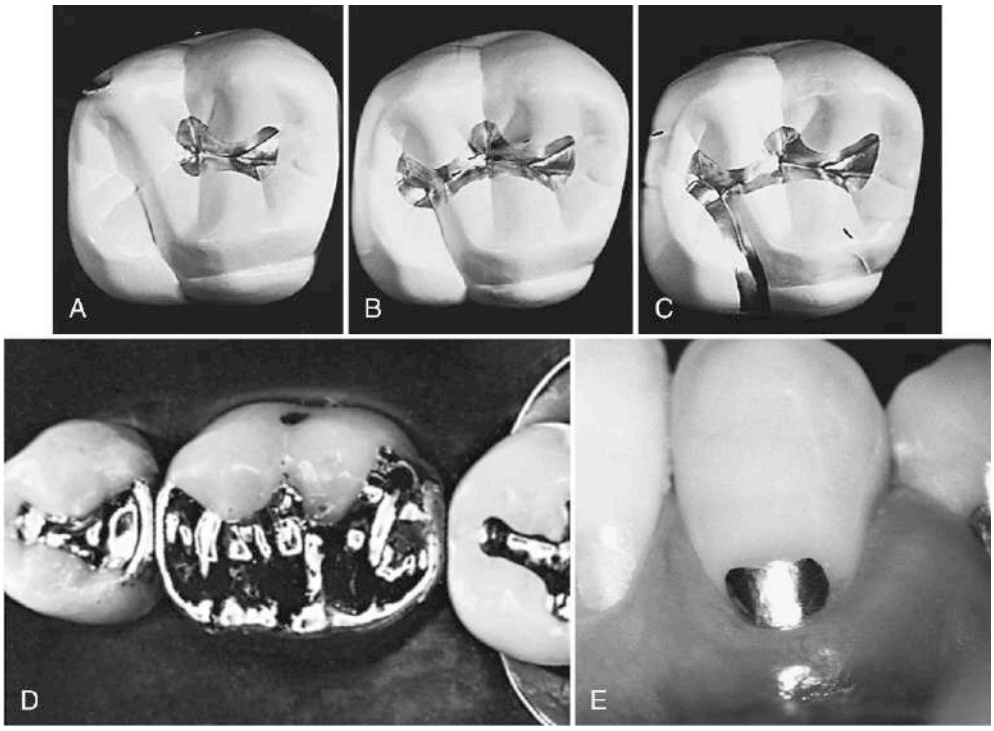


Figure 13.4 Amalgam restorations. (A–C) Class I; (D) complex and (E) Class V. (Most practitioners would restore all of these teeth with composite, except for the indication shown in (D))

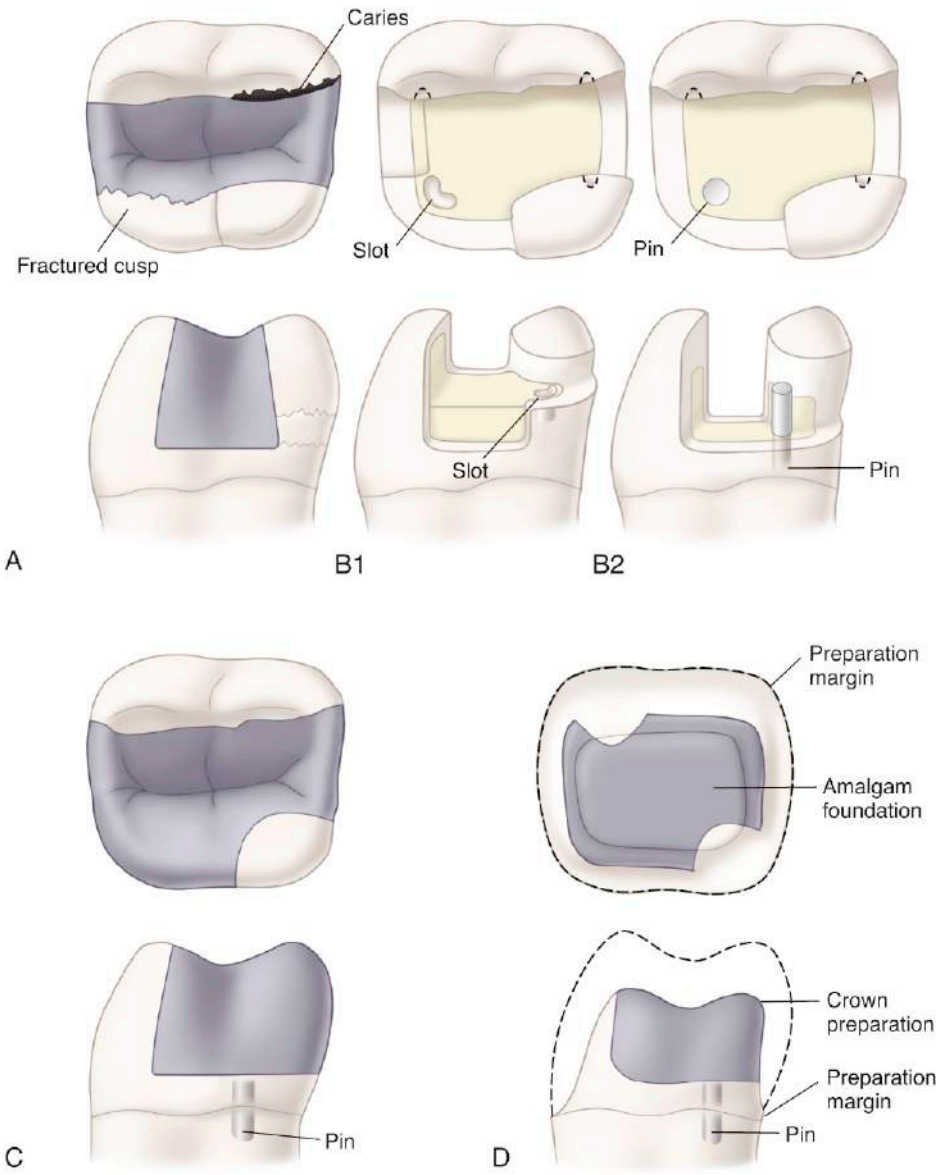


Figure 13.5 Amalgam foundation. (A) Defective restoration (defective amalgam, mesiolingual fractured cusp, distofaciocclusal caries); (B) tooth preparation with secondary retention, using slot (B1) and pin (B2); (C) amalgam foundation placed and (D) tooth with amalgam foundation prepared for crown. Note that *dashed lines* on the occlusal surface only indicate the position of the retention grooves on the facial and lingual walls of the proximal preparation. The grooves do not extend onto the occlusal surface.

in nonesthetic areas and areas with isolation problems (certain Class III and Class V situations).[26]

Contraindications

I. Amalgams are contraindicated in patients who are allergic to alloy components.
II. The use of amalgam in more prominent esthetic areas of the mouth may represent a relative contraindication. These areas include anterior teeth and, in some patients, premolars and molars.
III. Amalgam should not be used when composite resin would offer more conservation of tooth structure and equal clinical performance.

Advantages

Some of the advantages of amalgam restorations already have been stated, but the following list presents the primary reasons for the successful use of amalgam restorations over many years:

I. Ease of use
II. High compressive strength
III. Excellent wear resistance
IV. Favourable long-term clinical research results
V. Lower cost than for composite restorations

Disadvantages

The primary disadvantages of amalgam restorations relate to esthetics and increased tooth structure removal during

tooth preparation. The following is a list of these and other disadvantages of amalgam restorations:

I. Non-insulating
II. Nonesthetic
III. Less conservative tooth preparation than for composite restorations (more removal of tooth structure during tooth preparation)
IV. More difficult tooth preparation than for composite restorations
V. Initial marginal leakage

The following sections begin with the introduction of general clinical techniques and associated concepts to be considered when restoring caries lesions or defects with dental amalgam. The general technique is then followed by specific discussion addressing the creation of Class I and Class II restorations. Readers are referred to Chapter 14 for complex amalgam restorations and to Online Chapter 26 for more information on Class III and Class V amalgam restorations.

General Clinical Technique for Amalgam Restorations

Initial Clinical Procedures

Complete examination, diagnosis and treatment planning must be completed before a patient is scheduled for operative appointments (except in emergencies). A brief review of patient's dental record (including medical factors), treatment plan and radiographs should precede each restorative procedure. At the beginning of each appointment, the dentist should also carefully examine the operating site to confirm the diagnosis of the tooth or teeth scheduled for treatment.

Local Anaesthesia

Local anaesthesia is recommended for most operative procedures. Profound anaesthesia contributes to comfortable and uninterrupted operation and usually results in a marked reduction in salivation.

Isolation of the Operating Site

Complete instructions for the control of moisture are given in Chapter 8. Isolation for amalgam restorations may be accomplished with a rubber dam, cotton rolls, with or without a retraction cord.

Other Preoperative Considerations

I. Preoperative assessment of the occlusion should be made. This step should occur before rubber dam placement. The dentist should identify not only the occlusal contacts of the tooth to be restored but also the contacts on the opposing and adjacent teeth. Knowing the *preoperative location of occlusal contacts* is important in planning the restoration outline form and in establishing the proper occlusal contacts on the restoration. Remembering the location of the contacts on adjacent teeth provides guidance in determining when the restoration contacts have been correctly positioned and adjusted.
II. A wedge placed preoperatively in the gingival embrasure is useful when restoring a proximal surface. This step causes slight separation of the operated tooth from the adjacent tooth and may help protect the adjacent proximal surface, the rubber dam and the interdental papilla.
III. It is important to *preoperatively visualize the anticipated extension of the tooth preparation*. Because the tooth preparation requires specific depths, extensions and marginal forms, the connection of the various parts of the tooth preparation should result in minimal tooth structure removal (i.e. as little as is necessary), while maintaining the strength of the cuspal and marginal ridge areas of the tooth as much as possible (Fig. 13.6). The projected facial and lingual extensions of a proximal box should be visualized before preparing the occlusal outline form through the marginal ridge of the tooth, reducing the chance of overextension while maintaining a butt-joint form of the facial or lingual proximal margins.

General Concepts Guiding Preparation for Amalgam Restorations

A solid comprehension of concepts presented in Chapter 11 is essential prior to the learning of information presented in the following sections. For an amalgam restoration to be successful, numerous steps must be accomplished correctly. After an accurate diagnosis is made, the dentist must create a tooth preparation that not only removes the defect (e.g. caries lesion, old restorative material, malformed structure) but also leaves remaining tooth structure as strong as possible by leaving as much dentin support as possible. Tooth preparation that conserves tooth structure is strongly recommended because it limits pulpal irritation and preserves the integrity of both the tooth and the subsequent restoration.[27–33]

Discussion in this chapter will include principles, techniques and procedures using classic examples of amalgam preparations. However, clinically, the outline form of the preparation should always conform to the level of disease in the tooth and not necessarily to the examples presented here. Making the tooth preparation form appropriate for the use of amalgam as the restorative material is equally

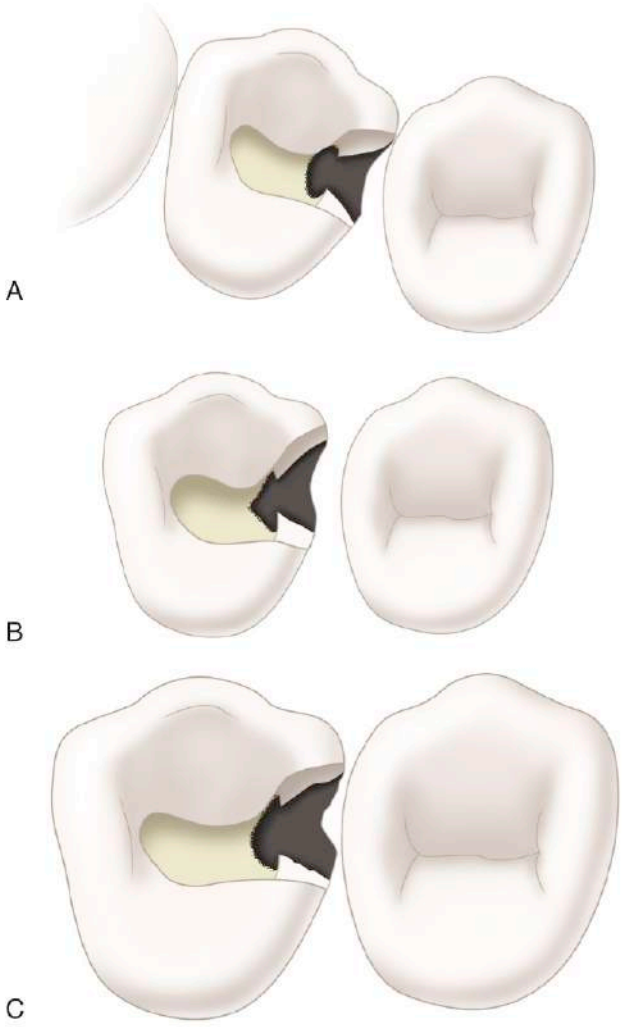


Figure 13.6 Preoperative visualization of tooth preparation extensions when a cavitated caries lesion is present gingival to the proximal contact and in the central groove area. (A) Rotated tooth (lingual extension owing to faulty central groove); (B) open proximal contact (preparation extended wider faciolingually to develop a proximal contact with appropriate physiologic proximal contours) and (C) normal relationship.

important. The physical properties of amalgam require that it be placed into a tooth preparation that provides for:

I. an approximately 90° or slightly greater occlusal cavosurface margin and 90° axial cavosurface margin (because of the amalgam's limited edge strength);
II. the preparation must allow a minimum thickness of 1.5–2 mm so that the amalgam will not flex and fracture when under occlusal load (most amalgams fail by bulk fracture) and
III. the amalgam must be placed into a prepared undercut form in the tooth so as to be mechanically retained (amalgam does not bond to tooth structure).

After appropriate tooth preparation, the success of the final restoration depends on proper insertion, carving and finishing of the amalgam material.

Principles

The basic principles of tooth preparation must be followed for amalgam tooth preparations to ensure clinical success. As presented in Chapter 11, the preparation for amalgam is discussed in two stages, for academic purposes, to facilitate student understanding of proper extension, form and caries lesion removal.

The initial stage:

I. places the tooth preparation extension into sound tooth structure at the marginal areas (not pulpally or axially);
II. extends the depth (pulpally, axially, or both) at the periphery of the preparation to a prescribed, uniform dimension;
III. provides an initial form consistent with amalgam retention in the tooth and
IV. establishes the preparation external walls in a form that will result in a 90° amalgam margin upon restoration.

The final stage of tooth preparation removes any remaining defect (caries lesion or old restorative material) and incorporates any additional preparation features (grooves, coves, slots, or pins) as needed to achieve the appropriate retention and resistance forms.

In academic institutions, assessing the tooth preparation after the initial preparation stage provides an opportunity to evaluate a student's knowledge and ability to extend the external walls properly and establish proper initial depth. If the student were to remove an extensive caries lesion in its entirety before any evaluation, the attending faculty would not know whether the prepared depths were obtained because of appropriate caries lesion removal or inappropriate overcutting of the tooth.

Initial Tooth Preparation

Step 1: Outline Form

Initial Depth

I. All initial depths of a tooth preparation for amalgam relate to the dentinoenamel junction (DEJ) except in the following two instances:
   i. When the occlusal enamel has become significantly thinner
   ii. When the preparation extends onto the root surface
II. *The initial depth pulpally is 0.2 mm inside (internal to) the DEJ or 1.5 mm as measured from the depth of the central groove* (Fig. 13.7), whichever results in the greater thickness of amalgam.
III. *The initial depth of the axial wall is 0.2 mm inside the DEJ when retention grooves are not used and 0.5 mm inside the DEJ when retention grooves are used* (Fig. 13.8).

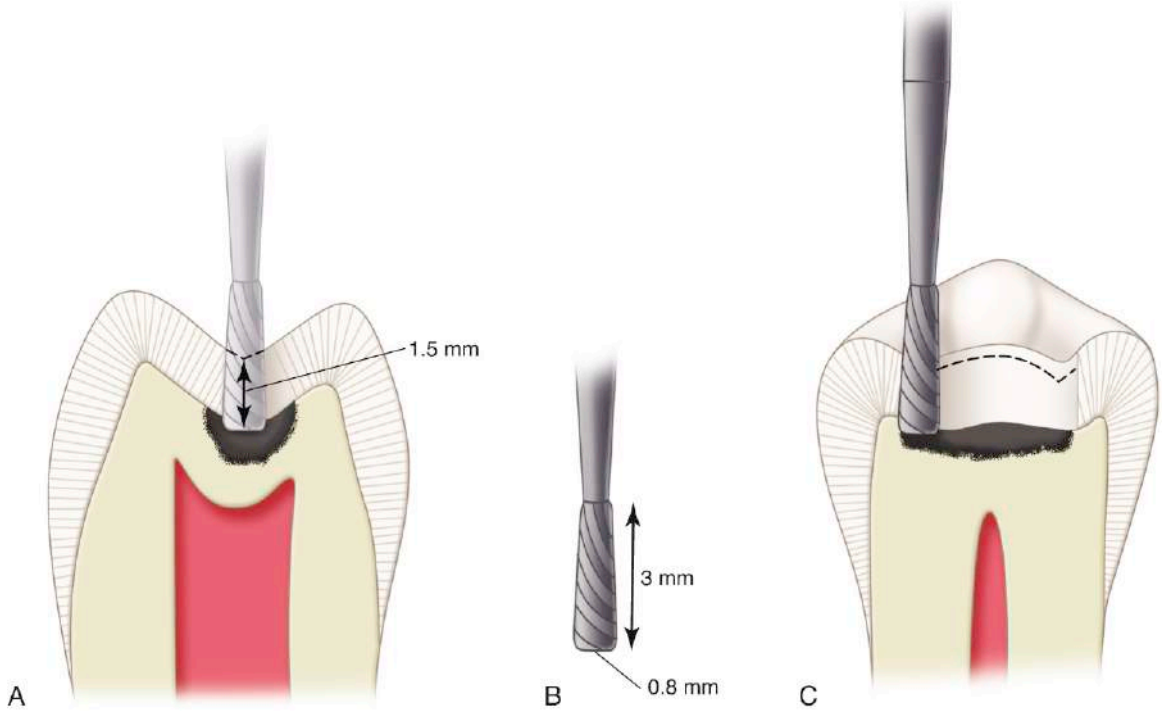


Figure 13.7 Pulpal floor depth. (A) Pulpal depth measured from central groove; (B) No. 245 bur dimensions and (C) guides to proper pulpal floor depth: (1) one-half the length of the No. 245 bur, (2) 1.5 mm, or (3) 0.2 mm inside (internal to) the dentinoenamel junction (DEJ).

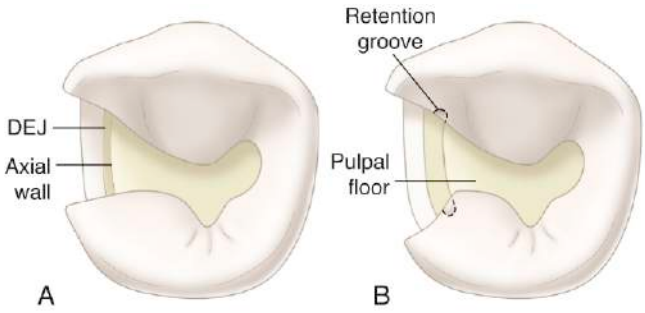


Figure 13.8 Axial wall depth. (A) If no retention grooves needed, axial depth 0.2 mm inside (internal to) the dentinoenamel junction (DEJ); (B) if retention grooves needed (i.e. proximal divergence secondary to wide F–L caries lesion extension), axial depth 0.5 mm inside (internal to) the DEJ. Note that *dashed lines* on the occlusal surface only indicate the position of the retention grooves on the facial and lingual walls of the proximal preparation. The grooves do not extend onto the occlusal surface.

IV. Axial depths on the root surface should be 0.75–1 mm deep so as to provide room for placement of retention grooves or coves.

Initial Extension

The initial extension of the tooth preparation should be *visualized preoperatively* by

I. the preparation extension is dictated primarily by the existing caries lesion, old restorative material, or defect
II. the preparation form requirements of the amalgam
III. adequate extension to provide access for tooth preparation, caries lesion removal, matrix placement and amalgam insertion also must be considered
IV. when making the preparation extensions, every effort should be made to preserve the dentinal support (i.e. the strength) of cusps and marginal ridges

Factors dictating the outline form are presented in greater detail in Chapter 11. They include caries lesion, old restorative material, inclusion of all of the defect, proximal or occlusal contact relationship and the need for convenience form.

Cavosurface Margin

Enamel cavosurface margins must be left at 90° or greater to limit the potential for enamel fracture. For enamel strength, the marginal enamel rods should be supported by sound dentin. These requirements for enamel strength must be combined with marginal requirements for amalgam (90° butt joint) when establishing the periphery of the tooth preparation (Fig. 13.9).

I. When viewed from the occlusal, the facial and lingual proximal cavosurface margins of a Class II preparation should be **90°** (i.e. perpendicular to a tangent drawn through the point of extension facially and lingually) (Fig. 13.9).
II. In most instances, preparation for proximal caries lesions result in facial and lingual proximal walls that

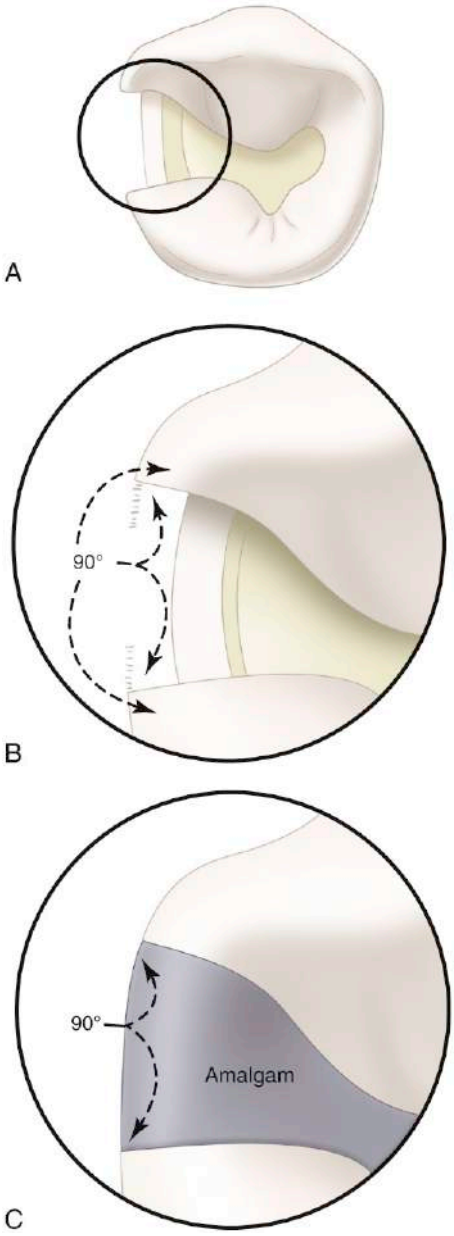


Figure 13.9 Proximal cavosurface margins. (A) Facial and lingual proximal cavosurface margins prepared at 90° angles to a tangent drawn through the point on the external tooth surface; (B) a 90° proximal cavosurface margin produces a 90° amalgam margin and (C) 90° amalgam margins.

must be extended into the facial or lingual embrasure. This extension provides adequate access for performing the preparation (with decreased risk of damaging the adjacent tooth), easier placement of the matrix band and easier condensation and carving of the amalgam. Such extension provides *clearance* between the cavosurface margin and the adjacent tooth (Fig. 13.10). Clearance also allows the operator to confirm that no voids exist at the proximal margins of the finished restoration. Occasionally, the level of disease may not require extension of the proximal margins beyond the proximal contact. This is especially helpful in areas that are more esthetically demanding.

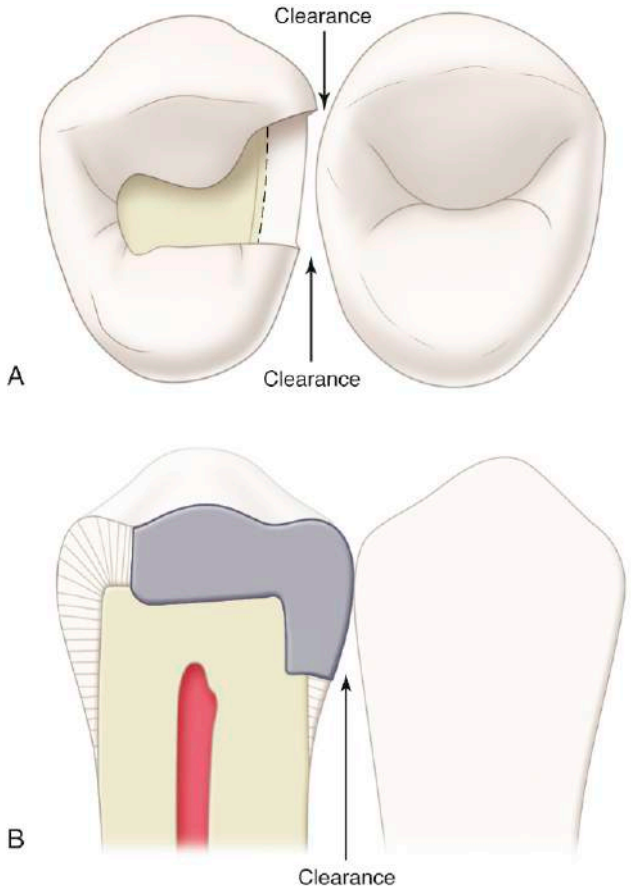


Figure 13.10 Proximal box preparation clearance of adjacent tooth. (A) Occlusal view and (B) lingual view of a cross-section through the central groove.

III. Preparation walls on the occlusal surface usually have obtuse enamel margins (representing the strongest enamel margin) and result in amalgam margins that are slightly less than 90° (Figs. 13.11 and 13.12). Rounding of the central groove area when carving the occlusal amalgam enables the marginal configuration to be closer to 90° (Fig. 13.12).

IV. Preparation walls on vertical aspect (in the occlusogingival axis) of the tooth (facial, lingual, mesial, or distal) should result in 90° enamel walls (representing a strong enamel margin) (Fig. 13.9) that meet the inserted amalgam at a *butt joint* (enamel and amalgam both having 90° margins).

Step 2: Resistance Form

Resistance form preparation features help the tooth and the restoration resist fractures caused by occlusal forces.

I. Resistance features that assist in preventing the tooth from fracturing include:
   i. maintaining as much tooth structure as possible (preserving the dentin supporting cusps and marginal ridges);

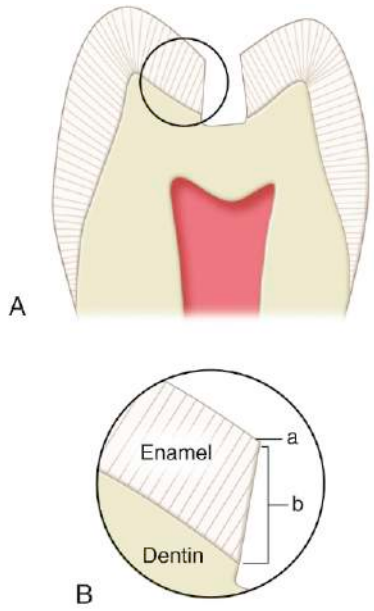


Figure 13.11 Occlusal cavosurface margins. (A) Tooth preparation and (B) occlusal margin representing the strongest enamel margin. Full-length enamel rods (*a*) and shorter enamel rods (*b*).

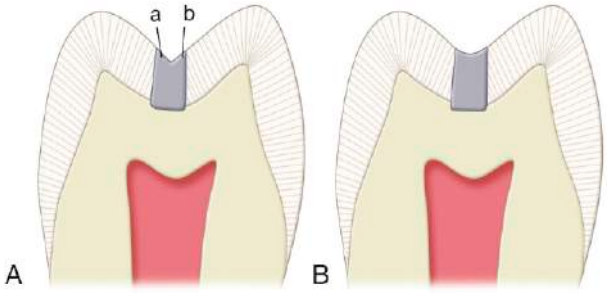


Figure 13.12 Amalgam form at occlusal cavosurface margins. (A) Amalgam carved too deep resulting in acute angles *a* and *b* and stress concentrations within the amalgam, increasing the potential for fracture; and (B) amalgam carved with appropriate anatomy, resulting in an amalgam margin close to 90°, although the enamel cavosurface margin is obtuse.

ii. having pulpal and gingival walls prepared perpendicular to occlusal forces, when possible;
iii. having rounded internal preparation angles;
iv. removing unsupported or weakened tooth structure and
v. placing slots and pins into the tooth as part of the final stage of tooth preparation, when indicated. The placement of slots and pins is considered a secondary resistance form feature and is discussed in Chapter 14.

II. Resistance form features that assist in preventing the amalgam from fracturing include
i. adequate thickness of amalgam (at least 1.5–2 mm in areas of occlusal contact and 0.75 mm in axial areas);

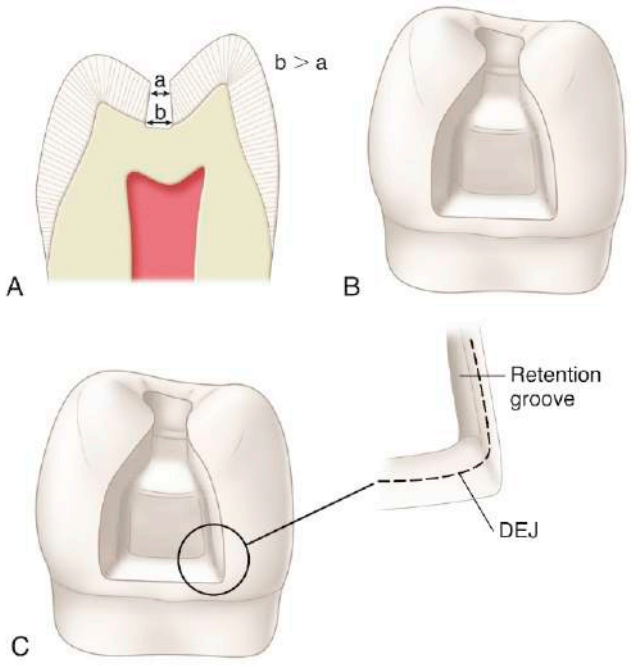


Figure 13.13 Typical amalgam tooth preparation retention form features. (A and B) Occlusal convergence of prepared walls (primary retention form) and (C) retention grooves in proximal box (secondary retention form) if the proximal preparation is wide F–L.

ii. amalgam margin of 90°;
iii. boxlike preparation form, which provides uniform amalgam thickness and
iv. rounded axiopulpal line angles in Class II tooth preparations. Many of these resistance form features may be achieved using the No. 330 or No. 245 bur.

Step 3: Retention Form

Retention form preparation features retain (i.e. 'lock') the restorative material in the tooth. For composite restorations, adhesion (bonding) provides most of the needed retention. Amalgam restorations must be mechanically retained in the tooth. Amalgam retention form (Fig. 13.13) is provided by

I. preparation of the vertical walls (especially facial and lingual walls) that converge occlusally (i.e. *primary retention*);
II. retention features such as grooves, coves, slots and pins that are placed during the final stage of tooth preparation (i.e. *secondary retention*), and
III. engagement of the inserted amalgam into any surface irregularities in the preparation that may exist. Primary retention form features are achieved by the orientation and type of the preparation instrument such as the No. 330 or No. 245 pear-shaped carbide bur or diamond (Fig. 13.7B). Secondary retention form features are discussed in subsequent sections.

Step 4: Convenience Form

Convenience form includes features that allow adequate access and visibility of the operating site to facilitate tooth preparation and restoration. Convenience form includes extension of the outline form so that

I. the caries lesion may be accessed for removal;
II. the matrix may be placed and
III. amalgam may be inserted, carved and finished—all while maintaining resistance form.

Final Tooth Preparation

Step 5: Removal of Defective Restorative Material and/or Soft Dentin

Any old restorative material or soft dentin remaining after the initial preparation will be located in the axial or pulpal walls (the extension of the peripheral preparation margins should have already been to sound tooth structure). Chapters 2 and 11 discuss:

I. when to leave or remove old restorative material,
II. how to remove remaining soft dentin and
III. what should be done to protect the pulp.

The removal of remaining soft dentin should be accomplished using the largest instrument that fits the carious area because it is least likely to penetrate the tooth in an uncontrolled manner. A hand instrument is more reliable than a rotating bur for judging the adequacy of removal of soft dentin. The removal of carious dentin should be stopped when the tooth structure feels firm unless exposure of the pulp becomes likely (Chapter 2). This situation often occurs before all lightly stained or discoloured dentin is removed.[34]

Pulp Protection

- *Ideal or shallow depth of preparation*: If the tooth preparation is of ideal or shallow depth, no liner or base is indicated.
- *Remaining dentin thickness (RDT) of 1.0–1.5 mm*: In deeper caries removal (where the RDT is judged to be 1–1.5 mm), a layer (i.e. 0.5–0.75 mm) of a resin-modified glass ionomer (RMGI) material should be placed (Fig. 13.14).[35,36] The RMGI insulates the pulp from thermal changes, bonds to dentin, releases fluoride, is strong enough to resist the forces of condensation and reduces micro-leakage.[36–38] The RMGI is applied only over the deepest portion of the caries removal. It should be placed in small increments and should flow when it is touched to the dentin surface. The entire dentin surface should not be covered. Dentin peripheral to the liner should be available for support of the restoration.[39]
- *RDT of <0.5 mm*: For pulpal protection in very deep caries removal (where the RDT is judged to be <0.5 mm and suspicion of potential microscopic pulpal exposure is increased), a thin layer (i.e. 0.5–0.75 mm) of a calcium hydroxide liner may be placed (Fig. 13.15). The calcium hydroxide liner may elicit tertiary dentin formation if the original odontoblasts are no longer vital. If the calcium hydroxide liner is used, it is placed by using the same instrument and the same technique as described for the RMGI liner. The calcium hydroxide liner should be placed only over the deepest portion of the caries removal (nearest the pulp). A layer of RMGI liner should be used to cover the calcium hydroxide.[40] The entire dentin surface should not be covered (Figs. 13.14 and 13.15). The RMGI liner is recommended to cover the calcium hydroxide to resist the forces of condensation and to prevent dissolution over time by sealing the deeply excavated area.[37]
- *Pulp exposure* : A pulpal exposure will require a direct pulp cap with calcium hydroxide or mineral trioxide aggregate, or endodontic treatment.[41]

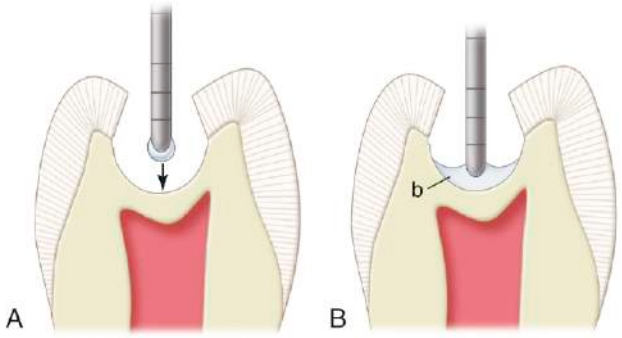


Figure 13.14 Liner/Base application. (A) Inserting resin-modified glass ionomer (RMGI) with periodontal probe and (B) in moderately deep caries removal, a base (*b*) thickness of 0.5–0.75 mm is indicated.

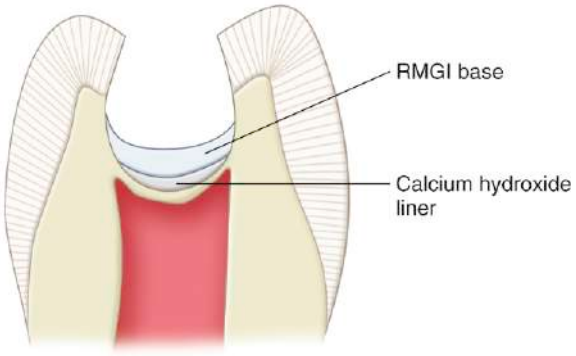


Figure 13.15 Placement of calcium hydroxide liner and RMGI base.

Step 7: Secondary Resistance and Retention Forms

Adequate resistance and retention form may no longer be present in the tooth preparation after caries lesion removal has resulted in excessive loss of vertical wall height. Therefore, strategic features (such as the placement of grooves, coves, slots, pins) that enhance the resistance and retention form are added. *Usually, the larger the tooth preparation, the greater the need for secondary resistance and retention forms*.

Step 8: Final Procedures: Debridement and Inspection

After the previous steps are performed, the completed tooth preparation should be cleaned of all debris (debridement) so that a careful inspection may be accomplished. The usual procedure in cleaning is to free the preparation of visible debris using short bursts of water spray from the air-water syringe followed by short bursts with the air syringe. If debris still clings to the preparation, it may be necessary to dislodge this material with an explorer, a small damp cotton pellet, or a commercially available applicator tip (i.e. a 'microbrush') moistened with water. Excess water should be removed so as to allow a clear field for inspection. However, it is important not to desiccate the tooth by overuse of air as this may damage the odontoblasts associated with the desiccated tubules. The preparation should be viewed from all angles. Careful assessment should be made to ensure that the depths are proper, the external walls provide for enamel support and correct orientation for amalgam (i.e. that the preparation is retentive and the resultant restoration margins will not be prone to fracture), and the caries lesion has been removed as indicated. Preparation walls should be smooth and transitions should be gently rounded.[42] The cavosurface margin should be precise and clearly visualized.

Preparation Designs

Typical tooth preparation for amalgam has generally been referred to as *conventional tooth preparation*. A 'box-only' tooth preparation is indicated if no occlusal caries lesion is present (i.e. only a proximal caries lesion is present). Fig. 13.16 illustrates various preparation designs. Appropriate details of specific tooth preparations are presented subsequently.

General Concepts Guiding Restoration with Amalgam

After tooth preparation, the tooth must be readied for the insertion of amalgam. Disinfectants may be used but are not considered essential.[43,44] However, it is highly recommended that a *dentin desensitizer* (current commercial formulations

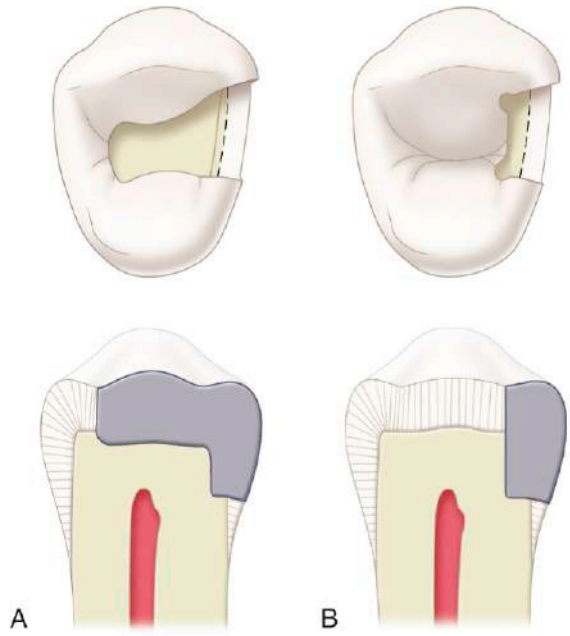


Figure 13.16 Types of amalgam tooth preparations. (A) Conventional and (B) box only.

contain 5% glutaraldehyde and 35% 2-hydroxyethyl methacrylate [HEMA] and water) be placed on the prepared dentin per manufacturer's instructions (Fig. 13.3). This type of dentin desensitizer cross-links tubular fluid proteins and, in the presence of HEMA, forms lamellar plugs in the dentinal tubules.[45] These plugs are thought to be responsible for reducing the potential for rapid tubular fluid movement and, thereby, the sensitivity of dentin. This step may occur before or after the matrix application.

Matrix Placement

A matrix primarily is used when a proximal surface is to be restored.

The *objectives* of a matrix are to

I. provide proper contact,
II. provide proper contour,
III. confine the restorative material,
IV. reduce the amount of excess material and
V. help protect the adjacent tooth from being damaged during preparation.

For a matrix to be effective, it should be easy to apply and remove, extend below the gingival margin enough that it can be engaged by a wedge, extend above the adjacent marginal ridge height so as to allow for proper condensation and resist deformation during material insertion. The various classifications of matrixes is given in Box 13.1.

Mixing (Triturating) the Amalgam

Because of its superior clinical performance, *high-copper amalgam* is recommended. Pre-proportioned, disposable

Box 13.1 Classification of matrixes

I. **Based on mode of retention**
   i. With retainer, e.g. Tofflemire matrix
   ii. Without retainer, e.g. Automatrix
II. **Based on type of band**
   i. Metallic nontransparent matrixes
   ii. Nonmetallic transparent matrixes
III. **Based on type of cavity for which it is used**
   i. **Matrix for class I cavity preparation**
      – Double banded Tofflemire (Barton matrix)
   ii. **Matrixes for class II cavity preparation**
      – Single banded Tofflemire matrix
      – Rigid material supported sectional matrix
      – Ivory matrix no. 1
      – Ivory matrix no. 8
      – Copper band matrix
      – Anatomical matrix
      – Automatrix
   iii. **Matrixes for Class III cavities**
      – Mylar strip matrix
      – S-shaped matrix
   iv. **Matrixes for class IV cavities**
      – Custom lingual matrix
      – Mylar strip matrix
      – Transparent crown form matrix
      – Modified S-shaped band matrix
   v. **Matrixes for class V cavities**
      – Window matrix
      – Cervical matrix

capsules are available in sizes ranging from 400 to 800 mg. Some pre-capsulated brands require activation of the capsules before trituration. The speed and time of trituration are factors that impact the setting reaction of the material. Alterations in either may cause changes in the properties of the inserted amalgam. Amalgam should be triturated (i.e. mixed) according to the manufacturer's directions. Correctly mixed amalgam should not be dry and crumbly; rather, it should retain sufficient 'wetness' so as to aid in achieving a homogeneous and well-adapted restoration.[46] It is often necessary to make several mixes to complete the restoration, particularly for large preparations.

Insertion of the Amalgam

An amalgam carrier is used to transfer amalgam to the tooth preparation. Increments extruded from the carrier should be smaller (often only 50% or less of a full-carrier tip) for a small preparation, particularly during the initial insertion. A *flat-faced, circular, or elliptic condenser* may be used to condense amalgam in the deeper areas of the preparation. The principal objectives during the insertion of amalgam are to condense the amalgam mass and to adapt it to the preparation walls and the matrix (when used).

Thorough and optimal condensation is essential:

I. To produce a restoration free of voids, and it helps to reduce marginal leakage.[47,48]
II. To minimize the mercury content in the restoration, which decreases corrosion and enhances restoration strength and marginal integrity.[48]

Spherical amalgam. This form of amalgam alloy is more easily condensed than admixed (lathe-cut) amalgam. Condensation of amalgam that contains spherical particles requires larger condensers than are commonly used for admixed amalgam. Smaller condensers tend to penetrate a mass of spherical amalgam, resulting in less effective force to compact or adapt the amalgam within the preparation.

Admixed amalgam. Many practitioners prefer the handling properties of the admixed type. Smaller condensers are indicated for the initial increments of admixed amalgam because it is more resistant to condensation pressure. This is because the area of a circular condenser face (nib) increases by the square of the diameter; doubling the diameter requires 4 times more force for the same pressure per unit area.

Each portion must be thoroughly condensed prior to placement of the next increment. Each condensed increment should fill only one-third to one-half the preparation depth. Each condensing stroke should overlap the previous condensing stroke to ensure that the entire mass is well condensed.

Lateral condensation (facially, lingually and proximally directed condensation) is important in the proximal box portions of preparations to ensure confluence of the amalgam with the margins, the elimination of voids and an adequate proximal contact. Generally, smaller amalgam condensers are used first, which allows the amalgam to be properly condensed into the internal line angles and secondary retention features. Subsequently, larger condensers are used. Amalgam preparations should be somewhat overfilled to ensure adequate condensation on the occlusal surface. The condensation of a mix should be completed within the time specified by the manufacturer (usually 2.5–3.5 min). Otherwise crystallization of the unused portion will be too advanced to react properly with the condensed portion. The mix should be discarded if it becomes dry, and another mix quickly made to continue the insertion.

Pre-carve Burnishing

To ensure that the marginal amalgam is well condensed before carving, the overpacked amalgam should be burnished immediately with a large burnisher, using heavy strokes mesiodistally and faciolingually, a procedure referred to as *pre-carve burnishing*. Pre-carve burnishing is useful to finalize the condensation, remove excess mercury-rich amalgam and initiate the carving process. To maximize its effectiveness, the burnisher head should be large enough that in the final strokes it contacts the cusp slopes but not the preparation margins. Pre-carve burnishing

produces denser amalgam at the margins of the occlusal preparations restored with high-copper amalgam alloys and initiates contouring of the restoration.[49,50]

Carving the Amalgam

The following discussion of carving (and shaping) of amalgam assumes the use of *sharp carving instruments*. All carving instruments dull with repeated use and sterilization cycles and, as a result, lose their efficiency. Carving a freshly condensed amalgam, which is setting and steadily getting harder, with a dull instrument requires the use of ever-increasing pressure on the instrument and increases the likelihood of losing control (slipping) and/or increasing the amount of time required to complete the carving. Always use sharp carving instruments.

The amalgam material selected for the restoration has a specific setting time. After pre-carve burnishing has been accomplished, the remainder of the accessible restoration must be contoured to achieve proper form and, as a result, function. The insertion (condensation) and carving of the material must occur before the material has hardened so much that it becomes uncarvable.

Occlusal Areas

A *discoid–cleoid instrument* may be used to carve the occlusal surface of an amalgam restoration. The rounded end (discoid) is positioned on the unprepared enamel adjacent to the amalgam margin and pulled parallel to the margin (Fig. 13.17). This removes any excess at the margin while not allowing the marginal amalgam to be carved below the preparation margins (i.e. 'submarginated'). The *pointed end* (*cleoid*) of the instrument may be used to define the primary grooves, fossae and cuspal inclines. The *Hollenback carver* is also useful for carving these areas.

The reproduction of grooves and fossae is necessary to provide appropriate mastication and sluiceways for the escape of food from the occlusal table. The mesial and distal fossae are carved to be inferior to the marginal ridge height, helping limit the potential for food to be wedged into the occlusal embrasure. Having rounded and relatively shallow occlusal anatomy also helps achieve a 90° amalgam margin on the occlusal surface and to ensure adequate occlusogingival dimension of the final amalgam restoration for strength (Fig. 13.12B).

For multiple surface restorations (which require use of a matrix), the initial carving of the occlusal surface should be rapid, concentrating primarily on the marginal ridge height and occlusal embrasure areas. Occlusal embrasure areas are developed with a thin explorer tip or carving instrument by mirroring the contours of the adjacent tooth. The explorer tip is pulled along the inside of the matrix band, creating the occlusal embrasure form. When viewed from the facial or lingual direction, the embrasure form created should be identical to that of the adjacent tooth, assuming that the adjacent tooth has appropriate

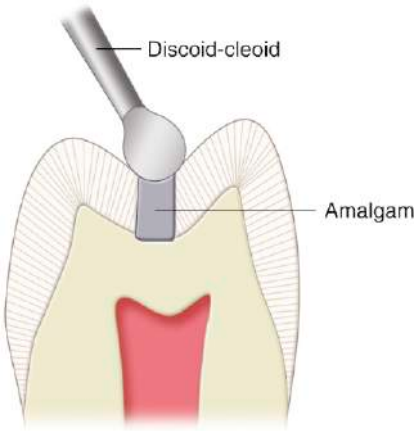


Figure 13.17 Carving the occlusal margins.



Figure 13.18 Defining the marginal ridge and the occlusal embrasure with an explorer.

contour. Likewise, the height of the amalgam marginal ridge should generally be the same as that of the adjacent tooth (Figs. 13.18 and 13.19). If both these areas are developed properly, the potential for fracture of the marginal ridge area of the restoration while checking the occlusion is significantly reduced. Placing the initial carving emphasis on the occlusal areas for a large restoration permits the operator to remove the matrix more quickly so as to carve any extensive axial surfaces of the restoration, especially the proximal areas. Some of these areas may be relatively inaccessible and must be carved while the amalgam material is still not fully set. Remaining necessary carving or contouring on the occlusal surface may be done later, and if the amalgam is too hard to carve, the use of rotary instruments in the handpiece may be required.

Facial and Lingual Areas

Most facial and lingual areas are accessible and may be carved directly. The side of an explorer tine may be very effective in creating correct contours when the amalgam is early in its setting reaction. A Hollenbeck carver or base of the amalgam knife (scaler 34/35) is also useful for carving these areas. With regard to the cervical areas, it is important to remove any excess and develop the proper

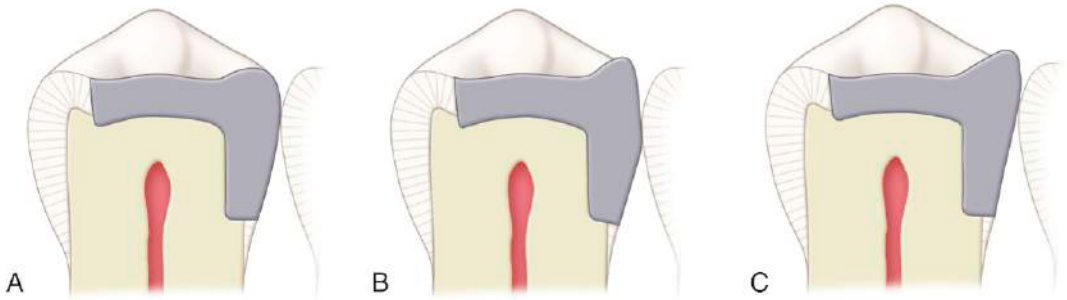


Figure 13.19 Proximal contour. (A) Correct proximal contour; (B) incorrect marginal ridge height and occlusal embrasure form and (C) the occlusogingival proximal contour is too straight, the contact is too high, and the occlusal embrasure form is incorrect.

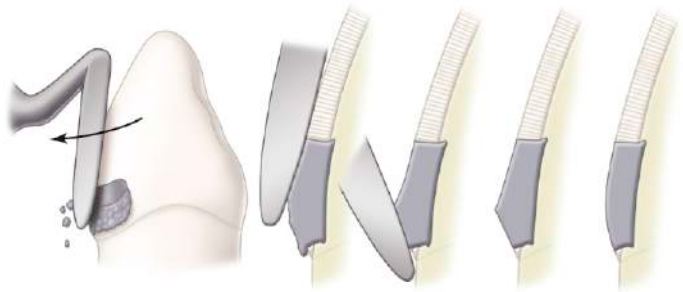


Figure 13.20 Positioning of the carving instrument to prevent over-carving amalgam and to develop the desired gingival contours.

contour of the restoration, which is usually convex. The convexity is developed by using the occlusal and gingival unprepared tooth structure as guides for initiating the carving (Fig. 13.20). The marginal areas are blended together, resulting in the desired convexity and providing a physiologic contour that promotes the health of adjacent gingival tissue.

Proximal Embrasure Areas

The initial development of the occlusal embrasure of the proximal surface while the matrix is still in place has already been described. After removal of the matrix, the amalgam knife (or scaler 34/35) is an excellent instrument for removing proximal excess and developing proximal contours and embrasures (Figs. 13.21 and 13.22). The knife is positioned below the gingival margin and drawn occlusally to carefully shave off excess, to refine the proximal contour (below the contact) and the gingival embrasure form. The sharp tip of the knife also is beneficial in developing the facial and lingual embrasure forms. Care must be exercised in not carving away any of the contour necessary to create the desired proximal contact.

Development of proper proximal contour and contact is important for the physiologic health of interproximal soft tissue. Likewise, ensuring a smooth junction between the tooth and the amalgam is important. An amalgam overhang (excess of amalgam) may result in compromised gingival health. Voids at the cavosurface margins may result in recurrent caries lesion formation.

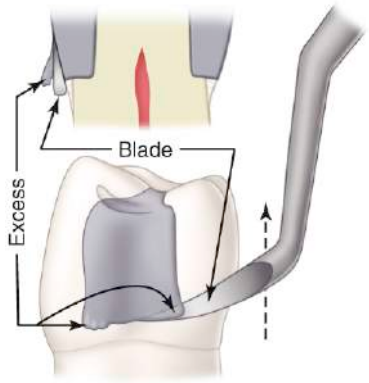


Figure 13.21 Gingival excess may be removed with amalgam knives.

The proximal portion of the carved amalgam is evaluated by *visual assessment* (reflecting light into the contact area from the lingual and occlusal perspectives to confirm the dimensions and location of the proximal contact) and placement of very *thin dental floss* through the contact area. If dental floss is used, it must be used judiciously, ensuring that amalgam in the contact area is not inadvertently removed. A piece of floss may be inserted through the contact and into the gingival embrasure area by initially wrapping the floss around the adjacent tooth and exerting pressure on that tooth rather than the restored tooth while moving the floss through the contact area. When the floss is into the gingival embrasure area, it is wrapped around the restored tooth and moved occlusally and gingivally to

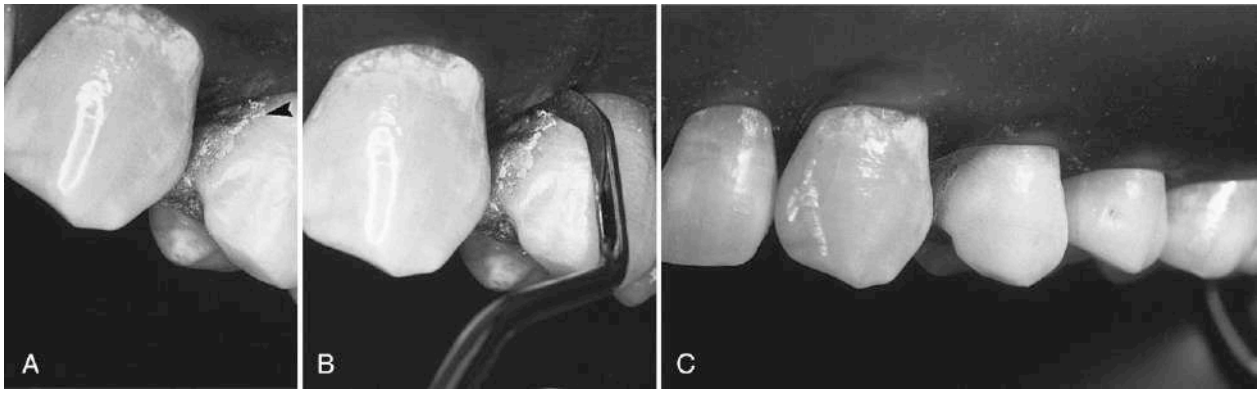


Figure 13.22 Removal of gingival excess of amalgam. (A) Excess of amalgam (*arrowhead*) at the gingival corner of the restoration; (B) use of the amalgam knife for removal of gingival excess and (C) gingival corner of restoration with excess removed.

determine whether excess exists and to smooth the proximal amalgam material. If excess material is detected along the gingival margin, the amalgam knife should again be used until a smooth margin is created.

Post-carve Burnishing

Some operators prefer to *post-carve* burnish the amalgam surface by using a *small burnisher*. Post-carve burnishing is done by lightly rubbing the carved surface with a burnisher of suitable size and shape to improve smoothness and produce a satin (not shiny) appearance. The surface should not be rubbed hard enough to produce grooves in the amalgam. Post-carve burnishing may improve the marginal integrity of low- and high-copper amalgams and may improve the smoothness of the restoration.[51–53]

Post-carve burnishing in conjunction with pre-carve burnishing may serve as a viable substitute for conventional polishing.[53]

Evaluation of Occlusal Contact Areas on the Restoration

After completion of the carving and the removal of the rubber dam, the occlusal contacts on the restoration must be evaluated. Even if the carving has been carefully accomplished, the restoration occasionally is 'high', indicating premature occlusal contact. A piece of articulating paper is placed over the restoration and adjacent teeth, and the patient is instructed to close gently into occlusion. *It is essential that the teeth be completely dry for proper marking with articulating paper*. The patient is advised not to bite firmly because of the danger of fracturing the restoration, which is weak at this stage. The operator should visually assess the contact potential of the restored tooth and the extent of closure. If local anaesthesia is still present, it may be difficult for the patient to tell when the teeth are in contact. After the patient has reopened the mouth and the articulating paper is removed, the following two features of the occlusal relationship suggest that the restoration is high:

I. cusp tips of adjacent teeth are not in occlusal contact when it is known from the preoperative occlusal assessment that they should be touching, and
II. an opposing cusp prematurely occludes with the new restoration.

Contact areas on the amalgam should be assessed by the colour imparted by the articulating paper. Deeply coloured areas (heavy contacts) and those with light-coloured centres (even heavier contacts) are reduced until all markings are uniformly of a light hue, and contacts are noted on adjacent teeth (Fig. 13.23). Heavy occlusal contacts are commonly referred to as 'prematurities' or 'high spots'. Observing the bite opening between nearby teeth when the restoration is in contact indicates how much occlusal adjustment is indicated. For example, if the adjacent teeth are 0.5 mm apart (by visual estimation), the high spot(s) should be reduced by approximately that amount. This observation may expedite the occlusal adjustment compared with making an insufficient adjustment and having to repeat closure-and-carving numerous times. The sequence of closure, observation and carving is repeated until the appropriate surfaces of opposing teeth are touching.

Occlusal contacts on a cuspal incline or ridge slope are undesirable because they cause a deflective lateral force on the tooth. These should be adjusted until the resulting contact is stable (i.e. the occlusal contact should be perpendicular to the occlusal load where possible in maximum inter-cuspation, or, in other words, the occlusal force should be parallel to the long axis of the tooth).

To this point in the procedure, the patient has been instructed to close vertically into maximum inter-cuspation (i.e. 'bite together on your back teeth'). After replacing the articulating paper over the tooth, the patient is asked to occlude lightly and to slide the teeth lightly from 'side to side' to identify excursive interferences and 'front to back' to identify protrusive interferences. Any additional occlusal marks are evaluated, and undesirable contact areas (interferences) are eliminated. Note that amalgam restorations carved completely out of occlusion (i.e. that are 'over-carved') may allow supra-eruptive tooth movement resulting in undesirable tooth contacts and this, too, should be

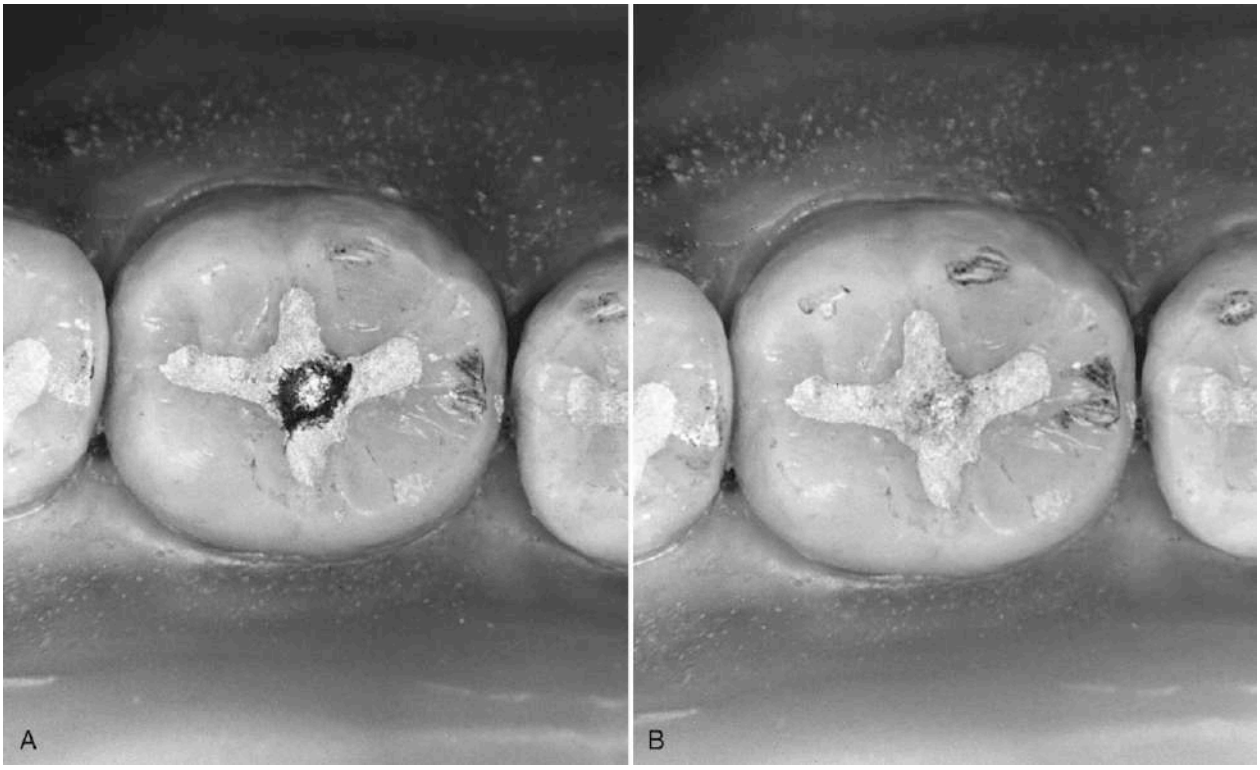


Figure 13.23 Occluding the restoration. (A) Heavy occlusal contacts on new amalgam should be adjusted. Articulating paper marks heavy contacts as dark areas, and it marks very heavy contacts as dark areas with light-coloured, potentially shiny, centres and (B) amalgam should not be carved out of occlusion. Rather, it should have light occlusal contact or contacts, as indicated by faint markings.

avoided. Finally, the patient should be cautioned to protect the newly condensed restoration by not using it to chew firm food for 24 h.

Finishing and Polishing of the Amalgam

Most amalgams do not require further finishing and polishing. These procedures are occasionally necessary, however, to:

I. Refine the anatomy, contours and marginal integrity and
II. Refine the surface texture of the restoration.

Additional finishing and polishing procedures for amalgam restorations are not attempted within 24 h of insertion because crystallization of the restoration is incomplete.[46] If used, these procedures are often delayed until all of the patient's amalgam restorations have been placed, rather than finishing and polishing periodically during the course of treatment. An amalgam restoration is less prone to tarnish and corrosion if a smooth, homogeneous surface is achieved.[54–56] *Polishing of high-copper amalgams is less important than it is for low-copper amalgams because high-copper amalgams are less susceptible to tarnishing and marginal breakdown.*[21,29,55–59]

The finishing procedure may be initiated by use of a *green carborundum* or *white alumina stone* (Fig. 13.24A). The green stone is more abrasive than the white stone; the tip of either stone may be blunted on a diamond wheel before use. This helps prevent marring the centre of the restoration while the margins are being adjusted. During the surfacing of amalgam, the stone's long axis is held at a 90° angle to the margins. Reduction of any occlusal contact should be avoided. After the stone is used, the margins should be re-evaluated with the tine of an explorer and any additional discrepancies removed. The surface may be smoothed further using light pressure with an appropriate finishing bur (Fig. 13.24B). A large, round finishing bur is generally used for this finishing step. If the groove and fossa features are not sufficiently defined, a *small round finishing bur* also may accentuate them without eliminating the occlusal contact areas. The long axis of the bur or stone should be at a ~45° angle to the margin to allow the unprepared tooth structure to guide the bur and prevent unnecessary removal of amalgam (Fig. 13.24C). A smooth surface should be achieved before the polishing procedure is initiated. The finishing bur should remove the minor scratches that resulted from use of the green or white stone. Often, however, these scratches can be removed only with the use of rubber abrasive points.

The polishing procedure is initiated by using a *coarse, rubber abrasive point* at low-speed and air-water spray to produce an amalgam surface with a smooth, satin appearance (Fig. 13.24D and E). If the amalgam surface does not exhibit

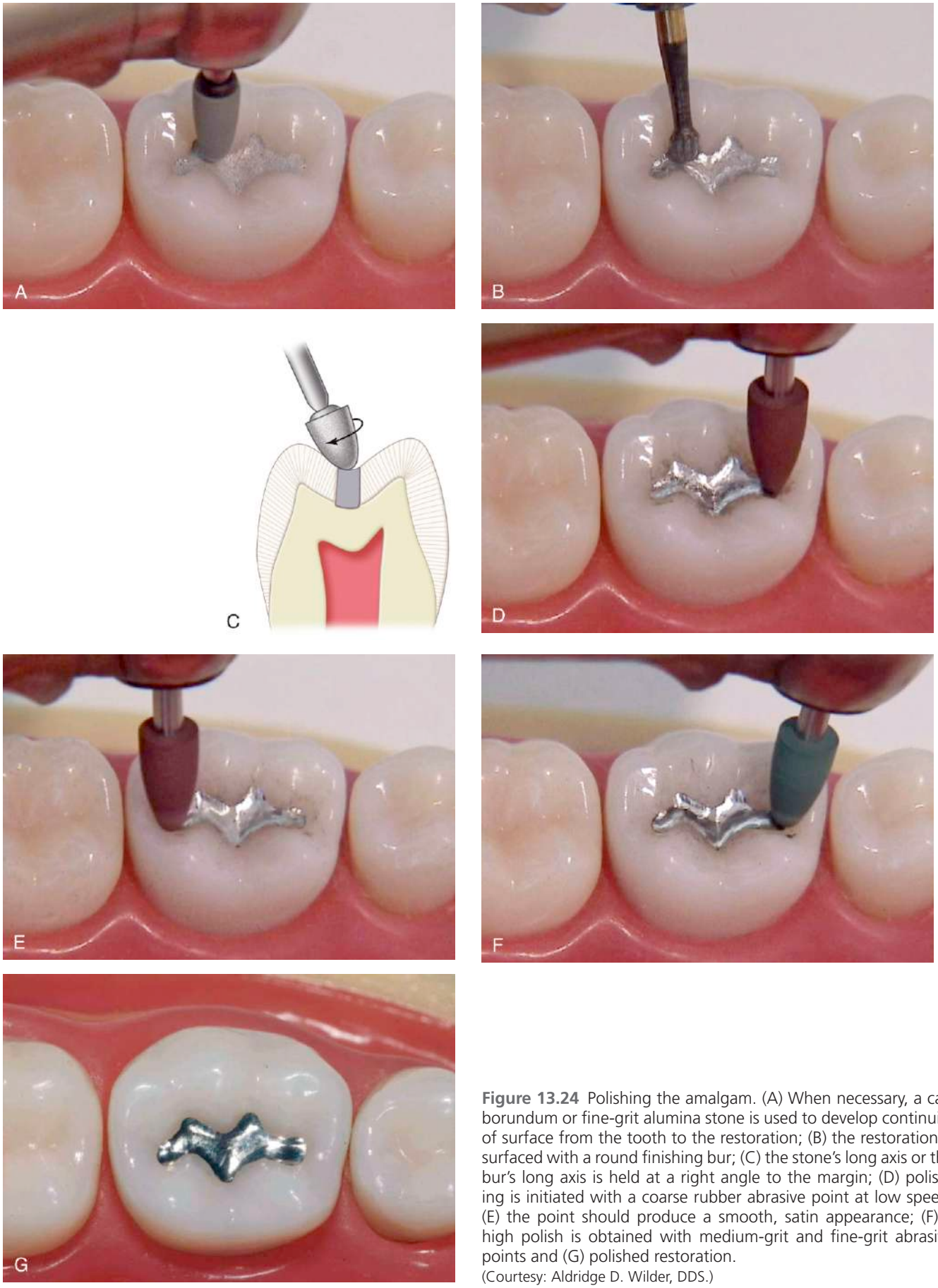


Figure 13.24 Polishing the amalgam. (A) When necessary, a carborundum or fine-grit alumina stone is used to develop continuity of surface from the tooth to the restoration; (B) the restoration is surfaced with a round finishing bur; (C) the stone's long axis or the bur's long axis is held at a right angle to the margin; (D) polishing is initiated with a coarse rubber abrasive point at low speed; (E) the point should produce a smooth, satin appearance; (F) a high polish is obtained with medium-grit and fine-grit abrasive points and (G) polished restoration.
(Courtesy: Aldridge D. Wilder, DDS.)

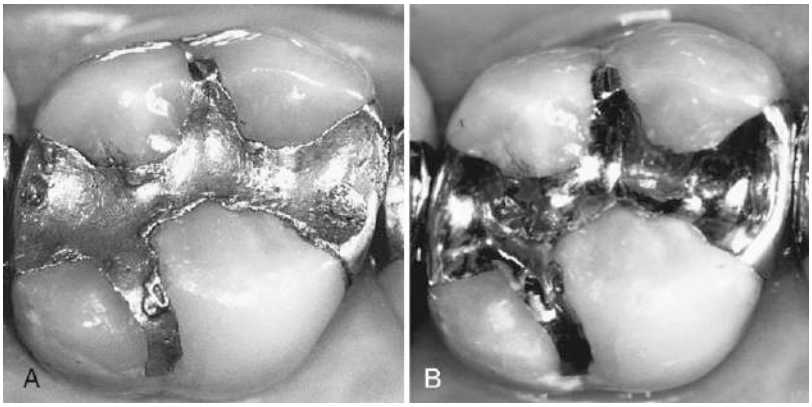


Figure 13.25 (A) Existing amalgam restoration exhibiting marginal deterioration and surface roughness and (B) same restoration after finishing and polishing.

this appearance after only a few seconds of polishing, the surface was too rough at the start. In this instance, resurfacing with a finishing bur is necessary, followed by the coarse, rubber abrasive point to develop the satiny appearance. It is important that the rubber points be used at *low speed* (≤6000 revolutions per minute [rpm]) or just above 'stall out' speed so as to limit the danger of point disintegration (which may occur at high rotational speeds) and the danger of frictional temperature elevation of the restoration and the tooth. Temperature above 140 °F [>60 °C] is able to cause irreparable damage to the pulp, the restoration, or both. When overheated, the amalgam surface appears cloudy, even though it may have a high polish. This cloudy appearance indicates that mercury has been brought to the surface, which results in increased corrosion of the amalgam and loss of strength.[46]

Polishing with the coarse abrasive rubber point will result in a moderately polished surface. No deep scratches should remain on the amalgam surface. After the area is washed free of abrasive particles and dried, a high polish may be imparted to the restoration with a series of medium-grit and fine-grit abrasive points (Fig. 13.24F). As with the more abrasive points, the finer abrasive points must be used at a low speed. If a high lustre does not appear within a few seconds, the restoration requires additional polishing with the more abrasive points. The system that is illustrated in Fig. 13.24 includes *coarse-grit*, *medium-grit* and *fine-grit rubber abrasive points*. Using these points in sequence, from coarse to fine, produces an amalgam surface with a brilliant lustre (Fig. 13.24G). As an alternative to rubber abrasive points, final polishing may be accomplished using a rubber cup with flour of pumice followed by a high-lustre agent such as precipitated chalk. Finishing and polishing of older, existing restorations may be performed to improve their contour, margins, surface, or anatomy, when indicated (Fig. 13.25).

These procedures should not leave the restoration undercontoured and should not alter the carefully designed occlusal contacts. The tip of an explorer should pass from the tooth surface to the restoration surface (and vice versa), without jumping or catching, thus verifying continuity of contour across the margin. Every effort to avoid removal of adjacent tooth structure should be made.

Repairing an Amalgam Restoration

If an amalgam restoration fractures during insertion, generally all of the inserted amalgam must be removed and new amalgam condensed. If a small portion of amalgam fractures during insertion and the amalgam is still carvable, it may be possible to apply and condense newly triturated amalgam to repair the affected area. If a small void in the amalgam is discovered after the matrix is removed, for example, and if the amalgam is still carvable and the area is accessible, any poorly condensed amalgam in the void should be removed with an explorer or other instrument, and the void repaired with newly triturated amalgam. In cases where a new mix of amalgam is added to carvable existing amalgam, the 'new' amalgam will adhere to the 'old'.

If a repair needs to be made to set, uncarvable amalgam, the defective area may be re-prepared as if it were a small restoration. Appropriate depth and retention form must be generated, sometimes entirely within the existing amalgam restoration. If necessary, another matrix must be placed. A new mix of amalgam may be condensed directly into the defect.

Clinical Technique for Class I Amalgam Restorations

This section describes the use of amalgam for Class I restorations. Class I restorations restore defects on the occlusal surfaces of posterior teeth, the occlusal thirds of the facial and lingual surfaces of molars, and the lingual surfaces of maxillary anterior teeth (Fig. 13.26).

The procedural description for a small (i.e. 'conservative') Class I amalgam restoration simply and clearly presents the basic information relating to the entire amalgam restoration technique, including tooth preparation and placement and contouring of the restoration. This basic procedural information may be expanded to describe Class I amalgam restoration of lesions/defects of all sizes. The maxillary first premolar is used for illustration in this section.

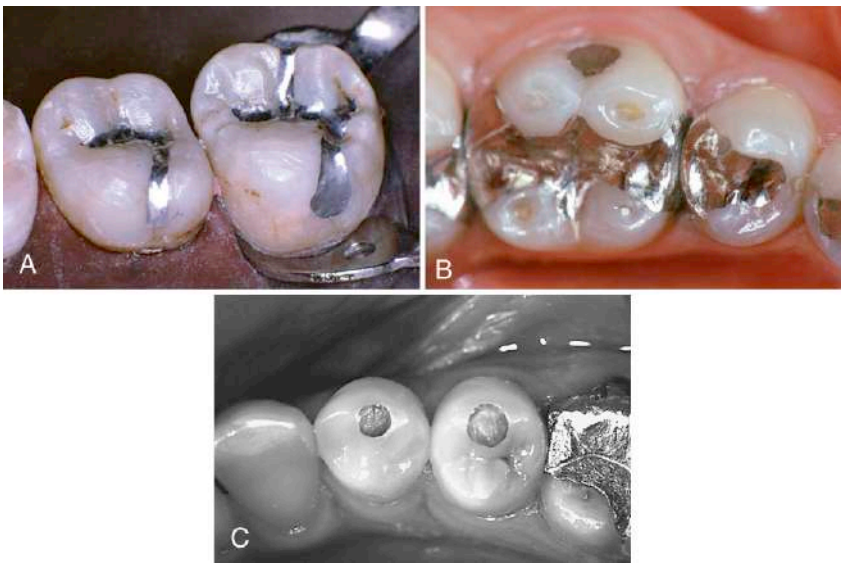


Figure 13.26 Clinical examples of Class I, II and VI amalgam restorations. (A) Class I amalgam in the occlusal surface of the first molar; (B) Class II amalgams in a premolar and molar and (C) Class VI amalgams in premolars.

Initial Clinical Procedures

A preoperative assessment of the occlusal relationship of the involved and adjacent teeth is necessary. After administration of local anesthetic, isolation of the operating field with the rubber dam is recommended.[46,60] The rubber dam may be applied in the few minutes necessary for onset of profound anaesthesia before initiating the tooth preparation. Rubber dam isolation is especially indicated when removing deep caries (judged to be <1 mm from the pulp), during amalgam condensation and for mercury hygiene. The rubber dam should be used for isolation of the operating site when a caries lesion is extensive. If caries removal exposes the pulp, pulp capping may be more successful if the site is isolated with a properly applied rubber dam. In addition, the dam prevents moisture contamination of the amalgam mix during insertion.[33] For a single maxillary tooth, where caries is not extensive, adequate control of the operating field may also be achieved with cotton rolls and high-volume evacuation.

Tooth Preparation for Class I Amalgam Restorations

This section describes the specific technique for preparing the tooth for a Class I amalgam restoration. It is divided into initial and final stages.

Initial Tooth Preparation

Initial tooth preparation is defined as establishing the outline form by extension of the external walls to sound tooth structure while maintaining a specified, limited depth (usually just inside the DEJ) and providing resistance and retention forms.

Outline Form

The outline form for the Class I occlusal amalgam tooth preparation should follow the following principles:

I. The conservative class I tooth preparation should have an outline form with *gently flowing curves* and *distinct cavosurface margins*.
II. A *faciolingual width of no more than 1–1.5 mm and a depth of 1.5–2 mm* are considered ideal, but this goal is subject to the extension of the caries.
III. The *pulpal floor*, depending on the enamel thickness, is almost *always in dentin*.

Resistance Form

The resistance form principles allow margins to be positioned in areas that are sound and subject to minimal occlusal loading while maintaining the strength and health of the tooth by conserving uncompromised tooth structure (Fig. 13.27A). The resistance principles are as follows:

I. Establishing an optimal, conservative depth of *relatively flat pulpal floor* in sound tooth structure to resist forces directed in the long axis of the tooth and to provide a strong, stable seat for the restoration.
II. *Sufficient depth* (i.e. 1.5–2.0 mm) that results in adequate thickness of the restoration, providing resistance to fracture and wear
III. *Minimal extension* of external walls:
    i. Extending around the cusps to conserve tooth structure and prevent the internal line angles from approaching the pulp horns too closely.
    ii. Extending the outline to include fissures, placing the margins on relatively smooth, sound tooth structure.

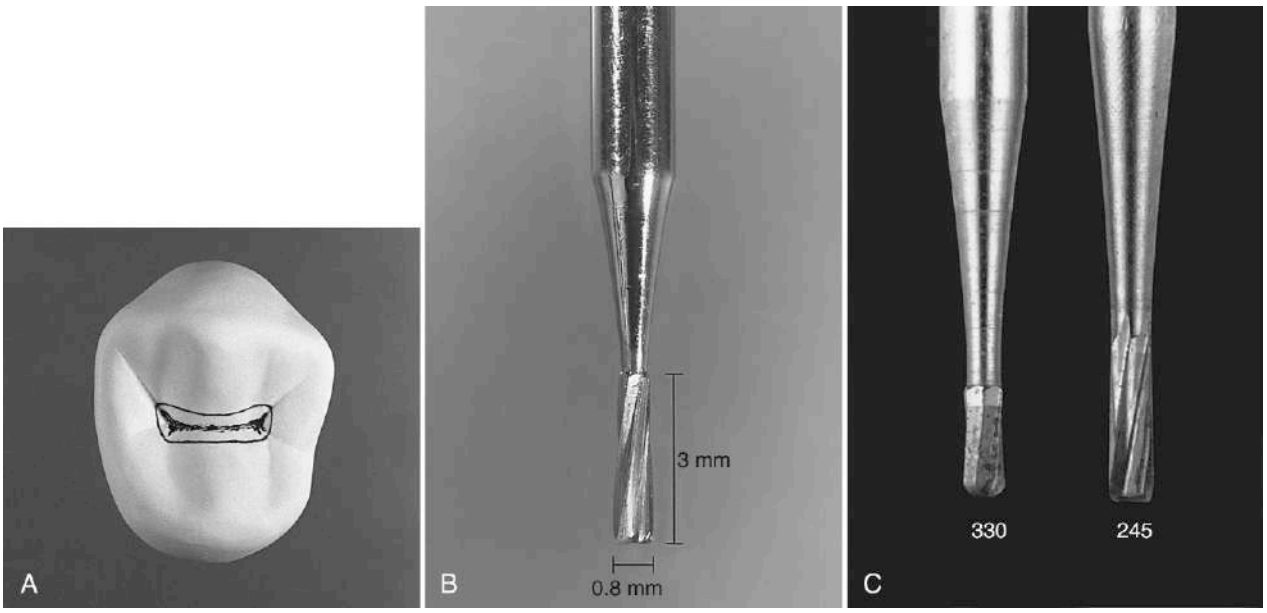


Figure 13.27 Class I preparation outline. (A) Ideal outline includes all carious occlusal pits and fissures; (B) dimensions of head of a No. 245 bur and (C) Nos. 330 and 245 burs compared.

iii. Keeping the *facial* and *lingual margin* extensions as minimal as possible between the central groove and the cusp tips.
iv. Minimally extending into the *marginal ridges* (only enough to include the defect) without removing dentinal support.

IV. Strong, ideal *enamel margins*:
i. Extending the outline form to include enamel undermined by caries.
ii. Cavosurface margin of 90° and extending a weak wall of enamel by joining two outlines that come close together (i.e. <0.5 mm apart).
iii. Using *enameloplasty* on the terminal ends of shallow fissures to conserve tooth structure.

V. Rounded internal preparation angles.

Retention Form

The *slight occlusal convergence* of two or more opposing, external walls provides the primary retention form.

Convenience Form

Conservation of tooth structure is important but convenience form requires that the extent of the preparation provides *adequate access* and *visibility*.

Clinical Technique

Armamentarium

A **No. 245 bur**, with a *head length of 3 mm and a tip diameter of 0.8 mm*, or a smaller **No. 330 bur** is recommended to prepare the Class I tooth preparation (Fig. 13.27B and C).

The silhouette of the No. 245 bur reveals sides slightly convergent towards the shank. This produces an occlusal convergence of the facial and lingual preparation walls, providing adequate retention form for the tooth preparation. The slightly rounded corners of the end of the No. 245 bur produce slightly rounded internal line angles that render the tooth more resistant to fracture from occlusal force.[61] The No. 330 bur is a smaller and pear-shaped version of the No. 245 bur and is also indicated for amalgam preparations (Fig. 13.27C).

Step 1: Punch Cut

- Class I occlusal tooth preparation is begun by entering the deepest or most carious pit with a 'punch cut' using the No. 245 carbide bur at high speed with air-water spray. A punch cut is performed by orienting the bur such that its long axis parallels the long axis of the tooth crown (Fig. 13.28A and B). The bur is inserted directly into the defective pit.
- When the pits are equally defective, the *distal pit* should be entered as illustrated. Entering the distal pit first provides increased visibility for the mesial extension. The bur should be positioned such that its distal aspect is directly over the distal pit, minimizing extension into the marginal ridge (Fig. 13.28C). The bur should be rotating when it is applied to the tooth and should not stop rotating until it is removed from the tooth. On posterior teeth, the approximate depth of the DEJ is located at 1.5–2 mm from the occlusal surface.
- As the bur enters the pit, an *initial target depth of 1.5 mm* should be established. This is one-half the length of the cutting portion of the No. 245 bur. The 1.5-mm pulpal depth is measured at the central fissure (Fig. 13.28D and E). The depth of the preparation is modified as

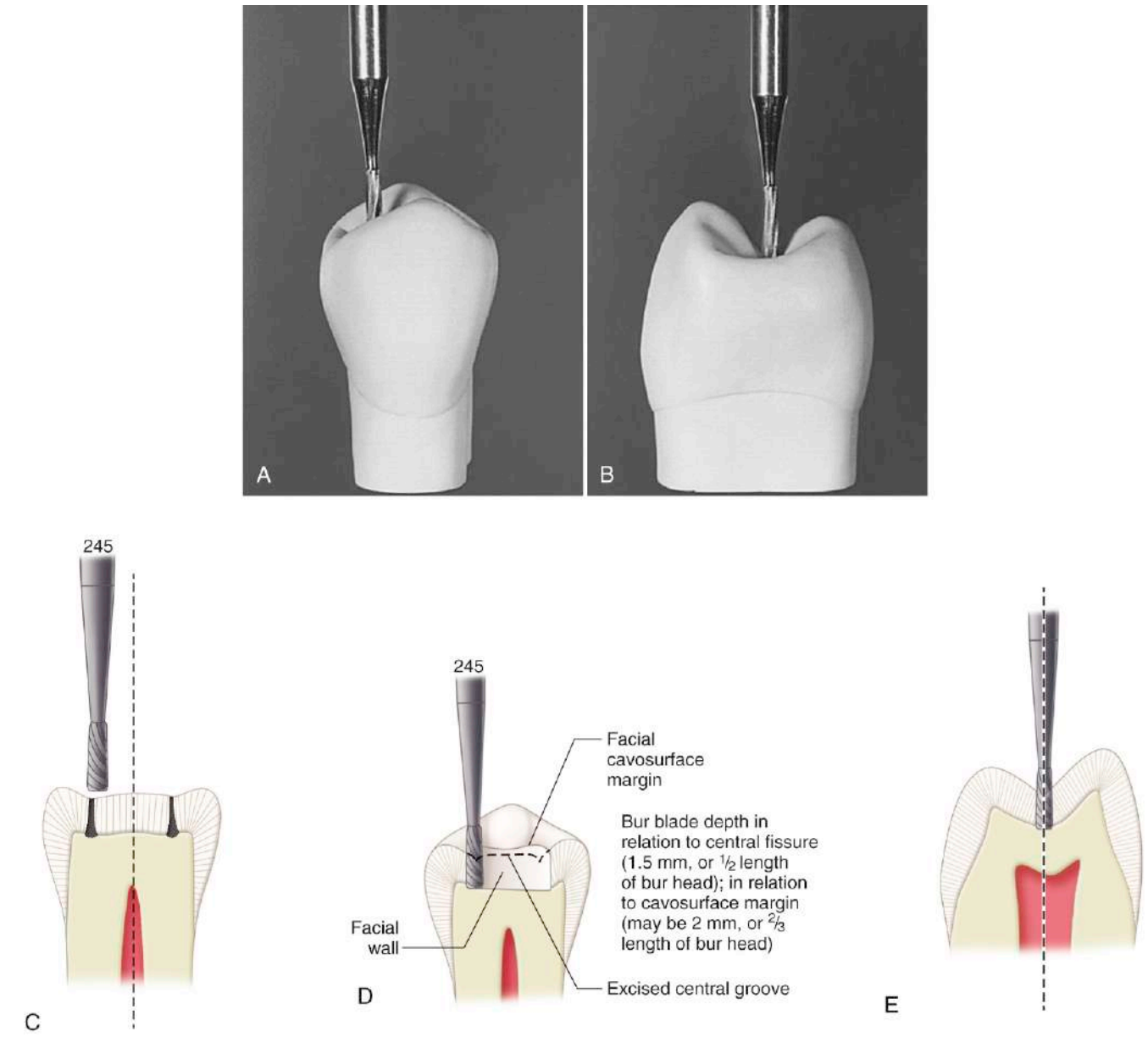


Figure 13.28 (A) No. 245 bur oriented parallel to long axis of tooth crown for entry as viewed from lingual aspect; (B) the bur positioned for entry as viewed from the distal aspect; (C) the bur is positioned over the most carious pit (distal) for entry. The distal aspect of the bur is positioned over the distal pit; (D) mesiodistal longitudinal section. Relationship of head of No. 245 bur to excised central fissure and cavosurface margin at ideal pulpal floor depth, which is just inside the DEJ and (E) faciolingual longitudinal section. *Dotted line* indicates the long axis of tooth crown and the direction of the bur.

needed so that the pulpal wall is established 0.1–0.2 mm into dentin. In other words, the depth of the prepared external walls should be 1.5–2 mm ('just inside the DEJ') (Fig. 13.28D and E), depending on the cuspal incline.

Step 2: Extension into the Involved Marginal Ridge

- Distal extension into the distal marginal ridge to include a fissure or caries occasionally requires a slight tilting of the bur distally (≤10°). This creates a *slight occlusal divergence* to the distal wall to prevent undermining the marginal ridge of its dentin support (Fig. 13.29A–C). Because the facial and lingual prepared walls converge, this slight divergence does not compromise the overall retention form.
- For premolars, the distance from the margin of such an extension to the proximal surface usually should not be less than ~1.6 mm, or two diameters of the end of the No. 245 bur (Fig. 13.29B) measured from a tangent to the proximal surface (i.e. the proximal surface height of

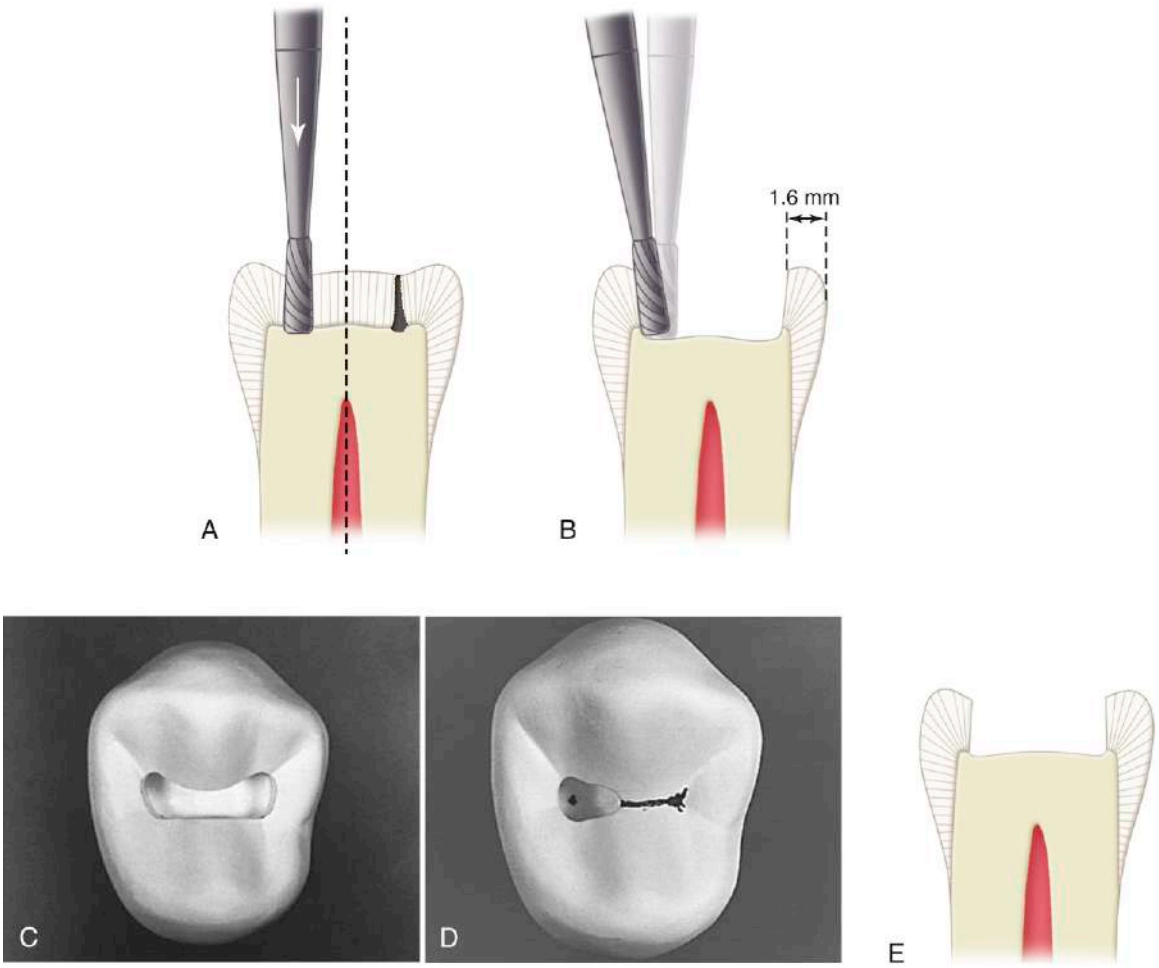


Figure 13.29 (A) Enter the pit with a punch cut to just inside the DEJ (depth of 1.5–2 mm or one-half to two-thirds the head length of the No. 245 bur). The 1.5-mm depth is measured at central fissure; the measurement of same entry cut at the prepared external wall is 2 mm. (B) Incline the bur distally to establish proper occlusal divergence to distal wall to prevent removal of the dentin supporting the marginal ridge enamel when the pulpal floor is in dentin, and distal extension is necessary to include a fissure or caries lesion. For such an extension on premolars, the distance from the margin to the proximal surface (i.e. imaginary projection) must not be less than 1.6 mm (i.e. two diameters of bur end). (C) Occlusal view of the initial tooth preparation that has mesial and distal walls that diverge occlusally. (D) Distofacial and distolingual fissures that radiate from the pit are included before extending along the central fissure. (E) Mesiodistal longitudinal section. The pulpal floors are generally flat but may follow the rise and fall of the occlusal surface.

contour). For molars, this minimal distance is ~2 mm. Minimal distal (or mesial) extension often does not require changing the orientation of the bur's axis from being parallel to the long axis of the tooth crown. In this case, the mesial and distal walls are parallel to the long axis of the tooth crown (or slightly convergent occlusally) (Fig. 13.29D and E).

- While maintaining the bur's orientation and depth, the preparation is extended distofacially or distolingually to include any fissures that radiate from the pit (Fig. 13.29D). When these fissures require extensions of more than a few tenths of a millimetre, however, consideration should be given to changing to a bur of smaller diameter or to using enameloplasty. Both of these approaches conserve the tooth structure and therefore minimize weakening of the tooth.

Step 3: Extension towards Central Fissure

- The bur's orientation and depth are maintained while extending along the central fissure towards the mesial pit, following the DEJ (Fig. 13.29D and E).
- When the central fissure has minimal caries, one pass through the fissure at the prescribed depth provides the desired minimal width to the isthmus. Ideally the *width of the isthmus should be just wider than the diameter of the bur.*
- It is well established that a *tooth preparation with a narrow occlusal isthmus is less prone to fracture.*[62,63]

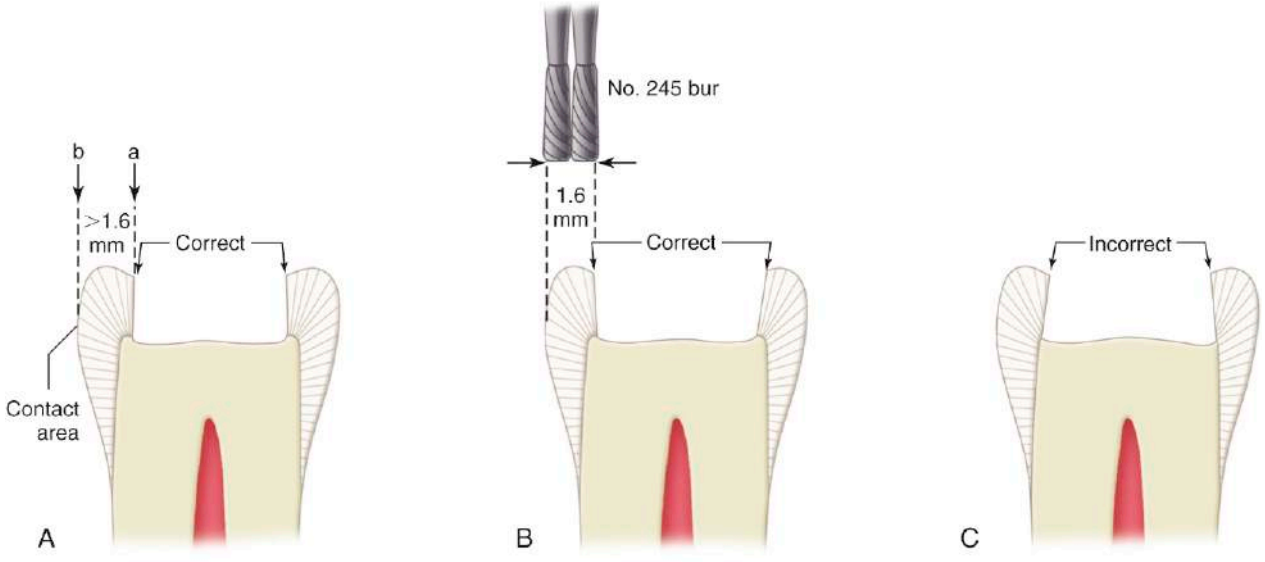


Figure 13.30 The direction of the mesial and distal walls is influenced by the remaining thickness of the marginal ridge as measured from the mesial or distal margin (*a*) to the proximal surface (i.e. imaginary projection of proximal surface) (*b*). (A) Mesial and distal walls should converge occlusally when the distance from (*a*) to (*b*) is greater than 1.6 mm; (B) when the operator judges that the extension will leave only 1.6-mm thickness (two diameters of No. 245 bur) of marginal ridge (i.e. premolars), the mesial and distal walls must diverge occlusally to conserve ridge-supporting dentin and (C) extending the mesial or distal walls to a two-diameter limit without diverging the wall occlusally undermines the marginal ridge enamel.

Step 4: Extension towards Opposing Marginal Ridge (If Required)

- As previously described for the distal margin, the orientation of the bur should not change as it approaches the mesial pit if the mesial extension is minimal. If the fissure extends farther onto the marginal ridge, the long axis of the bur should be changed to establish a slight occlusal divergence to the mesial wall if the marginal ridge would be otherwise undermined of its dentinal support. Fig. 13.30 illustrates the correct and incorrect preparation of the mesial and distal walls.

Step 5: Facial and Lingual Wall Extension (If Required)

- The remainder of any occlusal enamel defects is included in the outline, and the facial and lingual walls are extended, if necessary, to remove enamel undermined by the caries lesion.[64]
- The *strongest and ideal enamel margin should be composed of full-length enamel rods attached to sound dentin*, supported on the preparation side by shorter rods that are also attached to sound dentin (Fig. 13.31).

For the initial tooth preparation, the pulpal wall should remain at the initial ideal depth, even if any restorative material or soft caries lesion remains in the pulpal wall (Fig. 13.32). Remaining caries (and, if present, old restorative material) is removed during the final tooth preparation.

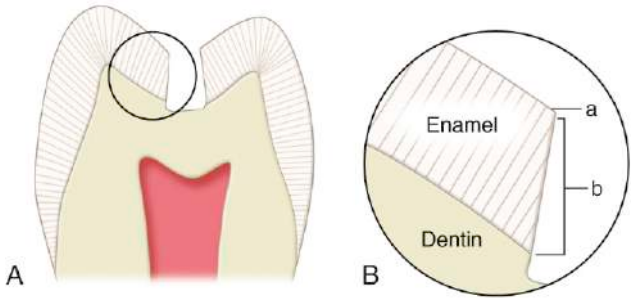


Figure 13.31 (A and B) The ideal and strongest enamel margin is formed by full-length enamel rods (*a*) resting on sound dentin supported on the preparation side by shorter rods, also resting on sound dentin (*b*).

Step 6: Enameloplasty (If Required)

When the remaining fissure is no deeper than one-quarter to one-third the thickness of enamel, enameloplasty is indicated. Enameloplasty refers to eliminating the developmental fault by removing it with the side of an appropriately shaped rotary finishing instrument, leaving a smooth surface (Fig. 13.33A–C). This procedure frequently reduces the need for further extension. The extent to which enameloplasty should be used cannot be determined exactly until the process of extending into the fissured area occurs, at which time the depth of the fissure into enamel may be observed. The surface left by enameloplasty should meet the tooth preparation wall, preferably with a *cavosurface angle no greater than approximately 100°*; this would produce a distinct margin for amalgam of no less than 80° (Fig. 13.33D). If enameloplasty is unsuccessful

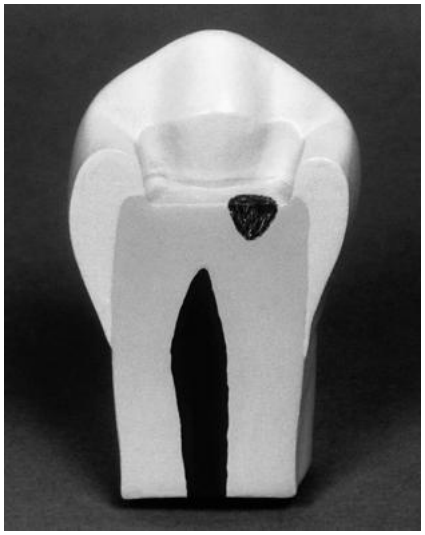


Figure 13.32 Mesiodistal longitudinal section showing example of the pulpal floor in dentin and the caries lesion that is exposed after the initial tooth preparation. The carious lesion is surrounded by sound dentin on the pulpal floor for the resistance form.

in eliminating a mesial (or distal) fissure that extends to the crest of a marginal ridge or beyond, three alternatives exist:

I. Make no further change in the outline form.
II. Extend through the marginal ridge when margins would be lingual to the contact (Fig. 13.34).
III. Include the fissure in a conservative Class II tooth preparation.

The first alternative usually should be strongly considered except in patients at high risk for caries. Enameloplasty is not indicated if an area of occlusal contact is involved. In this case, the choices are either to consider the preparation completed (an option for patients at low risk for caries) or to extend the preparation to include the fissure as previously described.



Figure 13.34 Mesial fissure that cannot be eliminated by enameloplasty may be included in the preparation if the margins are able to be lingual of the occlusal contact.

Extensive Caries Lesion Considerations

A tooth may be found to have a large or *extensive* Class I caries lesion. A lesion is considered extensive if

I. the distance between soft dentin and the pulp is judged to be less than 1 mm, or
II. when the faciolingual extent of the defect has involved much of the cuspal inclines.

In this case, selective removal of soft dentin and insertion of a liner, if needed, may precede the establishment of outline, resistance and retention forms. This approach protects the pulp as early as possible from any additional insult that may occur during tooth preparation. Normally, however, the outline, primary resistance and primary retention forms are established through proper orientation of the No. 245 bur and appropriate extension of the preparation. As is the case for smaller caries lesions, an initial depth to reach the DEJ (measured approximately 1.5 mm at any pit or fissure and 2 mm on the prepared external walls) should still be established

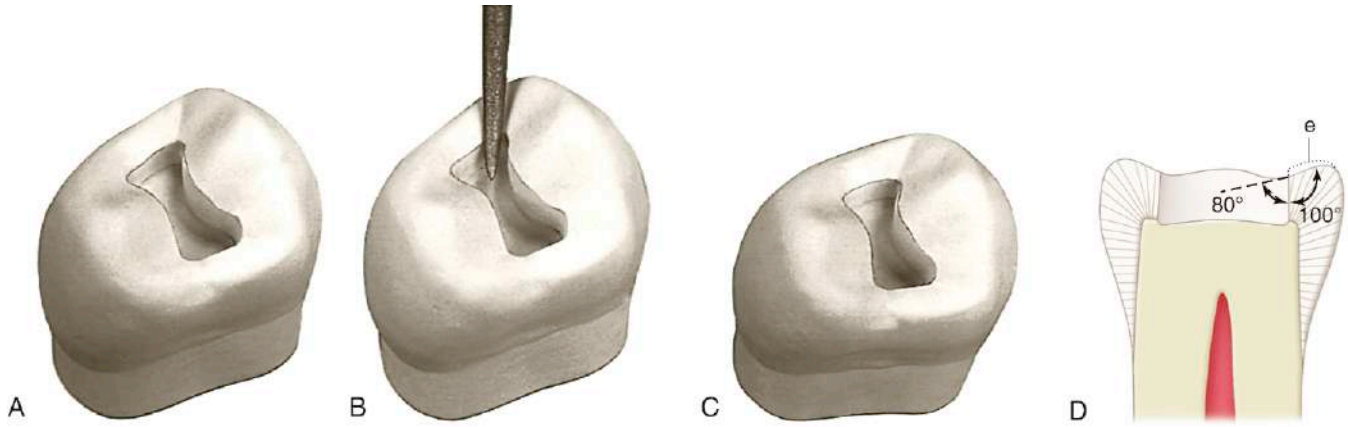


Figure 13.33 Enameloplasty. (A) Developmental defect at terminal end of fissure; (B) fine-grit diamond-finishing instrument in position to remove the defect; (C) smooth surface after enameloplasty and (D) the cavosurface angle should not exceed 100°, and the margin–amalgam angle should not be less than 80°. Enamel external surface (e) before enameloplasty.

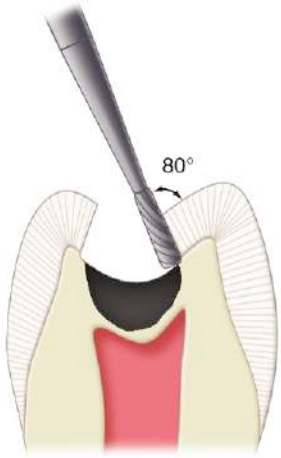


Figure 13.35 Initial tooth preparation with an extensive caries lesion. When extending laterally to remove enamel undermined by the caries lesion, the bur's long axis is altered to prepare a 90°–100° cavosurface angle. A 100° cavosurface angle on the cuspal incline results in an 80° marginal amalgam angle.

and maintained. The preparation is extended laterally at the DEJ to remove all enamel undermined by the caries lesion by alternately cutting and examining the lateral extension of the lesion.

For a caries lesion extending up the cuspal inclines, it may be necessary to alter the bur's long axis to prepare a 90–100° cavosurface angle while maintaining the initial depth (Fig. 13.35). If not, an obtuse cavosurface angle may remain (resulting in an acute, or weak, amalgam margin), or the pulpal floor (over the pulp horns) may be prepared too deeply.

Primary resistance form is obtained by extending the outline of the tooth preparation to include only undermined and defective tooth structure while preparing strong enamel walls and allowing strong cuspal areas to remain. Primary retention form is obtained by the occlusal convergence of the enamel walls. Secondary retention form may result from undercut areas that are occasionally left in dentin (and that are not filled in by a liner or base) after peripheral removal of soft dentin. When extending the outline form, enameloplasty should be used in any indicated area (as described previously and in detail in Chapter 11).

When the defect extends to more than one-half the distance between the primary groove and a cusp tip, reducing the cuspal tooth structure and restoring it with amalgam (also referred to as 'capping the cusp') may be indicated as discussed in Chapter 14 (complex amalgam restorations). When the distance is two-thirds, cusp reduction and coverage is usually required because of the risk of cusp fracture during subsequent functional occlusal loading. Fig. 13.36 illustrates examples of large (extensive) Class I amalgam preparation outlines.

Clinical Notes

- The Class I tooth preparation should have an outline form with gently flowing curves and distinct cavosurface margins.
- A faciolingual width of no more than 1–1.5 mm and a depth of 1.5–2 mm are considered ideal, but this goal is subject to the extension of the caries lesion.
- The pulpal floor, depending on the enamel thickness, is almost always in dentin (Fig. 13.29C and E).
- Although conservation of the tooth structure is important, the convenience form requires that the extent of the preparation be adjusted to provide adequate access and visibility.

Final Tooth Preparation

The final tooth preparation includes:

I. *Removal of remaining defective enamel and soft dentin on the pulpal floor as indicated*:

If several enamel pit-and-fissure remnants remain in the floor, or if a central fissure remnant extends over most of the floor, the floor should be deepened with the No. 245 bur to eliminate the defect or to uncover the caries lesion to a maximum preparation depth of 2 mm (Fig. 13.37). If these remnants are few and small, they may be removed with an appropriate carbide bur (Fig. 13.38). Removal of the remaining lesion (i.e. caries lesion that extends pulpally from the established pulpal floor) is best accomplished using a *discoid-type spoon excavator* or a *slowly revolving round carbide bur* of appropriate size (Fig. 13.39A and B).

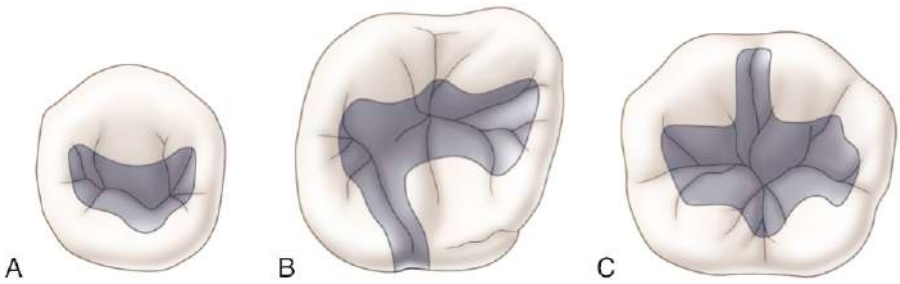


Figure 13.36 Examples of more extensive Class I amalgam tooth preparation outline forms. (A) Occlusal outline form in the mandibular second premolar; (B) occlusolingual outline form in the maxillary first molar and (C) occlusofacial outline form in the mandibular first molar.

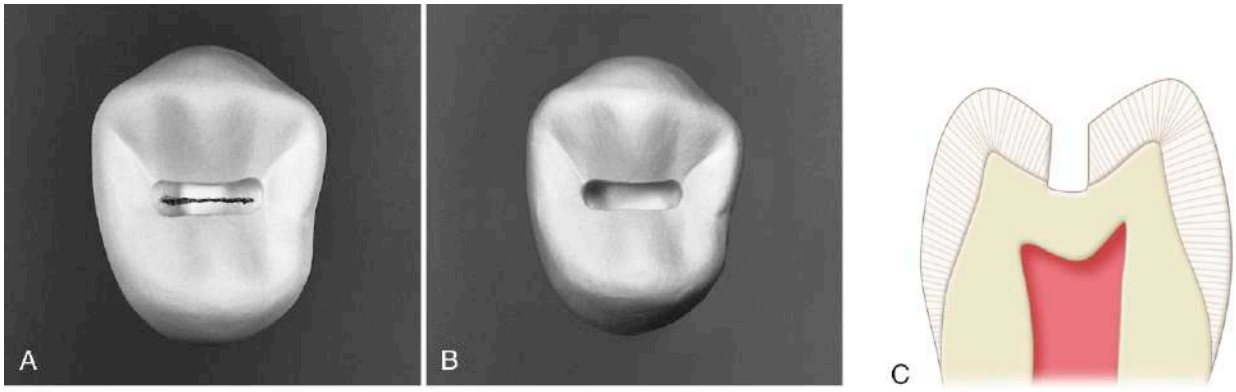


Figure 13.37 Removal of enamel fissure extending over most of the pulpal floor. (A) Full-length occlusal fissure remnant remaining on the pulpal floor after the initial tooth preparation and (B and C) the pulpal floor is deepened to a maximum depth of 2 mm to eliminate the fissure or uncover dentin caries lesion.

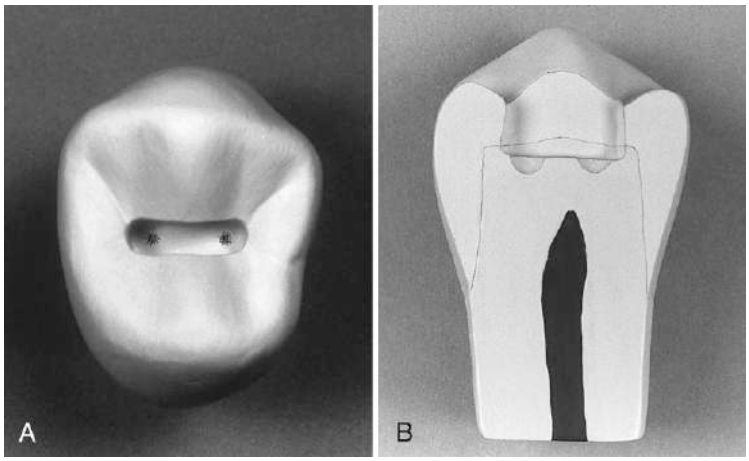


Figure 13.38 Removal of enamel pit and fissure and dentin caries lesion that is limited to a few small pit-and-fissure remnants. (A) Two pit remnants remain on the pulpal floor after the initial tooth preparation and (B) carious enamel and dentin caries lesion have been removed.

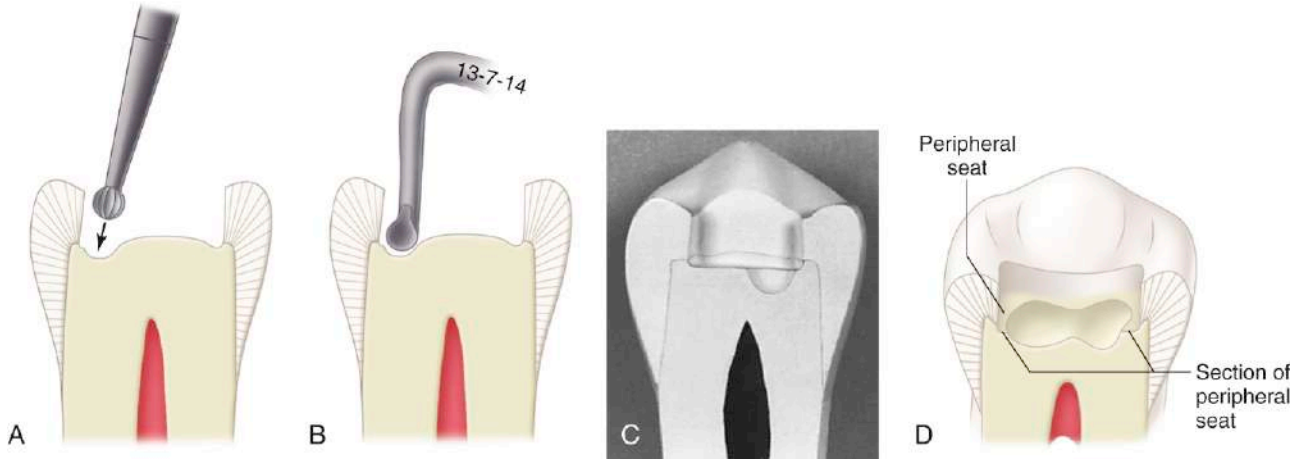


Figure 13.39 (A and B) Removal of dentin caries lesion is accomplished with round burs (A) or spoon excavators (B) and (C and D) the resistance form may be improved with a flat floor peripheral to the excavated area(s).

The removal of carious dentin will not affect the resistance form of the tooth because the periphery would not need further extension. In addition, it will not affect the resistance form of the restoration as it will rest on the pulpal wall peripheral to the excavated area or areas. The peripheral pulpal floor should be at the previously described initial pulpal floor depth just inside the DEJ (Fig. 13.39C and D). Usually, no secondary resistance or retention form features are necessary Class I amalgam preparations.

II. *Pulp protection, where indicated*

When using bases and liners, it should be remembered that the ultimate goal is to have a minimum of 2 mm of *remaining dentin thickness* (RDT) or a combination of tooth structure base and liner between the restorative material and the pulp.

- **Shallow preparation (≥2 mm of RDT)**

  In a shallow tooth excavation (which includes ≥2 mm of RDT), pulpal protection, other than in terms of chemical protection, is not necessary. For an amalgam restoration, the preparation is coated with *two thin coats of a varnish, a single coat of a dentin sealer or a dentin bonding system,* and then restored. In most cases, a *dentin sealer* is the material of choice (e.g. Gluma by Heraeus Kulzer, South Bend, IN; Hurriseal by Beutlich Pharmaceuticals, Waukegan, IL).
- **Moderately deep preparation (0.5–2 mm of RDT)**

  In a moderately deep tooth excavation for amalgam that includes some extension of the preparation toward the pulp so that a region includes less than ideal dentin protection, it may be judicious to apply a base *only at that site* using light cured RMGI.
- **Very deep preparation (<0.5 mm of RDT)**

  i. If the RDT is very small or if pulp exposure is a potential problem, calcium hydroxide is used to stimulate reparative dentin for any restorative material. A thickness of 0.5–1 mm of set calcium hydroxide liner is sufficient to treat a near or actual pulp exposure.

  ii. If extensive dentin is lost because of caries, and the tooth excavation extends close to the pulp, a *light cured RMGI base should be applied over the already placed calcium hydroxide liner.*

  iii. The sealer or bonding agent is not applied until after the base is in place.

III. *Procedures for finishing the external walls*

External walls may already have been finished during initial preparation steps; however, operator assessment of the external walls of the preparation is accomplished at this point, and any additional finish is accomplished as described previously and in Chapter 11. *An occlusal cavosurface bevel is contraindicated in the tooth preparation for an amalgam restoration.*[64] It is important to provide an approximate 90–100° cavosurface angle, which should result in 80–90° amalgam at the margins.[64] This *butt-joint marginal area* (which approximates 90° cavosurface marginal angles for both enamel and amalgam) allows strength for both. Amalgam is a brittle material with low-edge strength and tends to chip under occlusal stress if its angle at the margins is less than 80°. The transition between the external and internal walls should be gently rounded.[65]

IV. *Final procedures of debridement (cleaning) and inspecting the preparation*

The preparation should then be cleaned, carefully inspected as indicated previously in the General Concepts Guiding Preparation for Amalgam Restorations section and any final modifications completed

Other Class I Amalgam Preparations

Class I preparations, especially those in esthetically important areas, may be restored with composite because of their small size and the maximal thickness of enamel available for bonding. However, the following preparations may also be restored with amalgam:

I. Facial pit of the mandibular molar.
II. Lingual pit of the maxillary lateral incisor.
III. Occlusal pits of the mandibular first premolar.
IV. Occlusal pits and fissures of the maxillary first molar.
V. Occlusal pits and fissures of the mandibular second premolar.

Examples of some of these types of preparations and restorations are provided in Figs. 13.40–13.45.

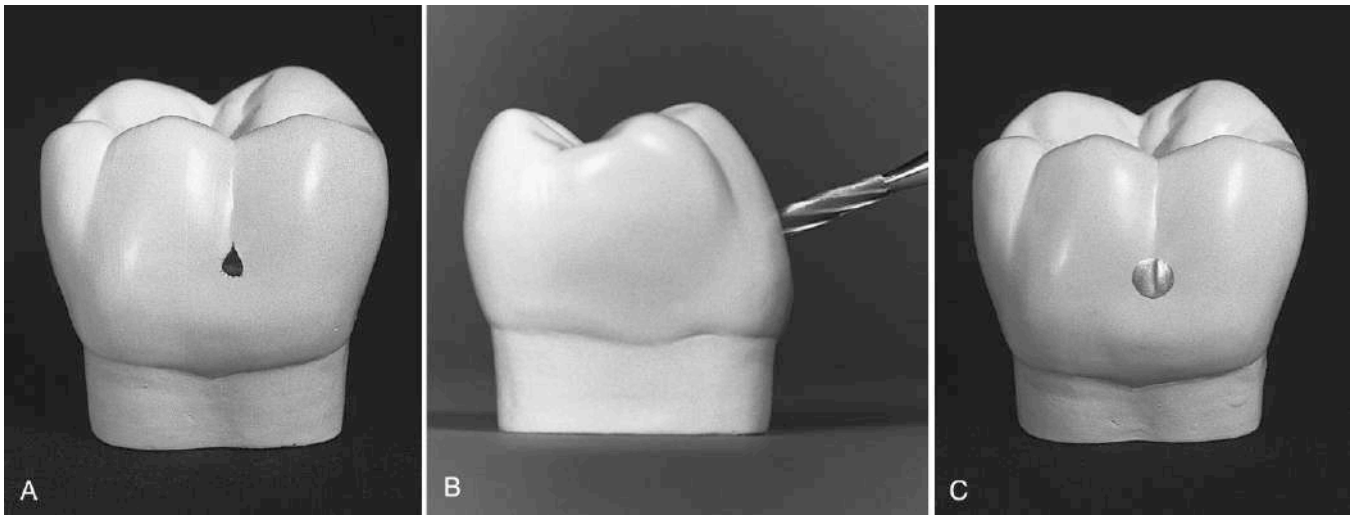


Figure 13.40 Mandibular molar. (A) Facial pit with a caries lesion; (B) the bur positioned perpendicular to the tooth surface for entry and (C) outline of restoration.

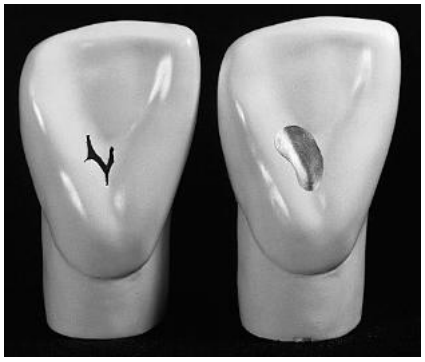


Figure 13.41 Carious lingual pit and fissure and restoration on the maxillary lateral incisor.

Restorative Technique for Class I Amalgam Preparations

Desensitizer Placement

A dentin desensitizer is placed in the preparation according to manufacturer recommendations before amalgam condensation (Fig. 13.46).[45]

Matrix Placement

Generally, matrices are unnecessary for a conservative Class I amalgam restoration except as specified in later sections.



Figure 13.42 Maxillary lateral incisor. (A) Preoperative radiograph of dens in dente and (B) radiograph of restoration after 13 years. (Courtesy: Dr. Ludwig Scott.)

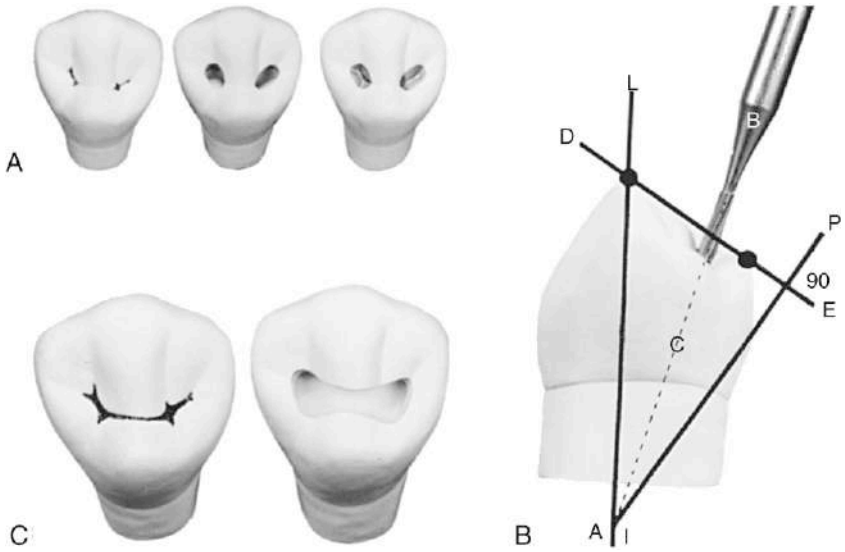


Figure 13.43 (A) Preparation design and restoration of carious occlusal pits on the mandibular first premolar. (B) Bur tilt for entry. The cutting instrument is held such that its long axis (*broken line*, *CI*) is parallel with the bisector (*B*) of the angle formed by the long axis of the tooth (*LA*) and the line (*P*) that is perpendicular to the plane (*DE*) drawn through the facial and lingual cusp points. This *dotted line* (*CI*) is the bur position for entry. (C) Conventional outline, including occlusal pits and central fissure.

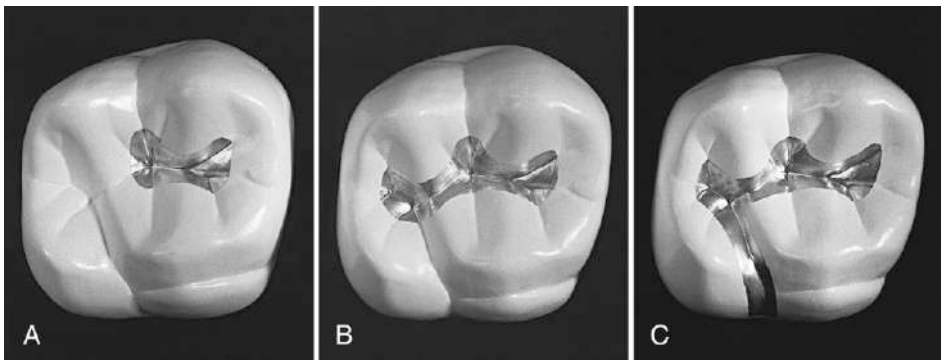


Figure 13.44 Maxillary first molar. (A) Outline necessary to include the carious mesial and central pits connected by the fissure; (B) preparation outline extended from outline in *A* to include distal pit and connecting deep fissure in oblique ridge and (C) preparation outline extended from outline in (B) to include distal oblique and lingual fissures.

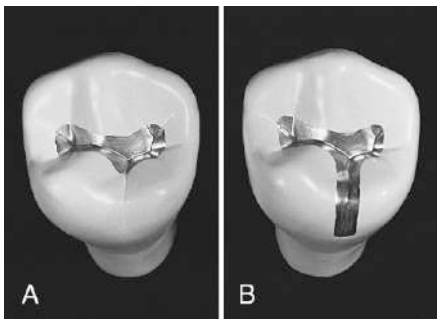


Figure 13.45 Mandibular second premolar. (A) Typical occlusal outline and (B) extension through the lingual ridge enamel is necessary when enameloplasty does not eliminate the lingual fissure.



Figure 13.46 Use of microbrush to apply the dentin desensitizer in the tooth preparation.
(Courtesy: Aldridge D. Wilder, DDS.)

Insertion and Carving of the Amalgam

The triturated amalgam (Fig. 13.47A) is placed into an amalgam well. The outline of the tooth preparation should be reviewed before inserting amalgam to establish a mental image that will later aid in carving amalgam to the cavosurface margin. Preoperative occlusal contact locations should be recalled (Fig. 13.47B). Amalgam should be carefully condensed into the pulpal line angles (Fig. 13.47C). The preparation should be overpacked 1 mm or more using heavy pressure (Fig. 13.47D). This ensures that the cavosurface margins are completely covered with well-condensed amalgam. Final condensation over cavosurface margins should be done perpendicular to the external enamel surface adjacent to the margins. The overpacked amalgam should then be pre-carve burnished.

Finishing and Polishing of the Amalgam

With care, carving may begin immediately after condensation. Sharp discoid–cleoid carvers of suitable sizes are useful for this. The larger end of the discoid–cleoid instrument (No. 3/6) is used first, followed by the smaller instrument (No. 4/5) in regions not accessible to the larger instrument. Alternatively, the Hollenback carver may be used, with care not to over-carve the groove or fossa areas. All carving should be done with the edge of the blade perpendicular to the margins as the instrument is moved parallel to the margins. Part of the edge of the carving blade should rest on the unprepared tooth surface adjacent to the preparation margin (Fig. 13.47F). Using this surface as a guide helps prevent over-carving amalgam at the margins and produces a continuity of surface contour across the preparation margins. During carving, thin amalgam should be removed from areas of enameloplasty. Thin amalgam left in these areas may fracture because of its low edge strength. Deep occlusal anatomy should not be carved into the restoration because these may thin the amalgam at the margins and weaken the restoration (Fig. 13.47G). Under-carving leaves thin portions of amalgam (subject to fracture) on the unprepared tooth surface. The thin portion of amalgam extending beyond the margin is referred to as *flash*. The mesial and distal fossae should be carved slightly deeper than the proximal marginal ridges (Fig. 13.47H).

After carving is completed, the outline of the amalgam margin should reflect the contour and location of the prepared

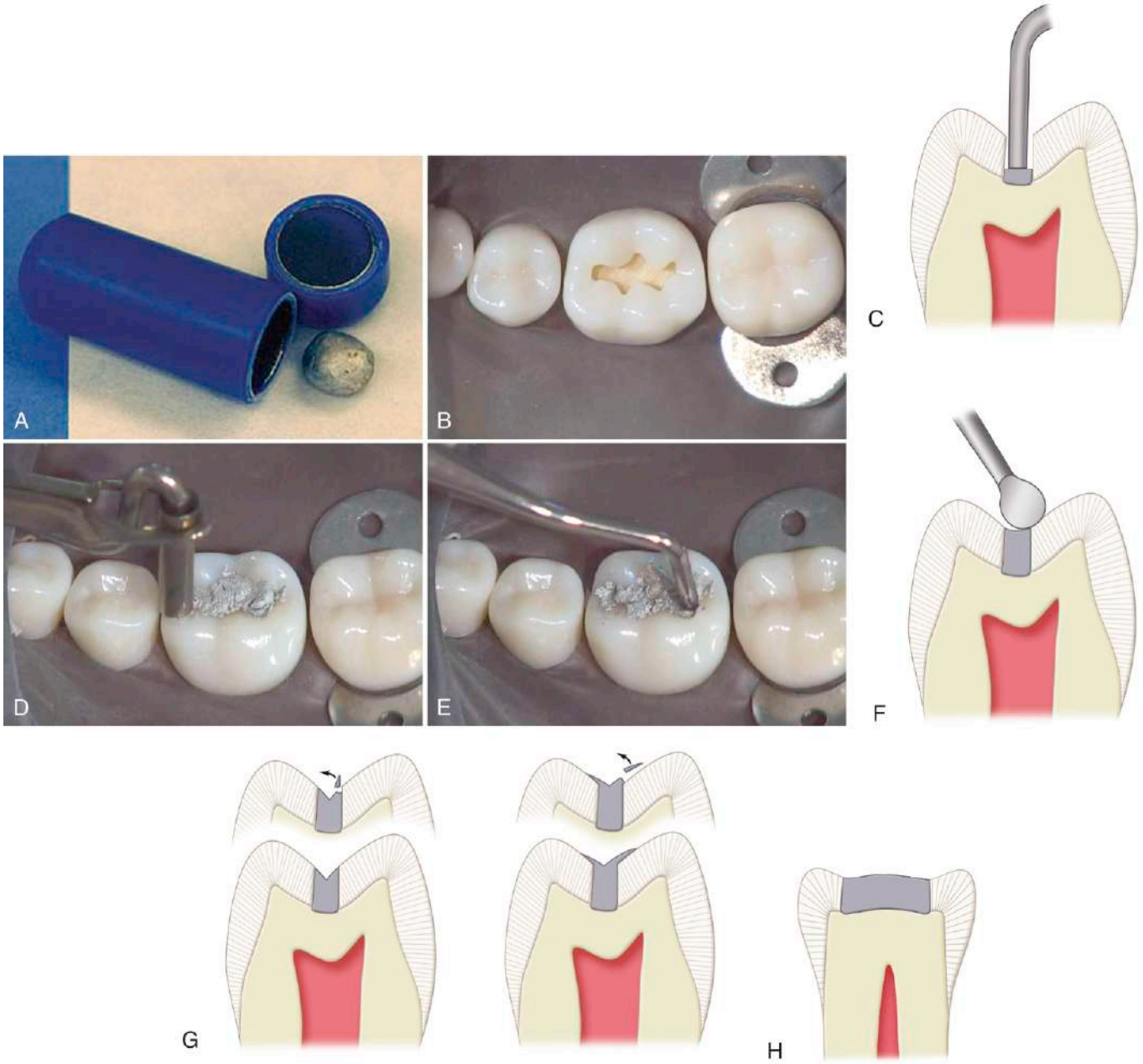


Figure 13.47 Restoration of occlusal tooth preparation. (A) Properly triturated amalgam is a homogeneous mass with slightly reflective surface. A properly mixed mass will flatten slightly if dropped on a tabletop. (B) The operator should have a mental image of the outline form of the preparation before condensing amalgam to aid in locating the cavosurface margins during the carving procedure. (C) Amalgam should be inserted incrementally and condensed with overlapping strokes. (D and E) The tooth preparation should be overpacked to ensure well-condensed marginal amalgam that is not mercury rich. (F) The carver should rest partially on the external tooth surface adjacent to the margins to prevent over-carving. (G) Deep occlusal grooves invite chipping of amalgam at the margins. Thin portions of amalgam left on the external surfaces soon break away, giving the appearance that amalgam has grown out of the preparation. (H) Carve fossae slightly deeper than the proximal marginal ridges.
(Part A, From Darby ML, Walsh MM: *Dental hygiene: theory and practice*, ed 3, St. Louis, Saunders, 2010; Parts B, D and E, Courtesy Aldridge D. Wilder, DDS.)

cavosurface margin, revealing a regular (i.e. not ragged) outline with gentle curves. An amalgam outline that is larger or irregular is under-carved and requires further carving or finishing (Fig. 13.48). An amalgam restoration that is more than minimally over-carved (i.e. a submarginal defect >0.2 mm) should be replaced.[66] If the total carving time is short enough, the smoothness of the carved surface may be improved by wiping with a small, damp ball of cotton held with the cotton pliers. Alternatively, post-carve burnishing may be utilized to refine the anatomy and leave a smooth, satin finish. All shavings from the carving procedure should be removed from the mouth with the aid of the oral evacuator.

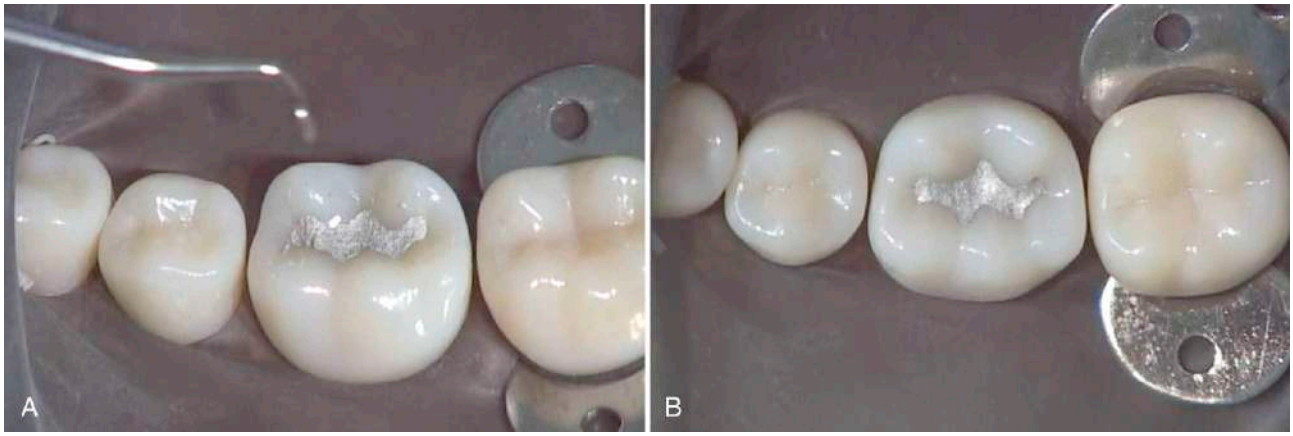


Figure 13.48 (A) Under-carved amalgam with flash beyond the margins. The restoration outline is irregular and larger than the preparation outline in Fig. 13.36B. (B) Correctly carved amalgam restoration.
(Courtesy: Aldridge D. Wilder, DDS.)

The occlusion on the restoration is then evaluated and adjusted so that all markings on the restoration and adjacent teeth are uniform (Fig. 13.23).

Extensive caries lesions require a more extensive restoration (which is a clear indication for amalgam as compared with composite resin). Use of amalgam in large Class I restorations provides good wear resistance and stable occlusal contact relationships. For very large Class I restorations, a bonding system may be used, although this book does not promote such use. The perceived benefits of bonded amalgams have not been substantiated.[18–21,67,68] Bonded amalgams have no advantage as compared with the conventional technique described above. Carving the extensive Class I restoration is often more complex because more cuspal inclines are included in the preparation. Appropriate contours, occlusal contacts and groove/fossa anatomy must be provided. Finishing and polishing indications and techniques are as described previously.

Tooth Preparation for Class I Occlusolingual and Occlusofacial Amalgam Restorations

Occlusolingual amalgam restorations may be used on maxillary molars when a lingual fissure connects with the distal oblique fissure and distal pit on the occlusal surface (Fig. 13.49). Composite may also be used as the restorative material in smaller lesions.

After local anaesthesia and evaluation of the occlusal contacts, the use of a rubber dam is generally recommended for isolation of the operating field. Alternatively, typical Class I preparations may be adequately isolated with cotton rolls.

Tooth Preparation

The initial tooth preparation involves the establishment of the outline, primary resistance and primary retention forms, as well as initial preparation depth. The accepted principles of the outline form (previously presented) should be observed with special attention to the following:

I. The tooth preparation should be no wider than necessary; ideally, the mesiodistal width of the lingual extension should not exceed 1 mm except for extension necessary to remove carious or undermined enamel or to include unusual fissuring.
II. When indicated, the tooth preparation should be more at the expense of the oblique ridge, rather than centreing over the fissure (weakening the small distolingual cusp).
III. Especially on smaller teeth, the occlusal portion may have a slight distal tilt to conserve the dentin support of the distal marginal ridge (Fig. 13.50).
IV. The margins should extend as little as possible onto the oblique ridge, distolingual cusp and distal marginal ridge.

These objectives help conserve the dentinal support (i.e. strength of the tooth) and aid in establishing an enamel cavosurface angle as close to 90° as possible (Fig. 13.51). They also help to minimize deterioration of the restoration

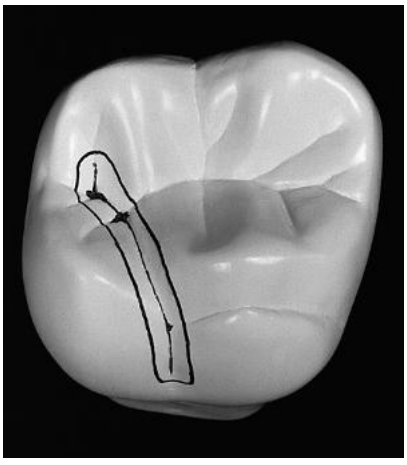


Figure 13.49 Outline of margins for occlusolingual tooth preparation.



Figure 13.50 Small distal inclination of the bur on smaller teeth may be indicated to conserve the dentinal support and the strength of the marginal ridge.

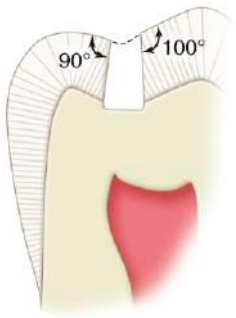


Figure 13.51 Enamel cavosurface angles of 90°–100° are ideal.

margins by locating the margins away from enamel eminences where occlusal forces may be concentrated.

The distal pit is identified with indirect vision and entered with the end of the No. 245 bur (Fig. 13.52A). The long axis of the bur usually should be parallel to the long axis of the tooth crown. The dentinal support and strength of the distal marginal ridge and the distolingual cusp should be conserved by positioning the bur such that it cuts more of the tooth structure mesial to the pit rather than distal to the pit (e.g. 70:30 rather than 50:50), if needed. The initial cut is to the level of the DEJ (a depth of 1.5–2 mm) (Fig. 13.52B). At this depth, the pulpal floor is usually in dentin. When the entry cut is made (Fig. 13.52C), the bur (maintaining the initial established depth) is moved to include any remaining fissures facial to the point of entry (Fig. 13.52D). The bur is then moved along the fissure towards the lingual surface (Fig. 13.52E). As with Class I occlusal preparations, a slight distal inclination of the bur is indicated occasionally (particularly in smaller teeth) to conserve the dentinal support and strength of the marginal ridge and the distolingual cusp. To ensure adequate strength for the marginal ridge, the distopulpal line angle should not approach the distal surface of the tooth closer than 2 mm. On large molars, the bur position should remain parallel to the long axis of the tooth, particularly if the bur is offset slightly mesial to the centre of the fissure. Keeping the bur parallel to the long axis of the tooth creates a distal wall with slight occlusal convergence, providing favourable enamel and amalgam angles. The bur is moved lingually along the fissure, maintaining a uniform depth until the preparation is extended onto the lingual surface (Fig. 13.52F). The pulpal floor should follow the contour of the occlusal surface and the DEJ, which usually rises occlusally as the bur moves lingually.

The mesial and distal walls of the occlusal portion of the preparation should converge occlusally because of the shape of the bur. If the slight distal bur tilt was required, the mesial and distal walls still should converge relative to each other (although the distal wall may be divergent

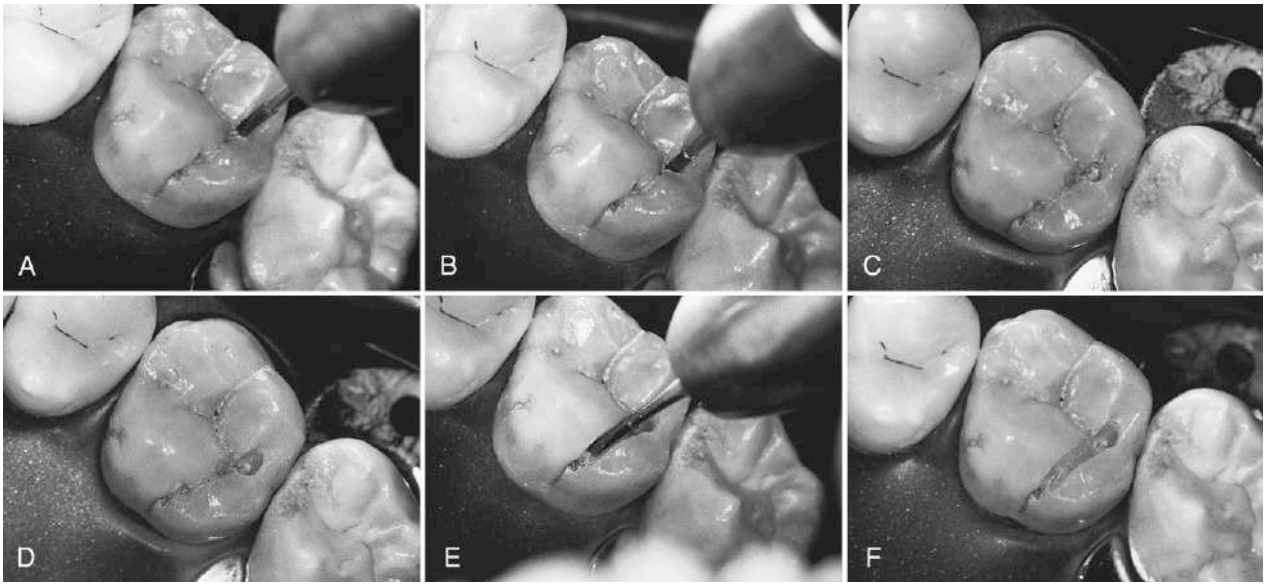


Figure 13.52 Occlusolingual tooth preparation. (A) No. 245 bur positioned for entry; (B) penetration to a minimal depth of 1.5–2 mm; (C) entry cut; (D) the remaining fissures facial to the point of entry are removed with the same bur and (E and F) a cut lingually along the fissure until the bur has extended the preparation onto the lingual surface.

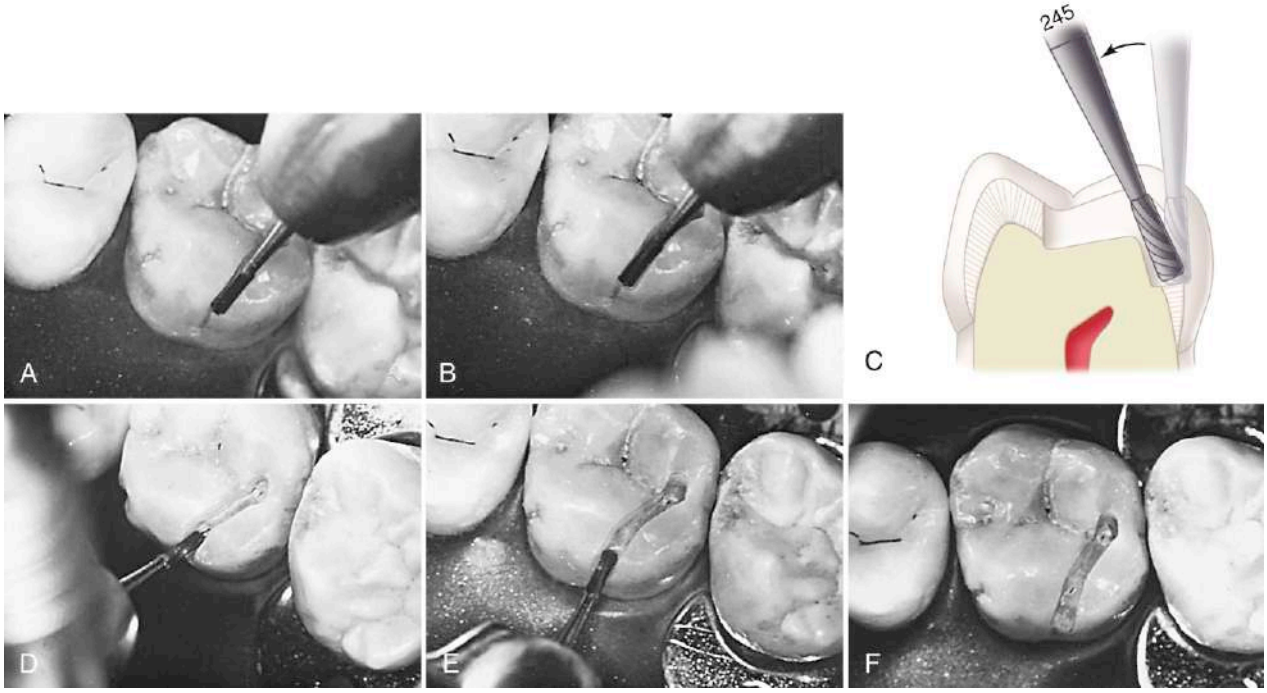


Figure 13.53 Occlusolingual tooth preparation. (A) Position of bur to cut the lingual portion; (B) initial entry of the bur for cutting the lingual portion; (C) the inclination of the bur is altered to establish the correct axial wall depth; (D and E) the bur is directed perpendicular to the axial wall to accentuate the mesioaxial and distoaxial line angles and (F) the axial wall depth should be 0.2–0.5 mm inside the DEJ.

occlusally, relative to the tooth's long axis). Convergence of the mesial and distal walls ensures occlusal retention form is adequate.

The lingual portion is prepared at this point by using one of two techniques. In one technique, the lingual surface is prepared with the bur's long axis parallel with the lingual surface (Fig. 13.53A and B). The tip of the bur should be located at the gingival extent of the lingual fissure. The bur should be controlled so that it does not 'roll out' onto the lingual surface, which may 'round over' or damage the cavosurface margin. The facial inclination of the bur must be altered as the cutting progresses to establish the axial wall of the lingual portion at a uniform depth of 0.5 mm inside the DEJ (Fig. 13.53C). The axial wall should follow the contour of the lingual surface of the tooth with an axial depth of 0.2 mm inside the DEJ at the preparation periphery. When the caries lesion affecting the lingual surface is wide mesiodistally the resultant mesial and distal external walls may diverge to the lingual. In this case, the amalgam will require retention grooves to prevent lingual displacement. An axial depth of 0.5 mm inside the DEJ is indicated if retentive grooves are required. A second, alternate approach to preparing the lingual aspect of the preparation may also be considered. In this case, the No. 245 bur is held perpendicular to the cusp ridge and then the lingual surface, as it extends the preparation from the occlusal surface gingivally (to include the entire defect). This technique also results in opposing preparation walls that converge lingually.

The No. 245 bur may be used with its long axis perpendicular to the axial wall to accentuate (i.e. refine) the mesioaxial and distoaxial line angles; this also results in the mesial and distal walls converging lingually because of the shape of the bur (Fig. 13.53D and E). During this step, the axial wall depth is not altered (Fig. 13.53F). The occlusal and lingual convergences usually provide a sufficient preparation retention form, and retention grooves are not needed.

The axiopulpal line angle must be rounded to limit the areas of stress concentration and ensure adequate preparation depth and amalgam thickness (Fig. 13.54). Initial tooth preparation of the occlusolingual preparation is now complete. As mentioned previously, enameloplasty may be performed to conserve the tooth structure and limit extension.

Additional retention in the lingual extension may be required if the extension is wide mesiodistally or if it was prepared without a lingual convergence. If additional retention is required, the No. 1/4 or No. 169 bur may be used to prepare grooves into the mesioaxial and distoaxial line angles (Fig. 13.55A). If these angles are in enamel, the axial wall must be deepened to 0.5 mm axially of the DEJ to allow the grooves to be prepared in dentin and to not undermine enamel. The depth of the grooves at the gingival floor is one-half the diameter of the No. 1/4 bur. The cutting direction for each groove is the bisector of the respective line angle. The grooves are prepared in the mesioaxial and distoaxial line angles

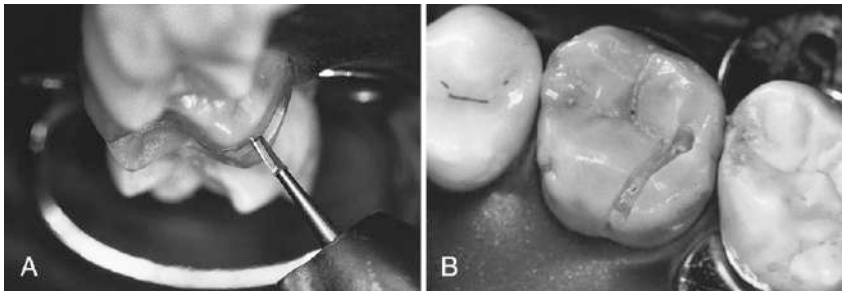


Figure 13.54 (A) Bur position for rounding the axiopulpal line angle and (B) axiopulpal line angle rounded.

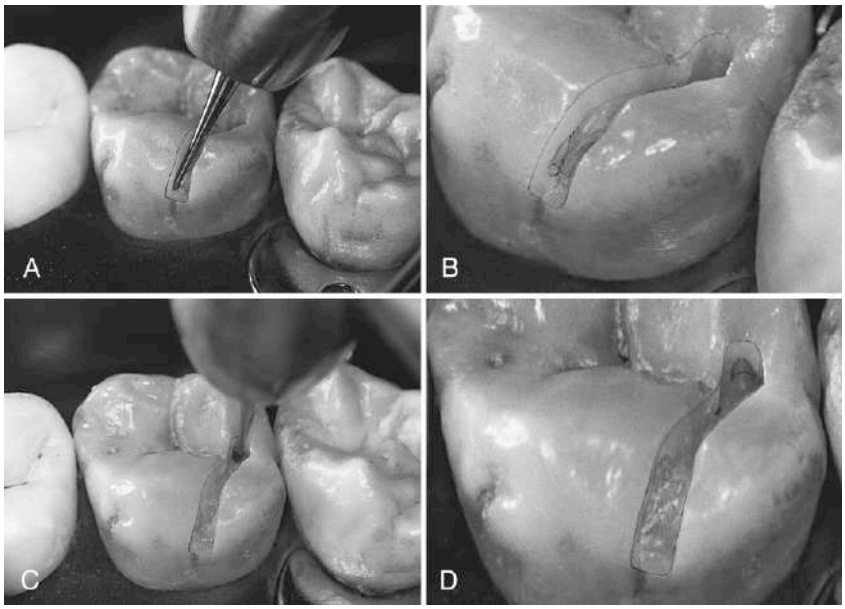


Figure 13.55 Secondary retention form. (A) Bur position for preparing groove in mesioaxial line angle; (B) completed groove is internal to the DEJ; (C) bur position for the retention cove in the faciopulpal line angle and (D) completed cove is internal to the DEJ.

of the ideally positioned axial wall and 0.2 mm axial to the DEJ. The grooves should diminish in depth towards the occlusal surface, terminating midway along the axial wall (Fig. 13.55B). The adequacy of the groove should be tested by inserting the tine of an explorer into the groove and detecting resistance to lingual movement. The depth of the groove should prevent the explorer from being withdrawn lingually. (See Secondary Resistance and Retention Forms section for a description of placing retentive grooves in the proximal boxes of Class II amalgam preparations; the techniques are similar.)

Extension of a facial occlusal fissure may have required a slight divergence occlusally to the facial wall to conserve support of the facial ridge. If so, and if deemed necessary, the No. 1/4 round bur may be used to prepare a *retention cove* in the faciopulpal line angle (Fig. 13.55C and D). The tip of the No. 245 bur held parallel to the long axis of the tooth crown also might be used to prepare this cove. Care should be taken so as not to undermine the occlusal enamel. This retentive cove is recommended only if occlusal convergence of the mesial and distal walls of the occlusal portion is absent or inadequate.

The final tooth preparation is accomplished by removal of the remaining caries lesion on the pulpal and axial walls (Fig. 13.56) with an appropriate round bur, a discoid-type spoon excavator, or both. A RMGI is placed in areas of deep caries removal for pulpal protection. The external walls are finished. Any irregularities at the margins may indicate weak enamel that may be removed by the side of the No. 245 bur rotating at low speed. The preparation should then be cleaned, carefully inspected as indicated in the General Concepts Guiding Preparation for Amalgam Restorations section and any final modifications completed.

Restorative Technique for Class I Occlusolingual or Occlusofacial Amalgam Preparations

Desensitizer Placement

A dentin desensitizer is placed over the prepared tooth structure and rinsed per manufacturer instructions. Excess moisture is removed using air stream without desiccating the dentin.

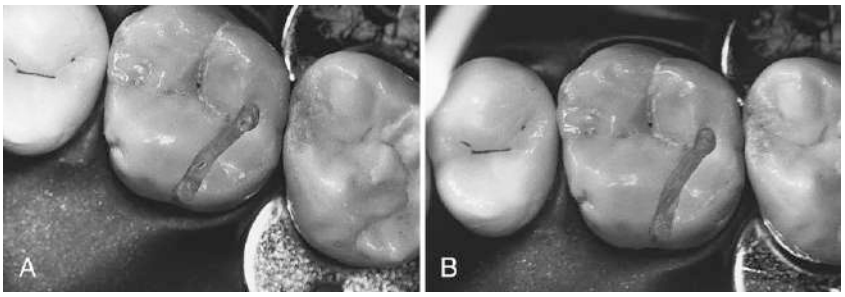


Figure 13.56 (A) Any remaining pit and fissure in enamel and soft dentin on established pulpal and axial walls are removed and (B) completed tooth preparation.

Matrix Placement (If Necessary)

Using a matrix to support the lingual portion of the restoration during condensation is occasionally necessary. A matrix is helpful to prevent 'land sliding' during condensation and to ensure marginal adaptation and strength of the restoration. The Tofflemire matrix retainer is used to secure a matrix band to the tooth (as described later). Because this type of matrix band does not intimately adapt to the lingual groove area of the tooth (Fig. 13.57A), an additional step may be necessary to provide a matrix that is rigid on the lingual portion of the tooth preparation. If so, a piece of stainless steel matrix material (0.05-mm thick and 8-mm wide) is cut to fit between the lingual surface of the tooth and the band already in place (Fig. 13.57B). The gingival edge of this segment of matrix material is placed slightly gingival to the gingival edge of the band to help secure the band segment. A quick-setting, rigid polyvinyl siloxane (PVS)-based material may be used between the sectional matrix and the Tofflemire matrix band, to prevent lingual displacement of the sectional matrix during condensation of the amalgam. Alternatively, green stick compound may be used. In this case, the end of a toothpick wedge is covered with softened (heated) compound. The compound-coated wedge is then immediately inserted between the Tofflemire band and the cut piece of matrix material (Fig. 13.57C). While the compound is still soft, a suitable burnisher is used to press the compound gingivally, securing the matrix tightly against the gingival cavosurface margin and the lingual surface of the tooth to provide a rigid, lingual matrix (Fig. 13.57D and E). This matrix for the occlusolingual amalgam restoration is referred to as the *Barton matrix*. Occasionally, the piece of strip matrix may be positioned appropriately by using only the wedge (without the rigid PVS or compound support).

Insertion of the Amalgam

Insertion of amalgam is accomplished as previously described in the General Concepts Guiding Restoration with Amalgam section and for the Class I occlusal tooth preparation. Condensation is begun at the gingival wall at the lingual component of the preparation. If a matrix is not used, care must be taken to ensure that 'land sliding' of the amalgam does not occur because two adjoining surfaces of the tooth are being restored. For this technique, the last increments of amalgam may be condensed on the lingual surface with the side of a large condenser. Its long, broadly rounded contour approximates the rectangular shape of the lingual groove preparation. Appropriate care should be taken (when condensing the occlusal surface) to prevent fracturing the lingual amalgam. Another technique is to have the assistant secure the condensed lingual surface with a broad condenser nib, while the operator completes the condensation of the occlusal surface. Regardless of the technique used, the amalgam must be well condensed.

Contouring and Finishing of the Amalgam Restoration

When the preparation is sufficiently overfilled, carving of the occlusal surface may begin immediately with a sharp discoid–cleoid instrument or a Hollenback carver. All carving should be done with the edge of the blade perpendicular to the margin and the blade moving parallel to the margin. To prevent over-carving, the blade edge should be guided by the unprepared tooth surface adjacent to the margin. An explorer is used to remove excess amalgam adjacent to the lingual matrix before matrix removal (Fig. 13.57F). After the occlusal carving is complete, the Tofflemire retainer is loosened from the band, and the band is removed with No. 110 pliers. The free ends of the band are pushed one at a time, lingually and occlusally, through the proximal contacts. After the lingual carving is complete (Fig. 13.57G), the rubber dam is removed and the restoration is adjusted to ensure proper occlusion. Most amalgams do not require finishing and polishing. Fig. 13.57H illustrates a polished occlusolingual restoration.

Class I Occlusofacial Amalgam Restorations

Occasionally, mandibular molars exhibit fissures that extend from the occlusal surface through the facial cusp ridge and onto the facial surface. The preparation and restoration of these defects are very similar to those described for Class I occlusolingual amalgam restorations. Although these may be restored with composite, an illustration of preparation and restoration with amalgam is provided in Fig. 13.58. The amalgam restoration may be polished after it is completely

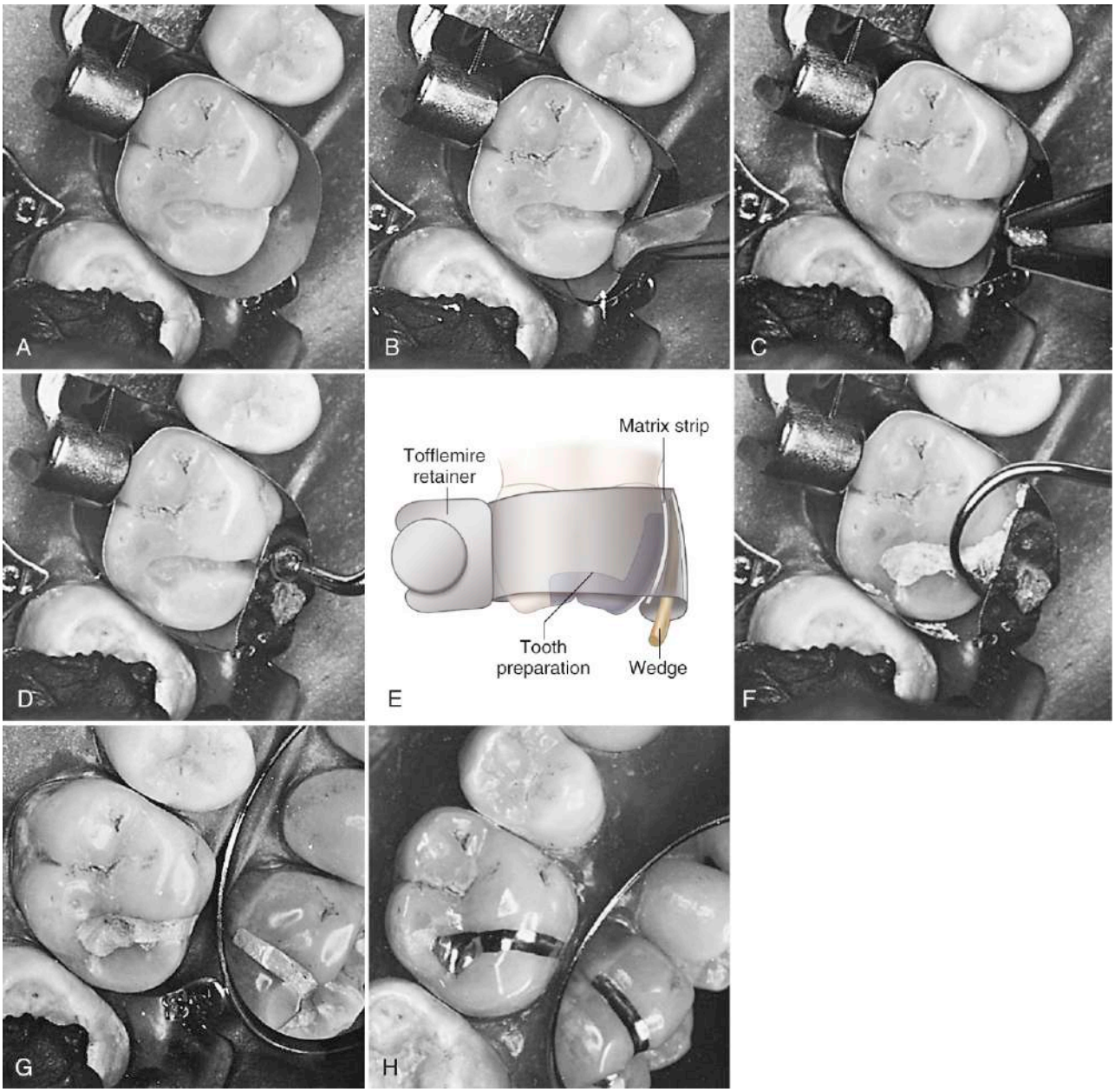


Figure 13.57 Matrix for occlusolingual tooth preparation. (A) Matrix band secured to the tooth with Tofflemire retainer; (B) positioning a small strip of stainless steel matrix material between the tooth and the band already in place; (C) inserting the wedge and the rigid supporting material; (D) compressing the rigid supporting material gingivally (or holding the steel strip in correct position while the material sets), which adapts the steel strip to the lingual surface; (E) cross-section of the tooth preparation and the matrix construction; (F) using the explorer to remove excess amalgam adjacent to the lingual matrix; (G) carving completed and (H) polished restoration.

set. The shape of abrasive points may need to be modified to allow optimal polishing (Fig. 13.58I and J).

Clinical Technique for Class II Amalgam Restorations

Class II restorations restore defects that affect one or both of the proximal surfaces of posterior teeth (Fig. 13.26B).

Amalgam restorations that restore one or both of the proximal surfaces of the tooth may provide years of service to the patient when (1) the operating field is isolated, (2) the tooth preparation is correct, (3) the matrix is suitable and (4) the restorative material is manipulated properly. Application of the principles discussed here will result in high-quality, durable Class II amalgam restorations.

Initial Clinical Procedures

Occlusal contacts should be marked with articulating paper before tooth preparation. A mental image of these contacts

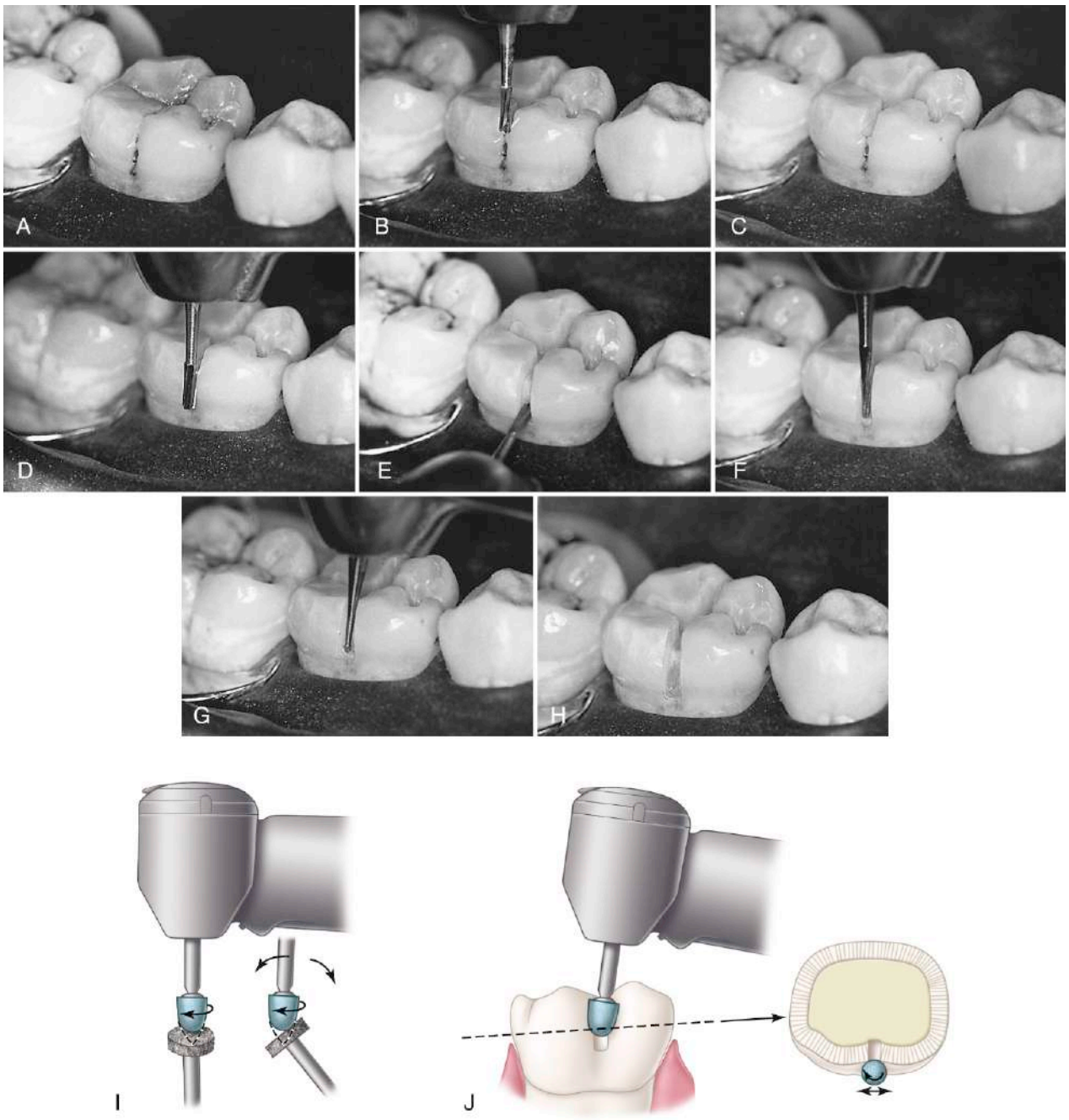


Figure 13.58 Fissure extension. (A) Facial occlusal fissure continuous with the fissure on the facial surface; (B) extension through the facial ridge onto the facial surface; (C) appearance of the tooth preparation after extension through the ridge; (D) the facial surface portion of the extension is cut with the side of the bur; (E) the line angles are sharpened by directing the bur from the facial aspect; (F) sharpening the line angles from the occlusal direction with a No. 169 L bur; (G) ensuring the retention form by preparing retention grooves with a No. 1/4 round bur; (H) completed tooth preparation; (I) the rubber polishing point may be re-shaped (blunted as indicated) on a coarse diamond wheel and (J) proper orientation of the rubber point when polishing the facial surface groove area.

will serve as a guide in tooth preparation and restoration. Any opposing '*plunging cusp*' (that is not aligned well with the occlusal plane) or other sharply pointed cusp may need to be re-contoured to reduce the risk of fracture of the new restoration.

Tooth Preparation for Class II Amalgam Restorations That Involve Only One Proximal Surface

This section introduces the principles and techniques of a Class II tooth preparation for an amalgam restoration involving a caries lesion on one proximal surface. A small,

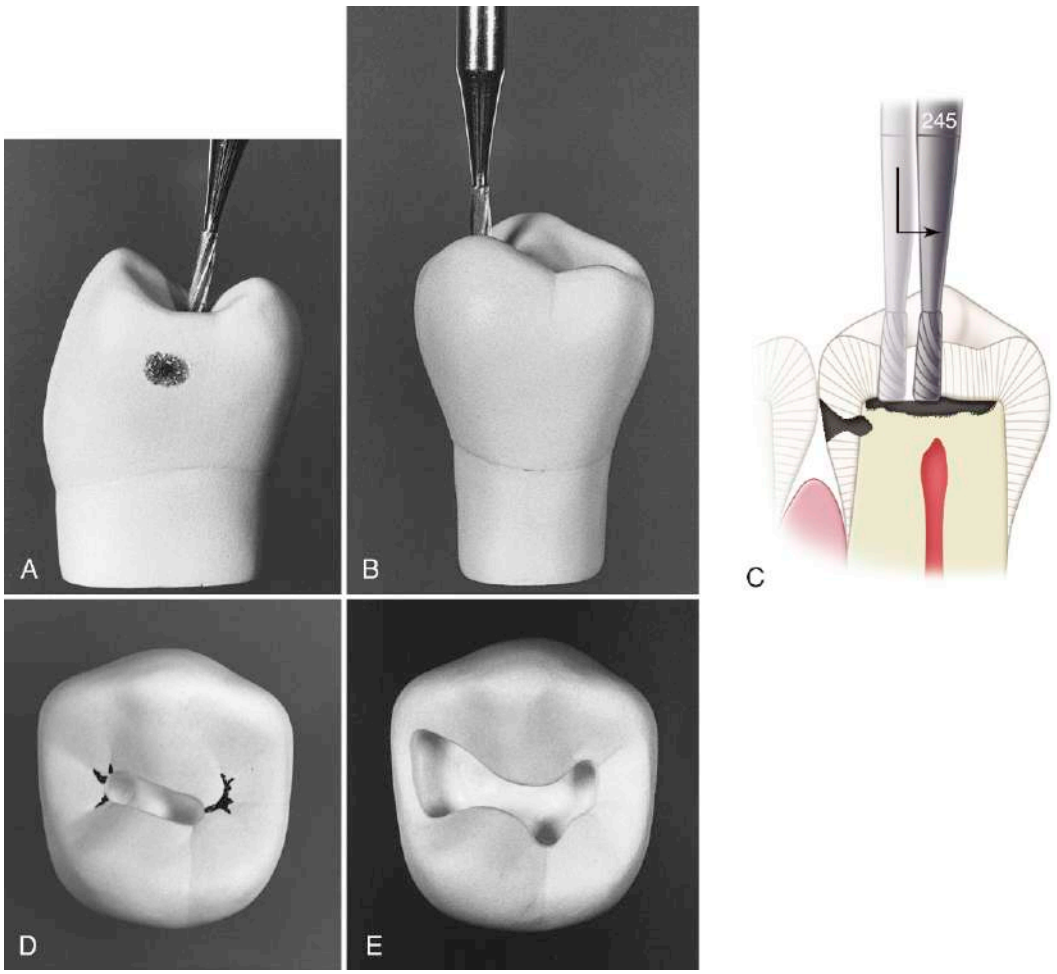


Figure 13.59 Entry and occlusal step. (A) Bur position for entry, as viewed proximally. Note the slight lingual tilt of the bur; (B) bur position as viewed lingually; (C) the tooth is entered with a punch cut, and extension is done distally along central fissure at a uniform depth of 1.5–2 mm (1.5 mm at fissure; because of the inclination of the unprepared tooth surface, the corresponding measurement on the prepared wall is greater); (D) occlusal view of (C) and (E) completed occlusal step.

conservative mesioocclusal tooth preparation on the mandibular second premolar is presented to illustrate basic concepts. Composite may also be an acceptable restorative material for this preparation. Preoperative placement of a wedge (*pre-wedging*) in the space adjacent to the proximal surface with the caries lesion will create space between the teeth. This lowers the risk of iatrogenic damage to the adjacent tooth during preparation.

Initial Tooth Preparation

Occlusal outline form (occlusal step)

Step 1: Punch cut

- The occlusal outline form of a Class II tooth preparation for amalgam is similar to that for a Class I tooth preparation. Using high speed with air-water spray, the operator enters the pit nearest the involved proximal surface with a punch cut using a No. 245 bur oriented as illustrated in Fig. 13.59A and B.
- The bur should be rotating when it is applied to the tooth and should not stop rotating until removed. Viewed from the proximal and lingual (facial) aspects, the long axis of the bur and the long axis of the tooth crown should remain parallel during the cutting procedures. The dentin caries lesion initially spreads at the DEJ; therefore the goal of the initial cut is to reach the DEJ. The DEJ location in posterior teeth is approximately 1.5–2.0 mm from the occlusal surface.
- As the bur enters the pit, a *target depth of 0.1–0.2 mm into dentin* (i.e. just into dentin) should be established. This depth is *one-half to two-thirds the length of the cutting portion of a No. 245 bur*, or approximately 1.5 mm as measured from the central fissure and 2 mm from the preparation external wall.

Step 2: Occlusal extension

- While maintaining the same depth and bur orientation, the bur is moved to extend the outline to include the carious central fissure and the opposite carious pit (Fig. 13.59C and D). The isthmus width should be as narrow as possible, preferably *no wider than one-quarter the intercuspal distance*.[27,28,69,70]

- Ideally, the preparation should be the *width of the No. 245 bur*.
- Narrow restorations provide a greater length of clinical service.[30,33] Ultimately, the extension of the caries lesion at the DEJ will determine the amount of preparation extension and resultant width. The pulpal floor of the preparation should follow the slight rise and fall of the DEJ along the central fissure in teeth with prominent triangular ridges.
- Maintaining the bur parallel to the long axis of the tooth crown creates facial, lingual and distal walls with a *slight occlusal convergence* as well as external wall orientation favourable for amalgam use. The facial, lingual and distal walls should be extended until a sound DEJ is reached. Proper extension will result in the formation of the peripheral seat, which aids in the primary resistance form. It may be necessary to tilt the bur to diverge occlusally at the distal wall if further distal extension would undermine the marginal ridge of its dentinal support.

Step 3: Occlusal dovetail (if required)

- During development of the distal pit area of the preparation, extension to include any distofacial and distolingual developmental fissures radiating from the pit may be indicated. The distal pit area (in this example) provides a dovetail retention form, which will prevent mesial displacement of the completed Class II restoration.
- However, the dovetail feature is *not required* in the occlusal step of a single proximal surface preparation unless a fissure emanating from an occlusal pit indicates it. This type of retention form (i.e. form that provides resistance to mesial displacement of the restoration) also is provided by any extension of the central fissure preparation that is not in a straight direction from pit to pit (Fig. 13.59E). A dovetail outline form in the distal pit is not required if radiating fissures are not present.[71,72] Enameloplasty should be performed, where indicated, to conserve the tooth structure.

Step 4: Proximal extension

- Before extending into the involved proximal marginal ridge (the mesial ridge, in this example), the final locations of the facial and lingual walls of the proximal box are estimated visually. Visual assessment prevents overextension of the occlusal outline form (i.e. occlusal step) where it joins the proximal outline form (i.e. proximal box). Fig. 13.60 illustrates visualization of the final location of the proximoocclusal margins before preparing the proximal box.
- *Reverse curve*. The reverse curve is an 'S'-shaped concave curve made by offsetting the outline form of the isthmus where it joins the outline form of the proximal box. Showing the view from the occlusal aspect, Fig. 13.61 illustrates a *reverse curve* in the occlusal outline of a Class II preparation, which often results when developing the mesiofacial wall perpendicular to the enamel rod direction while, at the same time, conserving as much of the facial cusp structure as possible.[70] The extension into the mesiofacial cusp is limited to that amount required to permit a 90° mesiofacial margin. Lingually, the reverse curve usually is minimal (if necessary at all) because of the relationship of the central fissure relative to the faciolingual position of the proximal contact.
- While maintaining the established pulpal depth and with the bur parallel to the long axis of the tooth crown, the preparation is extended mesially, stopping *approximately 0.8 mm short of cutting through the marginal ridge into the contact area*.
- The occlusal step in this region is made *slightly wider faciolingually* than in the Class I preparation because additional width is necessary for the proximal box. The proper depth of the occlusal portion of the preparation increases the strength of the restoration however, more than does faciolingual width (Fig. 13.59C and E).
- Although this extension includes part of the mesial marginal ridge, it also exposes the marginal ridge DEJ. The location of the DEJ is an important guide in the development of the proximal preparation.

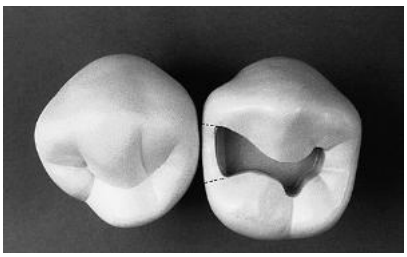


Figure 13.60 Visualize final location of proximoocclusal margins (*dotted lines*) before preparing the proximal box.

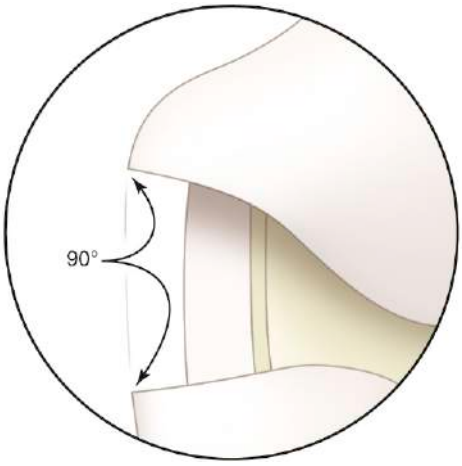


Figure 13.61 The reverse curve in the occlusal outline usually is created when the mesiofacial enamel wall is parallel to the enamel rod direction. Lingually, the reverse curve is very slight, often unnecessary.

Proximal outline form (the proximal 'box')

Before initiating the proximal outline form (i.e. the proximal 'box'), the operator should visualize the desired final

location of the facial and lingual walls relative to the contact. The objectives for the extension of the proximal margins are to:

I. include all caries lesion, defects, or existing restorative material;
II. create approximately 90° cavosurface margins (i.e. butt-joint margins);
III. establish (ideally) not more than 0.5-mm clearance with the adjacent proximal surface facially, lingually and gingivally.
IV. *primary flare* is the flare given from the isthmus region towards the proximal cavosurface margin. This is done to bring the facial and lingual proximal walls into the facial and lingual embrasures.

Step 1: Proximal ditch cut

- The initial procedure in preparing the outline form of the proximal box is the isolation of the proximal (i.e. in this case, mesial) enamel by the proximal ditch cut.
- While maintaining the same orientation of the bur, it is positioned over the DEJ in the pulpal floor next to the remaining mesial marginal ridge (Fig. 13.62A).
- The end of the 0.8-mm-diameter bur is allowed to cut a ditch gingivally along the exposed proximal DEJ:
  - *Two-thirds at the expense of enamel*: 0.5–0.6 mm into enamel
  - *One-third at the expense of dentin*: 0.2–0.3 mm into dentin
- Pressure is directed gingivally and lightly towards the mesial surface to keep the bur against the proximal enamel, while the bur is moved facially and lingually along the DEJ.

Step 2: Gingival extension of proximal box

- The ditch is extended gingivally *just beyond the caries lesion or the proximal contact*, whichever is *greater* (Fig. 13.62B). Axial wall dentinal depths will vary based on the gingival extension of the preparation (Fig. 13.62C). The axial wall follows the faciolingual contour of the proximal surface and the DEJ (Fig. 13.62D).
- The proximal ditch cut may diverge gingivally (i.e. *converge occlusally*) to ensure that the faciolingual dimension at the gingival aspect is greater than at the occlusal aspect (Fig. 13.62G). The shape of the No. 245 bur will provide this divergence. The gingival divergence of the proximal box contributes to:
  - The retention form of the proximal box,
  - Provides for the desirable extension of the facial and lingual proximal margins to include defective tooth structure or old restorative material at the gingival level,
  - Conserves the marginal ridge and
  - Provides for 90° amalgam at the margins on this ridge.[28]
- It is necessary to visualize the completed mesiofacial and mesiolingual margins as right-angle projections of the facial and lingual limits of the ditch to establish the proper faciolingual ditch extension (Fig. 13.62E). When preparing a tooth with a small lesion, these margins may clear the adjacent tooth by only 0.2–0.3 mm.[70] A guide for the gingival extension is the visualization that the finished gingival margin will be only slightly gingival to the gingival limit of the ditch. This gingival margin will likely clear the adjacent tooth by only *0.5 mm* when treating an early cavitated proximal lesion (Fig. 13.62F).[71] Recall that the caries lesion develops immediately gingival to the proximal contact and therefore the preparation will always result in gingival clearance, and that the amount of clearance will depend on the size of the caries lesion.
- Clearance of the proximal margins (i.e. mesiofacial, mesiolingual, gingival) *greater than 0.5 mm is excessive*, unless indicated to include the caries lesion, undermined enamel, or existing restorative material. The location of the final proximal margins (i.e. facial, lingual, gingival) should be established with hand instruments (i.e. chisels, hatchets, trimmers) and not the No. 245 bur so as to prevent unnecessary overextension (Fig. 13.62E).
- Extending gingival margins into the gingival sulcus should be *avoided*, where possible, because sub-gingival margins are more difficult to restore and may be a contributing factor to periodontal disease.[73–75]

Step 3: Preparation of axial wall

- Since the proximal enamel becomes thinner from the occlusal to the gingival aspect, the end of the bur comes closer to the external tooth surface as the cutting progresses gingivally (Fig. 13.62B, C and I). Premolars may have proximal boxes that are shallower pulpally than are molars because premolars typically have thinner enamel.
  - In the tooth crown, the ideal dentinal depth of the axial wall of the proximal boxes of premolars and molars should be the same (two-thirds to three fourths the diameter of the No. 245 bur [or **0.5–0.6 mm**]).[70]
  - When the extension places the gingival margin in cementum, the initial pulpal depth of the axiogingival line angle should be **0.7–0.8 mm** (the diameter of the tip end of the No. 245 bur is 0.8 mm). The bur may shave the side of the wedge that is protecting the rubber dam and the underlying gingiva (Fig. 13.62C).
  - The depth of the axial dentinal wall should be adjusted to approximately *0.5 mm if retention grooves are deemed necessary* (i.e. if caries lesion removal requires placement/orientation of facial and lingual walls that are very divergent). This will allow the grooves to be prepared into the axiolingual and axiofacial line angles without undermining the proximal wall enamel.
  - Occasionally it is permissible *not to extend the outline of the proximal box facially or lingually beyond the proximal contact to conserve the tooth structure in the following scenarios*:[71]

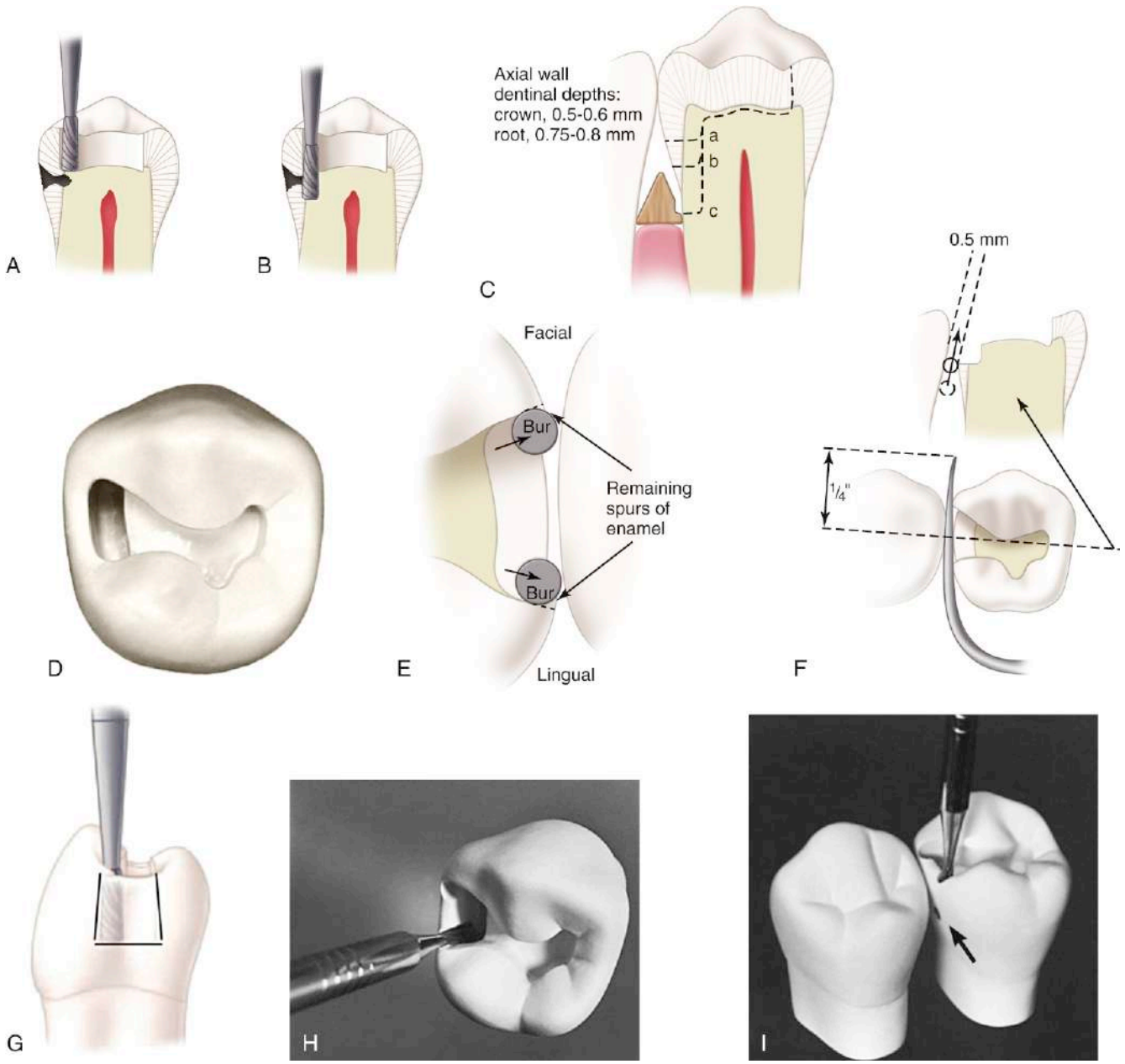


Figure 13.62 Isolation of proximal enamel. (A) Bur position to begin the proximal ditch cut. (B) The proximal ditch is extended gingivally to the desired level of the gingival wall (i.e. floor). (C) Variance in the pulpal depth of the axiogingival line angle as the extension of the gingival wall varies: (*a*) at minimal gingival extension; (*b*) at moderate extension; (*c*) at extension that places gingival margin in cementum, whereupon the pulpal depth is 0.75–0.8 mm and the bur may shave the side of wedge. (D) The proximal ditch cut results in the axial wall that follows the outside contour of the proximal surface. (E) The position of the proximal walls (i.e. facial, lingual, gingival) should not be overextended with the No. 245 bur, considering additional extension will occur when the remaining spurs of enamel are removed. (F) When a small lesion is prepared, the gingival margin should clear the adjacent tooth by only 0.5 mm. This clearance may be measured with the side of the explorer. The diameter of the tine of a No. 23 explorer is ~0.5 mm at 6 mm from its tip. (G) The faciolingual dimension of the proximal ditch is greater at the gingival level than at the occlusal level to provide occlusal convergence of the facial and lingual proximal box walls. (H) To isolate and weaken the proximal enamel further, the bur is moved towards and perpendicular to the proximal surface. (I) The side of the bur may emerge slightly through the proximal surface at the level of the gingival floor (*arrow*).

- narrow proximal lesion where broad proximal contact is present in a patient with low risk for caries.
- If it is necessary to extend 1 mm or more just to arbitrarily create clearance (i.e. to 'break the contact'), the proximal margin is left in contact with the adjacent proximal surface.
- The mesiofacial margin is left within the contact area for esthetic reasons in premolars.

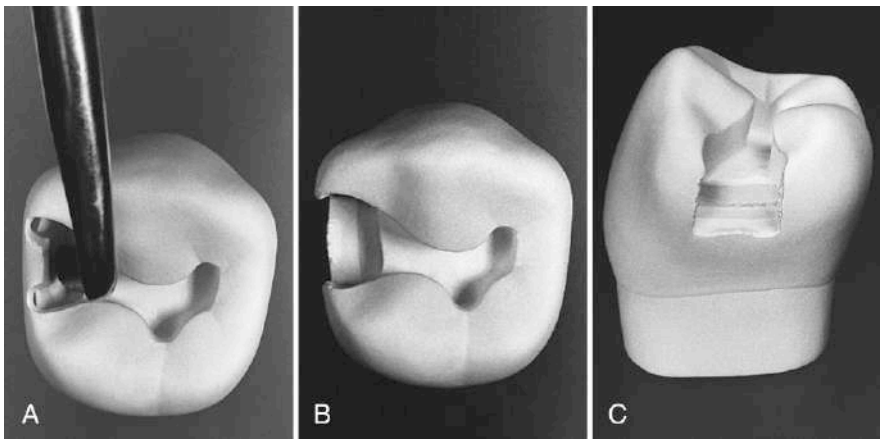


Figure 13.63 Removing isolated enamel. (A) Using a spoon excavator to fracture the weakened proximal enamel; (B) occlusal view with the proximal enamel removed and (C) proximal view with the proximal enamel removed.

- If the *proximal ditch cut is entirely in dentin, the initial axial wall is too deep*.

Clinical Notes

The gingival extension of the proximal ditch may be measured by first noting the depth of the nonrotating bur in the ditch. The dentist removes the bur from the preparation and holds it in the facial embrasure at the same level to observe the relationship of the end of the bur to the proximal contact. A periodontal probe also may be used.

Step 4: Proximal extension of proximal box

- The proximal extensions are completed when two cuts, one starting at the facial limit of the proximal ditch and the other starting at the lingual limit, extending towards and perpendicular to the proximal surface (until the bur is nearly through enamel at the contact level), are made (Fig. 13.62H).
- The side of the bur may emerge slightly through the surface at the level of the gingival floor (Fig. 13.62I); this weakens the remaining enamel by which the isolated portion is held. If this level is judged to be insufficiently gingival, additional gingival extension should be accomplished using the isolated proximal enamel, that is still in place, as a mental barrier to help the operator guide the bur.

Step 5: Breaking the proximal wall

- Ensuring the presence of the proximal enamel surface helps to limit the likelihood of iatrogenically damaging the proximal surface of the adjacent tooth. At this stage, however, the remaining wall of enamel often breaks away during cutting, especially when high speed is used. At such times, if additional use of the bur is indicated, a matrix band may be used around the adjacent tooth to limit potential marring of its proximal surface; however, the relative amount of protection afforded by the metal matrix material is very limited and serves more as a visual guide than an actual physical barrier. A No. 245 bur, rotating at between 200,000 and 400,000 rpm, will rapidly cut through any matrix material and damage the adjacent proximal surface.
- The isolated enamel, if still in place, may be fractured with a *spoon excavator* (Fig. 13.63) or by mesial movement with the side of the nonrotating bur.
- To protect the gingiva and the rubber dam when extending the gingival wall apically, a wooden wedge should already be in place in the gingival embrasure to depress soft tissue and the rubber dam.[28]
  - A *round toothpick wedge* is preferred unless a deep gingival extension is anticipated (Fig. 13.64A).
  - A *triangular (i.e. anatomic) wedge* is more appropriate for deep gingival extensions because the greatest cross-sectional dimension of the wedge is at its base; as the gingival wall is cut,

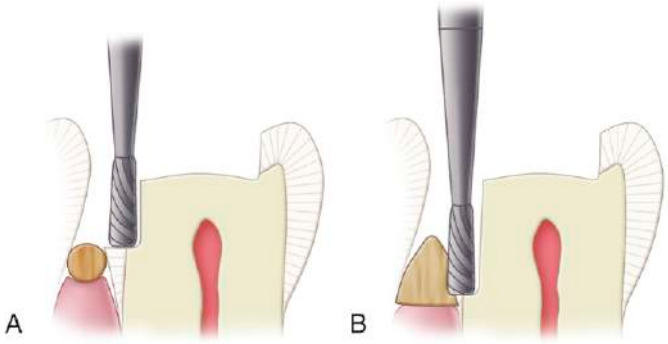


Figure 13.64 Wedging. (A) A round toothpick wedge placed in the gingival embrasure protects the gingiva and the rubber dam during preparation of the proximal box; (B) a triangular wedge is indicated when a deep gingival extension of the proximal box is anticipated because the wedge's greatest cross-sectional dimension is at its base. Consequently it more readily engages the remaining clinical tooth surface. Wedging increases the interproximal space and limits the potential for iatrogenic damage of adjacent tooth (or restoration) surfaces.

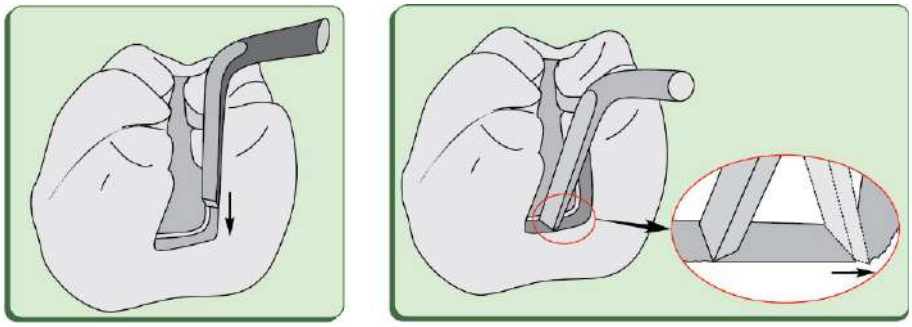


Figure 13.65 Removing the remaining undermined proximal enamel with an enamel hatchet on the proximal wall (A) and the gingival wall (B).
(From Gopikrishna V: *Preclinical manual of conservative dentistry*, ed 2, 2014, Elsevier).

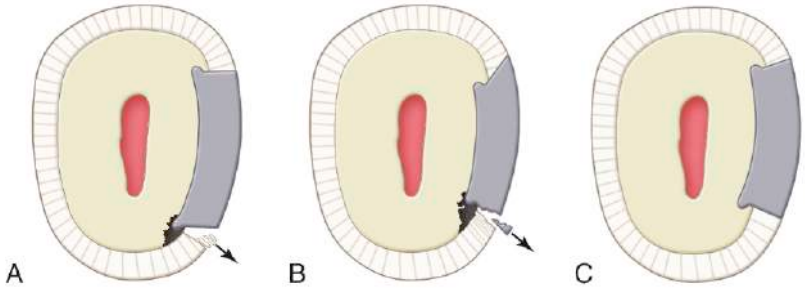


Figure 13.66 Direction of mesiofacial and mesiolingual walls. (A) Failure caused by a weak enamel margin; (B) failure caused by a weak amalgam margin and (C) proper direction to the proximal walls results in full-length enamel rods and 90° amalgam at the preparation margin. Retention grooves have been cut 0.2 mm inside the DEJ, and their direction of depth is parallel to the DEJ.

the bur's end corner may shave the wedge slightly (Fig. 13.64B).

- With a *sharp enamel hatchet* (10–7–14), bin-angle chisel (12–7–8), or both, the dentist cleaves away any remaining undermined proximal enamel (Fig. 13.65), establishing the proper orientation of the mesiolingual and mesiofacial walls. Proximal walls that result in cavosurface angles of 90° are desired.[28]
- Cavosurface angles of 90° ensure that no undermined enamel rods remain on the proximal margins and that the maximal edge strength of amalgam is maintained. The cutting edge of the instrument should not be aggressively forced against the gingival wall because this may cause a craze line (i.e. fracture) that extends gingivally in enamel, perhaps to the cervical line. Fig. 13.66 shows the importance of the correct direction of the mesiofacial and mesiolingual walls, dictated by enamel rod direction and the physical properties of amalgam. *If hand instruments were not used to remove the remaining spurs of enamel, the proximal margins would have undermined enamel.*
- To create 90° facial and lingual proximal margins with the No. 245 bur, the proximal margins would have to be significantly overextended for an otherwise conservative preparation. The weakened enamel along the gingival wall is removed by using the enamel hatchet in a *scraping motion* (Fig. 13.65).
- When a rotary instrument is used in a proximal box after the proximal enamel is removed, there is increased risk that the instrument may either mar the adjacent proximal surface or 'crawl out' of the box into the gingiva or across the proximal margins. The latter mishap produces a rounded cavosurface angle, which results in a weak amalgam margin of less than 90° if not corrected.

Primary resistance form

The primary resistance form is provided by

I. the pulpal and gingival walls being relatively *flat* (i.e. perpendicular to forces directed along the long axis of the tooth);
II. restricting the *extension of the walls* to allow sufficient dentin support to remain (and therefore strong cusps and ridge areas) while at the same time establishing the peripheral seat;

III. restricting the *occlusal outline form* (where possible) to areas receiving minimal occlusal contact;[30]
IV. use of a *reverse curve* to optimize the strength of the amalgam and tooth structure at the junction of the occlusal step and proximal box;
V. slight *rounding of the internal line angles* to reduce stress concentration in the tooth structure (automatically created by bur design except for the axiopulpal line angle) and
VI. providing enough *thickness of the restorative material* to prevent its flexure and resultant fracture from the forces of mastication.

Primary retention form

The primary retention form is provided by

I. the *occlusal convergence* of the facial and lingual walls and
II. by the *dovetail design* of the occlusal step, if present.

After completing the initial tooth preparation, the adjacent proximal surface should be evaluated. An adjacent proximal restoration may require re-contouring to re-establish the normal anatomic convex shape; this may be done with abrasive finishing strips, disks, finishing burs, or a combination of these. Any iatrogenic damage that compromises the convex adjacent proximal surface should be corrected by re-contouring or restoration.

Clinical Notes

Generally, the amount of remaining tooth structure is more important to restoration longevity than is the restorative material used.[76]

Final tooth preparation

Removal of any remaining defective enamel and infected carious dentin

- Removing enamel pit-and-fissure remnants and soft dentin on the pulpal wall in Class II preparations is accomplished in the same manner as in the Class I preparation.
- Soft dentin is removed with a slowly revolving round bur of appropriate size, a discoid-type spoon excavator, or both. In general, the caries lesion removal should stop when a hard or firm feel, with an explorer or small spoon excavator, is achieved; this often occurs before all of the stained or discoloured dentin is removed. The exception to this is when using the *selective caries removal protocol* (Chapter 2).
- Removing enamel pit-and-fissure remnants and carious dentin should not affect the resistance form. To achieve an enhanced resistance form, the occlusal step should have pulpal seats at the initial preparation depth, perpendicular to the long axis of the tooth in sound tooth structure and peripheral to the excavated area (Fig. 13.67).
- Carious dentin in the axial wall is removed with appropriate round burs, spoon excavators, or both so as to conserve as much tooth structure as possible (Fig. 13.68).

Pulp protection

- Any remaining old restorative material (including base and liner) may be left if no evidence of a recurrent caries lesion exists, if its periphery is intact and if the tooth has been asymptomatic (assuming the pulp is vital). This concept is particularly important if removal of all remaining restorative material may increase the risk of pulpal exposure. Appropriate steps to protect the pulp should be taken as indicated in the General Concepts Guiding Preparation for Amalgam Restorations section.
- After completion of the minimal gingival extension (i.e. the gingivoaxial line angle is in sound dentin), a remnant of the enamel portion of a caries lesion may remain on the gingival floor (wall), seen in the form of a decalcified (i.e. white, chalky) or faulty area bordering the margin (Fig. 13.69). This situation dictates further extension of a part or all of the gingival floor to place it in sound tooth structure. Extension of the entire gingival wall to include a large caries lesion may place the gingival margin so deep that proper matrix application and wedging become extremely difficult.

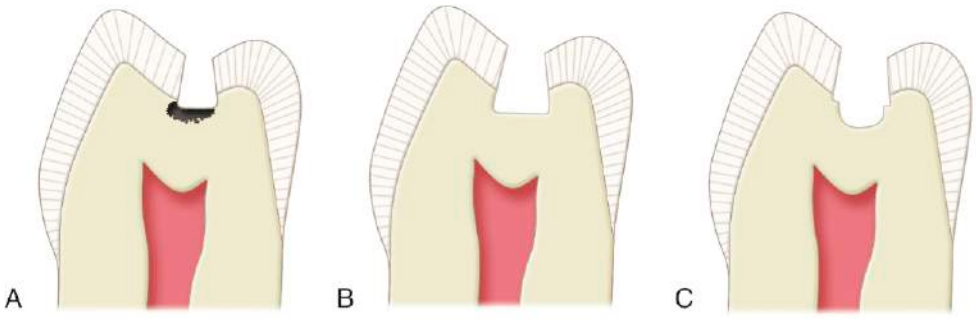


Figure 13.67 Management of small-to-moderate-sized carious lesion on the pulpal wall. (A) Soft dentin extending beyond the ideal pulpal wall position; (B) incorrect lowering of the pulpal wall to include soft dentin; (C) correct extension facially and lingually beyond the soft dentin. Note the caries removal below the ideal pulpal wall level and the facial and lingual seats at the ideal pulpal wall level.

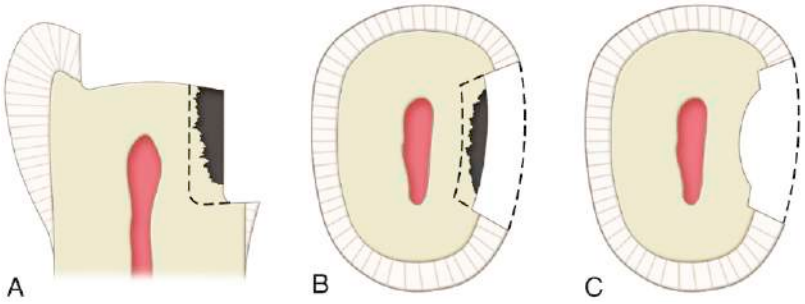


Figure 13.68 Management of a moderate to extensive carious lesion. (A and B) Soft dentin on the axial wall does not call for the preparation of the whole axial wall towards the pulp (*dotted lines*) and (C) soft dentin extending pulpally from the ideal axial wall position is conservatively removed with a round bur.

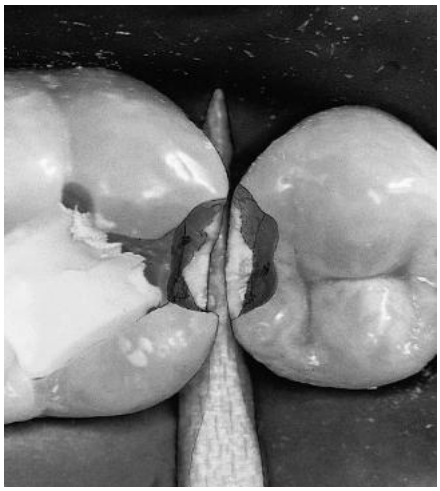


Figure 13.69 Remnant of caries lesion bordering the enamel margin after insufficient gingival extension. Such a lesion indicates extending part or all of the gingival floor to place it in sound tooth structure.
(Courtesy: Dr. C. L. Sockwell.)

Fig. 13.70A illustrates an outline form that extends gingivally in the central portion of the gingival wall to include a caries lesion that is deep gingivally, although leaving the facial and lingual gingival corners at a more occlusal position. This partial extension of the gingival wall permits wedging of the matrix band where otherwise it may be difficult and damaging to soft tissue. In this instance, the gingival wedge may not tightly support a small portion of the band. Special care must be exercised by placing small amounts of amalgam in this area first and thoroughly condensing with light pressure. In addition, care is exercised in carving the restoration in this area to remove any excess that may have extruded gingivally during condensation.

- Fig. 13.70B illustrates removal of a caries lesion facially and gingivally beyond the conventional margin position. Such minor variations from the ideal preparation form permit conservation of healthy tooth structure.

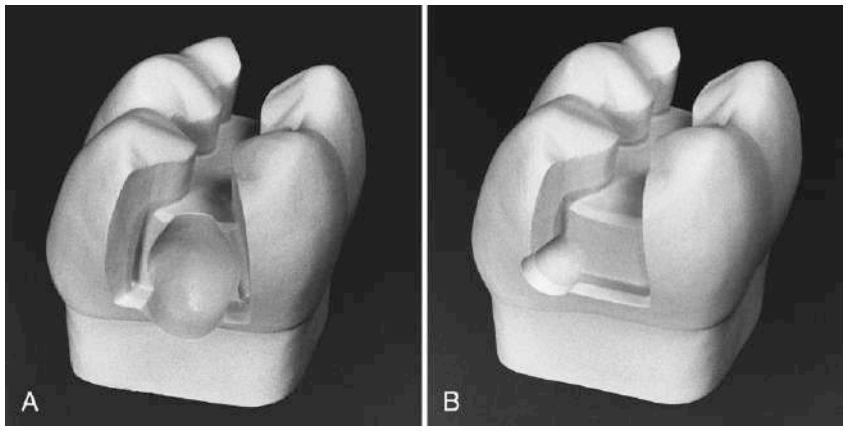


Figure 13.70 (A) Outline form that permits extension of the centre portion of the gingival wall to facilitate proper matrix construction and wedging in situations where the caries lesion extends deep gingivally and (B) outline form that permits partial wall extension facially and gingivally to conserve the tooth structure.

A partial extension of a facial or lingual wall is permissible if
- the entire wall is not weakened,
- the extension remains accessible and visible,
- sufficient gingival seats remain to support the restoration and
- a butt-joint fit at the amalgam–enamel margin (90° amalgam angle and 90° cavosurface angle) is possible.

Secondary resistance form

- Conserving as much tooth structure as possible by *limiting the extensions of external walls* helps to provide secondary resistance by limiting the risk of future tooth fracture from oblique forces.
- Ensuring that *all internal line angles are rounded* is necessary to limit areas of stress concentration in the remaining tooth structure and in the new restoration and improves secondary resistance of both.
- Use of the gingival margin trimmer or a bur to *round the axiopulpal line angle* (Fig. 13.71) helps to increase the bulk of restorative material and decrease the stress concentration within the restorative material.

Secondary retention form

- The secondary retention forms for the occlusal and proximal portions of the preparation should be independent of each other.
- The occlusal *convergence* of the facial and lingual walls provides a sufficient retention form to the occlusal portion of the tooth preparation.

Proximal retention grooves

- To enhance the retention form of the proximal portion, proximal grooves may be indicated to counter proximal displacement of the amalgam restoration.[28,77–79]
- *The use of retention grooves is recommended in tooth preparations with extensive proximal boxes.*
- It also has been reported that proximal retention grooves are *unnecessary* in
  - preparations that include dovetails, as part of the occlusal preparation, when high-copper amalgam is used;[78,80]
  - preparations that are conservative with narrow proximal boxes.[79]
- Location of retention grooves in proximal boxes:
  - It has been reported that *proximal retention grooves in the axiofacial and axiolingual line angles* may increase the fracture resistance and significantly strengthen the isthmus of a Class II amalgam restoration and that these grooves are significantly superior to the axiogingival grooves in increasing the restoration's fracture strength.[81–84]
  - Other investigators have suggested, however, that *retention grooves located occlusal to the axiopulpal line angle* provide more resistance than do conventional grooves.[77]
- Depth of the retentive groove:
  - Generally, the depth of the grooves should be approximately half of the diameter of the tip of the No. 169 or No. 1/4 round bur (i.e. *~0.25–0.5 mm*).
  - The depth of retention grooves in extensively wide proximal boxes may need to be *0.5 mm or greater* at the gingival aspect.
- Retention grooves are prepared with a No. 169 L or No. 1/4 round bur with air coolant only (to improve visualization) and reduced speed (to improve tactile 'feel' and control). The bur is placed in the properly positioned axiolingual line angle and directed (i.e. translated) to bisect the angle (Fig. 13.72) approximately parallel to the DEJ (Fig. 13.73). This positions the retention groove 0.2 mm inside the DEJ, maintaining the enamel support.
- The bur is tilted to allow cutting to the depth of the diameter of the end of the bur (which is initially positioned at the axiolinguogingival point angle) and to permit the groove to diminish in depth occlusally. The facial groove in the axiofacial line angle is prepared in a similar manner. When the axiofacial and axiolingual line angles are less than 2 mm in length, the tilt of the bur is reduced slightly so that the proximal grooves are extended occlusally to disappear midway between the DEJ and the enamel margin (Fig. 13.72B and C).

The four characteristics or determinants (Fig. 13.73) of proximal grooves are:

I. Position. *Position* refers to the axiofacial and axiolingual line angles of initial tooth preparation (0.5 mm axial to the DEJ).
II. Translation. *Translation* refers to the direction of movement of the axis of the bur.
III. Depth. *Depth* refers to the extent of translation (i.e. depth of the retention groove).
IV. Occlusogingival orientation. *Occlusogingival orientation* is considered when using the No. 169 L bur and refers to the tilt of the bur. The tilt dictates the occlusal

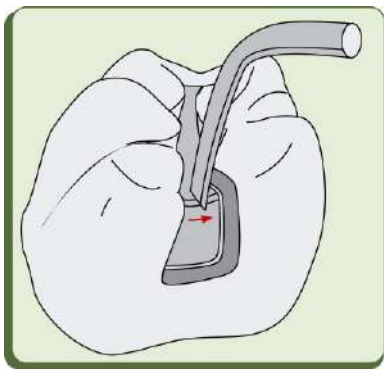


Figure 13.71 Rounding the axiopulpal line angle.
(From Gopikrishna V: *Preclinical manual of conservative dentistry*, ed 2, 2014, Elsevier).

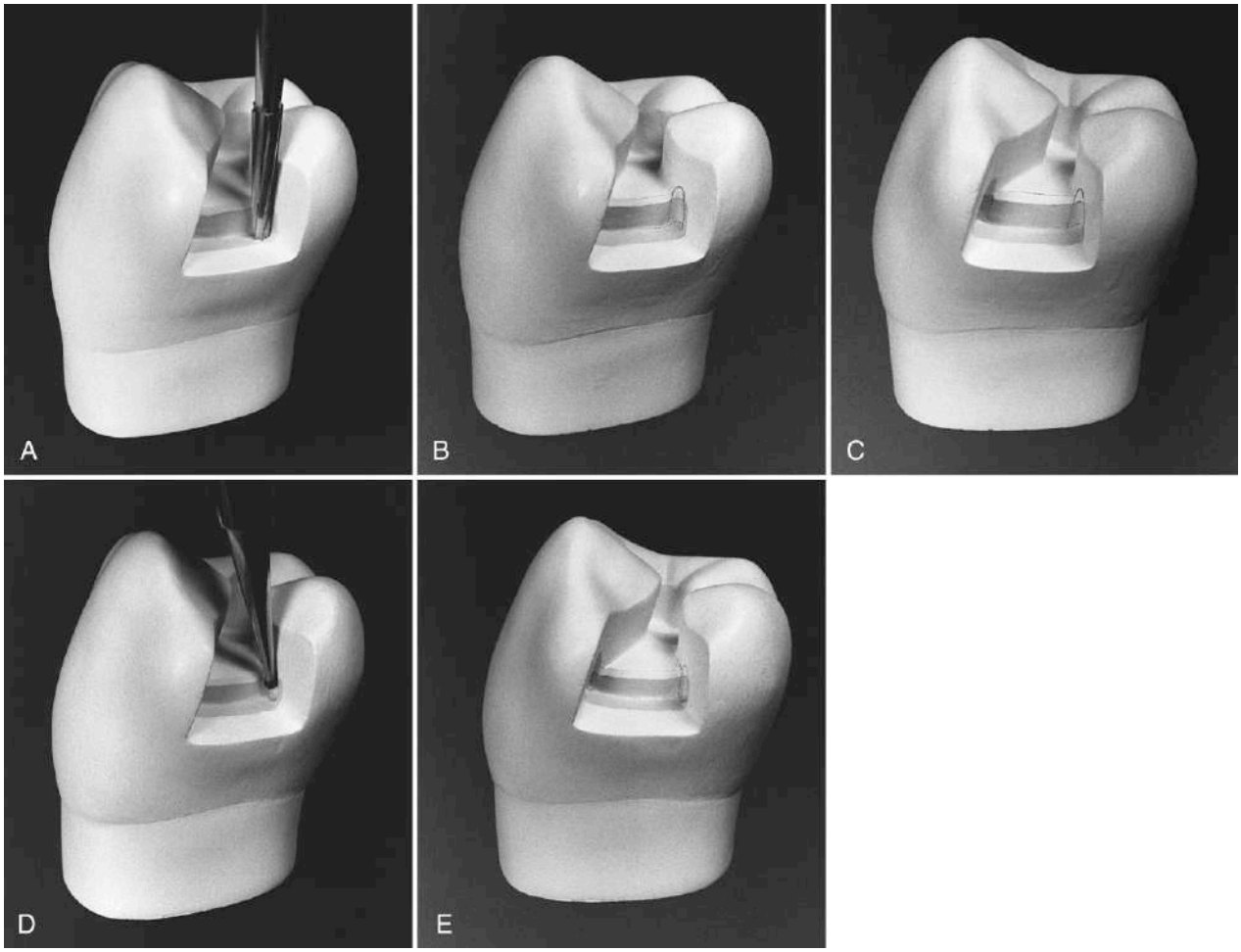


Figure 13.72 Proximal retention grooves. (A) Position of the No. 169 L bur to prepare the retention groove as the bur is moved lingually and pulpally; (B) lingual groove. Note the dentin support of the proximal enamel; (C) completed grooves; (D) grooves prepared with a No. 1/4 round bur and (E) completed grooves.

height of the groove, given a constant depth. When using the No. 1/4 bur to prepare the proximal groove, the rotating bur is carried into the axiolinguogingival (or axiofaciogingival) point angle, then moved parallel to the DEJ to the depth of ~0.25–0.5 mm. It is then drawn occlusally along the axiolingual (or axiofacial) line angle, allowing the groove to become shallower and to terminate at the axiolinguopulpal (or axiofaciopulpal) point angle (or more occlusally if the line angles are <2 mm in length) (Fig. 13.72D and E).

Regardless of the method used in placing the grooves, extreme care is necessary to prevent the removal of dentin that immediately supports the proximal enamel. In addition, it is essential that the grooves are not prepared entirely in the axial wall (i.e. incorrect translation [moving the bur only in an axial direction]) because groove placement in this area will not affect any resistance to proximal displacement of the amalgam (i.e. will not increase retention) and will increase the risk of iatrogenic pulpal involvement.

An improperly positioned axiofacial or axiolingual line angle must not be used as a positional guide for the proximal groove. If the axial line angle is too shallow, the groove may remove the necessary dentinal support of enamel. If the line angle is too deep, preparation of the groove may result in exposure of the pulp.

Retention grooves, when used, always should be placed 0.2 mm inside the DEJ of the facial and lingual proximal walls regardless of the depth of the axial wall and associated line angles.

Finishing the proximal walls

The preparation walls and margins should not have unsupported enamel and marginal irregularities.[85]

No occlusal cavosurface bevel is indicated in the tooth preparation for amalgam. Ideally, a 90° cavosurface angle (maximum of 100°) should be present at the proximal margin. The occlusal line angle may be 90°–100°. This angle aids in obtaining a marginal amalgam angle of 90° (≥80°). Clinical experience has established that this *'butt-joint'* relationship of enamel and amalgam creates the strongest

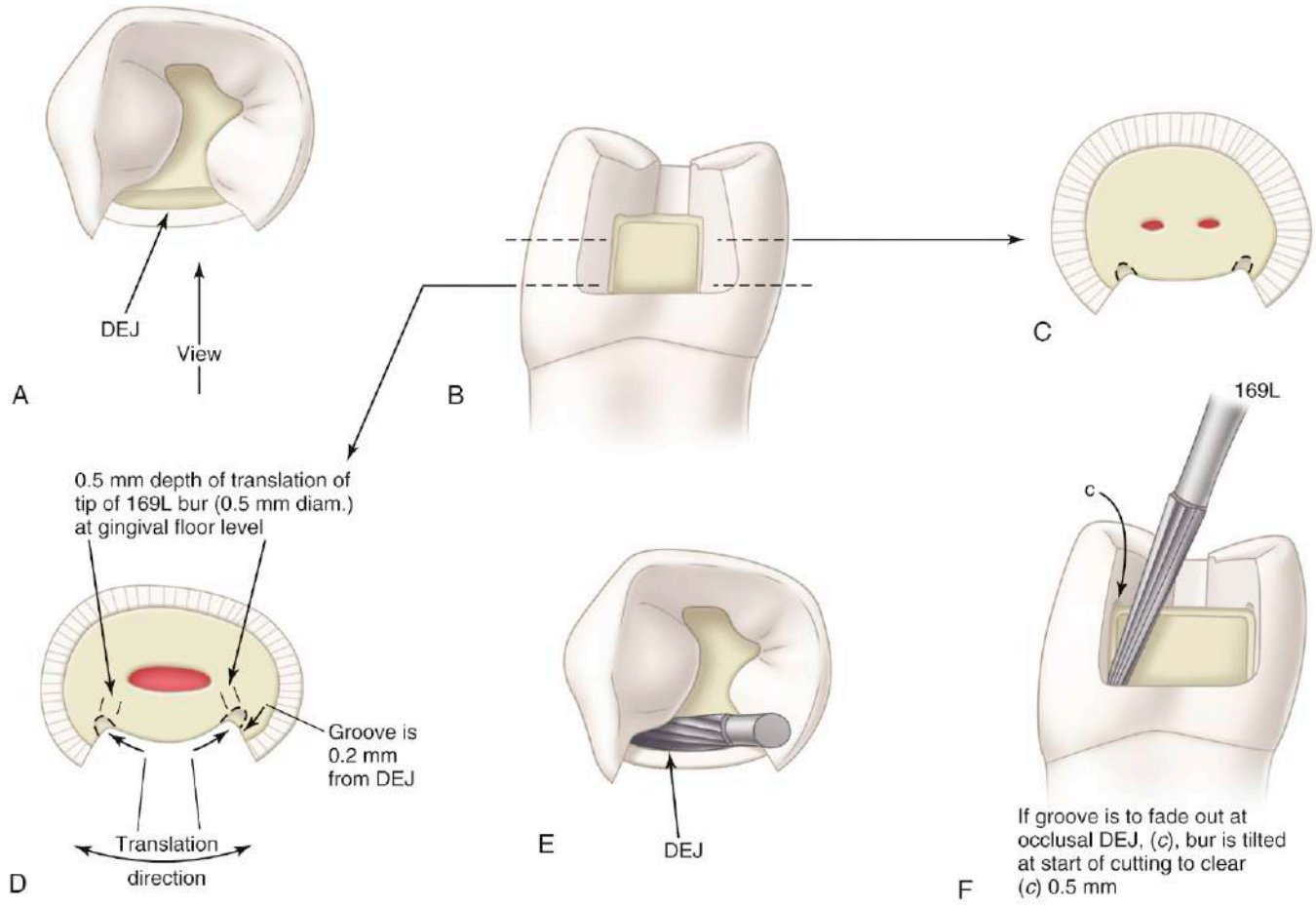


Figure 13.73 Four characteristics of retentive grooves. (A) Occlusal view of the mesioocclusal preparation before placement of the retention grooves; (B) proximal view of the mesioocclusal preparation; (C and D) position, translation and depth and (E and F) occlusogingival orientation.

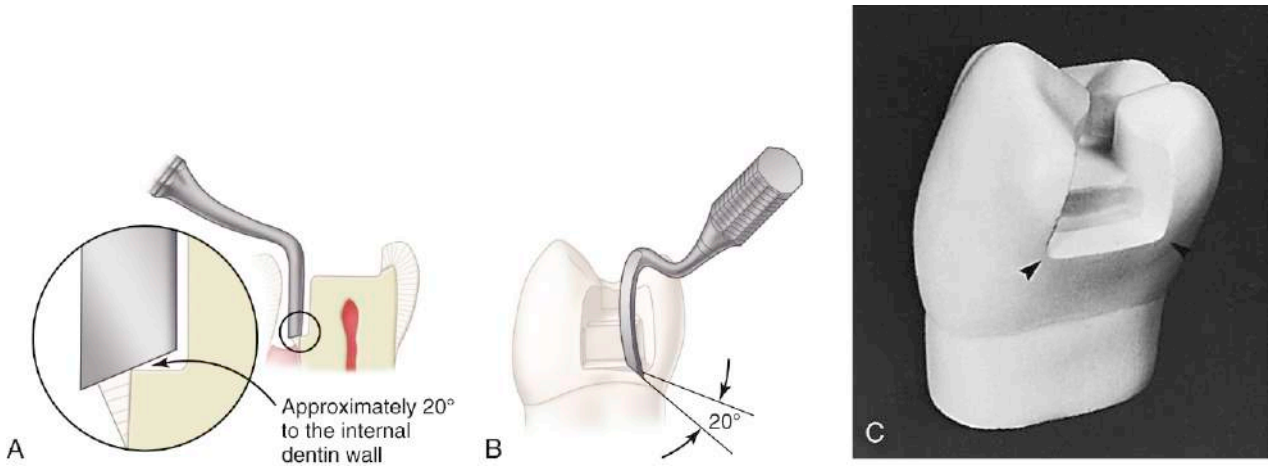


Figure 13.74 (A) The marginal configuration of the enamel portion of the gingival wall is established with a gingival margin trimmer to ensure full-length enamel rods forming the gingival margin and (B and C) The sharp angles at the linguogingival and faciogingival corners are rounded by rotational sweeping with a gingival margin trimmer.

margin.[28] Amalgam is a brittle material and may fracture under occlusal load if its angle at the margin is less than 80°.

A sharp mesial gingival margin trimmer (13–85–10–14, R and L) is used to remove unsupported gingival enamel by the establishment of a 90° gingival cavosurface angle at the gingival margin (Fig. 13.74). The resulting enamel gingival margin has a 6-centigrade (20°) gingival declination. The marginal configuration is angled no more than necessary to ensure that full-length enamel rods form the gingival margin and is no wider than the enamel (Fig. 13.74).

A sharp distal gingival margin trimmer (13–95–10–14, R and L) is similarly used for distal proximal preparations. Alternatively, the side of an explorer tine may be used to remove any friable enamel at the gingival margin. The tine is placed in the gingival embrasure apical to the gingival margin. With some pressure against the prepared tooth, the tine is moved occlusally across the gingival margin to 'trim' the margin of friable, unsupported enamel.

Clinical Notes

When the gingival margin is positioned gingival to the cementoenamel junction (CEJ) (i.e. on the tooth root), the gingival cavosurface configuration is 90° to the external surface of the root.[28]

Variations of Proximal Surface Tooth Preparations

This section describes variations in tooth preparation for some small Class II amalgam restorations. In most clinical situations, the restoration of these preparations would likely be with composite unless the patient has high caries risk. If amalgam is used, the features presented should be considered in the tooth preparation portion of the procedure.

Mandibular First Premolar

The technique of the Class II tooth preparation for amalgam for the mandibular first premolar must be modified because the morphologic structure of this tooth is different from other posterior teeth. The relationship of the pulp chamber to the DEJ and the relatively small size of the lingual cusp are illustrated in Fig. 13.75. (This figure also illustrates the correct position of the pulpal wall and how it differs in position and orientation compared with the second premolar.) Incorrect preparation of the central groove area may weaken the lingual cusp, and excessive extension in a facial direction may approach or expose the facial pulp horn. When preparing the occlusal portion, the bur is tilted slightly lingually to establish the correct pulpal wall direction (Fig. 13.43B).

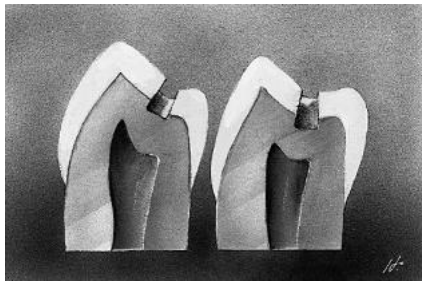


Figure 13.75 The mandibular first and second premolars are compared. Note differences in the sizes of the pulp chambers, lingual cusps and direction of pulpal walls.

The mandibular first premolar presents with unique occlusal morphology, most notably the presence of a large transverse ridge of enamel. Most often the coalescence of the enamel lobes is complete and no caries-prone central fissure results. Therefore, a caries lesion in the mesial pit or distal pit may be included as part of the Class II preparation, but with an outline form that does not extend to or across the transverse ridge (Fig. 13.76A). The level of divergence of the facial and lingual proximal walls (Fig. 13.76B) is such that secondary retention grooves are indicated (Fig. 13.76C). If the opposite pit or proximal surface is faulty, it is restored with a separate restoration so as to not compromise the resistance form of the tooth.

For a preparation that does not cross the transverse ridge, the proximal box is prepared before the occlusal portion to prevent removing the tooth structure that will form the isthmus between the occlusal dovetail and the proximal box. The pit adjacent to the involved proximal surface is entered with the No. 245 bur. Immediately after the entry, the bur is directed into the proximal marginal ridge and

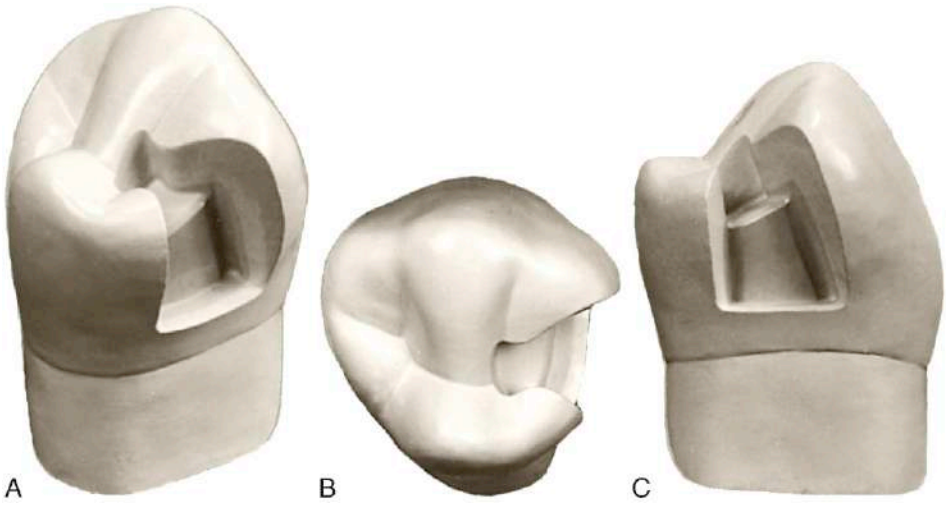


Figure 13.76 The mandibular first premolar with a sound transverse ridge. (A) A two-surface tooth preparation that does not include the opposite pit; (B) occlusal outline form and (C) proximal view of the completed preparation.

then gingivally until the proximal DEJ is visible. The bur axis for the proximal ditch cut should be parallel to the tooth crown, which is tilted slightly lingually for mandibular posterior teeth (Fig. 13.76C). The proximal enamel is isolated and the proximal box completed as previously described for the mandibular second premolar. The bur is then returned to the area of entry, and the occlusal step is prepared with a dovetail, if needed. When preparing the occlusal portion, the bur is tilted slightly lingual to establish the correct pulpal wall direction (which maintains the dentin support for the small lingual cusp and prevents encroachment on the facial pulp horn). The primary difference in tooth preparation on this tooth, compared with the preparation on other posterior teeth, is the facial inclination of the pulpal wall. The isthmus is broadened as necessary, but maintains the dovetail retention form. Fig. 13.76B illustrates the correct occlusal outline form. Removing any remaining caries lesion (if present) and inserting necessary liners, bases, or both precede the placement of proximal grooves and the finishing of the enamel margins (Fig. 13.76C).

Maxillary First Molar

When mesial and distal proximal surface amalgam restorations are indicated on the maxillary first molar that has an unaffected oblique ridge, separate two-surface tooth preparations are indicated (rather than a mesioocclusodistal preparation). The use of two preparations is preferred because the *strength of the tooth crown is significantly greater when the oblique ridge is intact.*[28]

The *mesioocclusal tooth preparation* is generally uncomplicated (Fig. 13.77A). Occasionally, extension through the oblique ridge and into the distal pit is necessary because of the extent of the caries lesion. The outline of the occlusolingual pit-and-fissure portion is similar to that of the Class I occlusolingual preparation. Fig. 13.77B and C illustrates a mesioocclusal preparation extended to include the distal pit and the outline form that includes the distal oblique and lingual fissures.

When the occlusal fissure extends into the facial cusp ridge and cannot be removed by enameloplasty, the defect should be eliminated by extension of the tooth preparation. Occasionally, this may be accomplished by tilting the bur to create an occlusal divergence of the facial wall while maintaining the dentin support of the ridge. If it is not possible to eliminate this fault without extending the margin to the height of the cusp ridge or undermining the enamel margin, the preparation should be extended facially through the ridge (Fig. 13.77D). The pulpal wall of this facial extension may have remaining enamel, but a depth of 1.5–2 mm is necessary to provide sufficient bulk of amalgam material for adequate strength (adequate resistance to flexure under load). For the best esthetic results, minimal extension of the proximal mesiofacial margin is indicated.

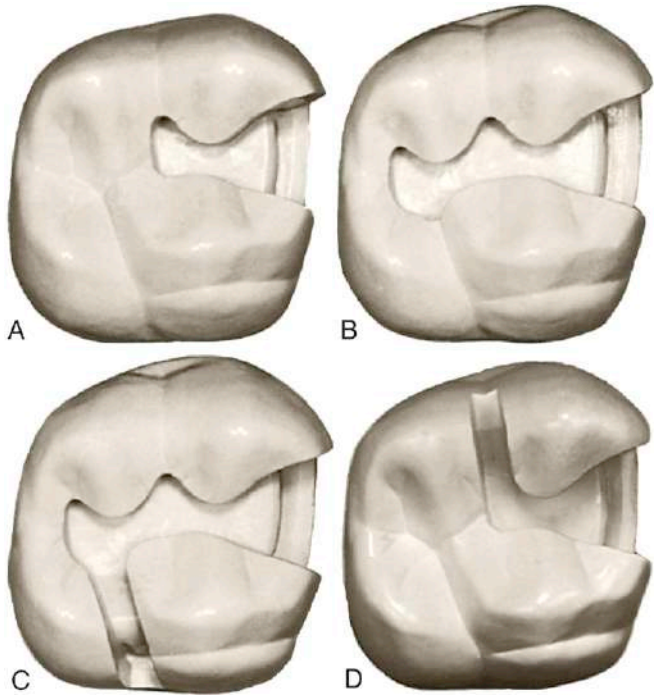


Figure 13.77 Maxillary first molar. (A) Conventional mesioocclusal preparation; (B) mesioocclusal preparation extended to include the distal pit; (C) mesioocclusolingual preparation, including the distal pit and the distal oblique and lingual fissures and (D) mesioocclusal preparation with facial fissure extension.

The *distoocclusal tooth preparation* may take one of several outlines, depending on the occlusal anatomy. The occlusal outline is determined by the pit-and-fissure pattern and by the amount and extension of the caries lesion. An extension onto the lingual surface to include a lingual fissure should be prepared only after the distolingual proximal margin is established. This approach may allow conservation of more tooth structure between the distolingual wall and the lingual fissure extension, resulting in more strength of the distolingual cusp. It is accomplished by preparing the lingual fissure extension more at the expense of the mesiolingual cusp than the distolingual cusp. Nevertheless, the distolingual cusp on many maxillary molars may be weakened during the distoocclusolingual tooth preparation because of the small cuspal portion remaining between the lingual fissure preparation and the distolingual proximal wall. In addition, removal of the caries lesion may reveal a weakened distolingual cusp. Reduction of the distolingual cusp and coverage with amalgam is often necessary to provide the proper resistance form (Chapter 14).

Maxillary First Premolar

A Class II amalgam tooth preparation involving the mesial surface of a maxillary first premolar requires special attention because the mesiofacial embrasure is esthetically prominent. The occlusogingival preparation of the

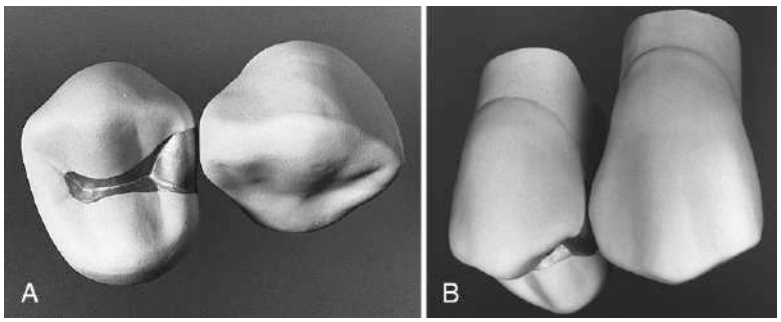


Figure 13.78 To produce an inconspicuous margin on the maxillary first premolar, the mesiofacial wall does not diverge gingivally, and facial extension with a No. 245 bur should be minimal so that the mesiofacial proximal margin of the preparation minimally clears the contact as the margin is finished. (A) Occlusal view and (B) facial view.

facial wall of the mesial box should be parallel to the long axis of the tooth instead of converging occlusally to minimize an unesthetic display of amalgam in the faciogingival corner of the restoration. In addition, the facial extension of the mesiofacial proximal wall should be minimal so that the mesiofacial proximal margin of the preparation only minimally clears the contact as the margin is finished with an appropriate enamel hatchet or chisel (Fig. 13.78).

If the mesial proximal involvement (1) is limited to a fissure in the marginal ridge that is at risk for caries, (2) is not treatable by enameloplasty and (3) does not involve the proximal contact, then the proximal portion of the tooth preparation is prepared by extending through the fault with the No. 245 bur so that the margins are lingual to the contact. Often this means that the proximal box is the faciolingual width of the bur, and the gingival floor may be at the same depth as the pulpal floor. The retention form for this extension is provided by the slight occlusal convergence of the facial and lingual walls (Fig. 13.34).

If the proximal caries lesion is limited to the mesiolingual embrasure, the mesial proximal contact should not be included in the tooth preparation. If only the lingual aspect of the mesial proximal contact is carious, the mesiofacial wall may be left in contact with the adjacent tooth (reducing the display of amalgam). A Class II tooth preparation involving the distal surface of the maxillary first premolar is similar to the preparation of the mandibular second premolar described earlier.

Box-Only Preparation

When restoring a small, cavitated proximal lesion in a tooth with neither occlusal fissures nor a previously inserted occlusal restoration, a proximal box preparation without an occlusal step has been recommended.[27,28] This preparation should be limited to:

I. a proximal surface with a narrow proximal contact (allowing minimal facial and lingual extensions) and
II. when the involved marginal ridge has no occlusal contact.

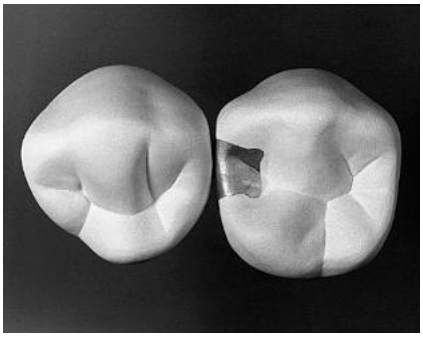


Figure 13.79 A simple box restoration without the occlusal step is permissible when restoring a small proximal lesion in the tooth without either occlusal fissures or previously inserted occlusal restoration and when the involved marginal ridge does not support occlusal contact. The proximal grooves extend to the occlusal surface.

To maximize retention, the box-only preparations should have the following characteristics:

I. Facial and lingual walls that almost oppose each other are recommended.
II. Facial and lingual proximal walls converge occlusally.
III. Retention grooves are *mandatory* in box-only preparations.[86] The proximal retention grooves should have a 0.5-mm depth at the gingival point angle and extend occlusally to be visible in the occlusal outline form (Fig. 13.79).

Tooth Preparation for Class II Amalgam Restorations Involving Both Proximal Surfaces

Perhaps the best indications for the use of amalgam restorations are moderate and large Class II defects that include both proximal surfaces and much of the occlusal surface. This section describes factors to consider when amalgam is used for moderate and large Class II restorations. The tooth preparation techniques presented for a two-surface Class II restoration apply to larger Class II restorations as well. When the defect is large, however, certain modifications in tooth preparation may be necessary.

Occlusal Extensions

Larger Class II defects may require greater extension of the occlusal surface outline form. This may require extending into grooves that are fissured, extending the outline form up the cuspal inclines, or reducing and covering the cusps that have inadequate dentin support. These alterations may be accomplished easily by following the principles presented previously. The depth and extension of the initial preparation of fissures occurs at the same initial pulpal floor depth (i.e. at the level of the DEJ) and follows the DEJ as the preparation is extended in a facial or lingual direction.

If an occlusal outline form extends up a cuspal incline, the extension also should maintain the pulpal floor at the level of the DEJ. This extension usually requires some alteration in the orientation of the No. 245 bur—a slight lingual tilt when extending in a facial direction and a slight facial tilt when extending in a lingual direction (Fig. 13.35). Maintaining the correct pulpal floor depth preserves tooth structure and reduces the potential for pulpal encroachment. The prepared facial (or lingual) cavosurface margin still should result in a 90° amalgam margin. If it is necessary to extend through the cusp ridge onto the facial or lingual surface, the preparation is accomplished as described for the occlusolingual Class I restoration.

When the occlusal outline form extends from a primary fissure to within two-thirds of the distance to a cusp tip, that cusp is usually sufficiently weakened so as to require reduction and coverage with the restoration. Leaving the cusp in a weakened state may be acceptable if the cusp is very large. Routine preparations in some teeth may predispose some cusps for reduction and coverage with amalgam:

- small distal cusp of the mandibular first molars,
- distolingual cusp of maxillary molars,
- lingual cusp of some mandibular premolars (especially first premolars) may be weakened when normal preparations of surrounding areas of the tooth are included. See Chapter 14 for information on when and how to reduce and cover cusps with amalgam.

Proximal Extensions

Larger Class II caries lesions often require larger proximal box preparations. These may include not only increased faciolingual or gingival extensions but also extension around a facial or lingual line angle. Large proximal box preparations may also need secondary retention features (i.e. retention grooves) for adequate retention form because the proximal external walls are excessively divergent. Extensive proximal boxes are usually prepared in the same fashion as more conservative proximal boxes but may require modifications. For increased faciolingual extensions, it may be necessary to tilt the No. 245 bur to include extensive proximal faults that are gingival to the contact area. Tilting the bur lingually when extending a facial proximal wall, or facially when extending a lingual proximal wall, conserves more of the marginal ridge and cuspal tooth structure. Although this action enhances preservation of some tooth structure strength, it results in a more occlusally convergent wall, which increases the difficulty of amalgam condensation in the gingival corners of the preparation.

When proximal extension around a line angle is necessary, it usually is associated with a reduction of the involved cusp (Chapter 14). Such proximal extension is necessary because of a severely defective (or fractured) cusp or a cervical lesion that extends from the facial (or lingual) surface into the proximal area. Often these areas are included in the preparation by extending the gingival floor of the proximal box around the line angle, using the same criteria for preparation as the typical proximal box:

I. facial (or lingual) extension results in an occlusogingival wall that has a 90° cavosurface margin and
II. the axial depth is 0.5 mm inside the DEJ.

When the proximal defect is extensive gingivally, isolation of the area, tooth preparation, matrix placement, and condensation and carving of amalgam are more difficult. If the proximal box is extended onto the *root surface*, the axial wall depth is no longer dictated by the DEJ.

- Any root surface preparation for amalgam should result in an initial axial wall depth of approximately 0.8 mm.
- This axial depth provides appropriate strength for amalgam, limits any potential pulp compromise and creates enough dimension for the placement of ~0.25–0.5 mm deep retention grooves while preserving the strength of the adjacent, remaining marginal dentin and cementum.
- The extent of the preparation onto the root surface, the contour of the tooth, or both may require that the bur be tilted towards the adjacent tooth when preparing the gingival floor of the proximal box. This tilting may result in an axial wall that has two planes, the more gingival plane angled slightly internally. It also may cause more difficulty in retention groove placement. The more occlusal part of the axial wall may be overreduced if the bur is not tilted.

Examples of Moderate Class II Amalgam Tooth Preparations That Involve Both Proximal Surfaces

Mandibular Second Premolar

A moderate mesioocclusodistal tooth preparation in a mandibular second premolar is illustrated in Fig. 13.80. Note the similarity with the two-surface mesioocclusal preparation.

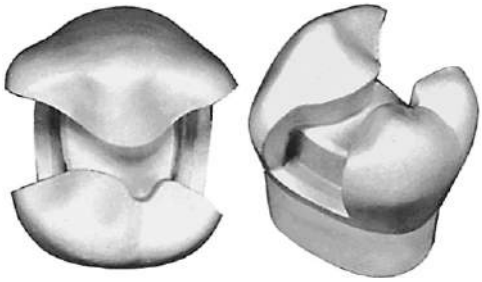


Figure 13.80 Mesioocclusodistal (*MOD*) preparation on the mandibular second premolar that had a moderately sized MOD carious lesion.

Mandibular First Premolar

When a mesioocclusodistal amalgam tooth preparation is needed for the mandibular first premolar, the support of the small lingual cusp may be conserved by preparing the occlusal step more at the expense of tooth structure facial to the central groove than lingual. In addition, the bur is tilted slightly lingually to establish the correct pulpal wall direction. Despite these precautions, the lingual cusp may need to be reduced for coverage with amalgam if the lingual margin of the occlusal step extends more than two-thirds the distance from the central fissure to the cuspal eminence (Chapter 14).

Maxillary First Molar

The mesioocclusodistal tooth preparation of the maxillary first molar may require extending through the oblique ridge to unite the proximal preparations with the occlusal step. Extending the preparation through the oblique ridge is indicated only if

I. the ridge is undermined by a caries lesion,
II. it is crossed by a deep fissure, or
III. occlusal portions of the separate mesioocclusal and distoocclusal outline forms leave less than 0.5 mm of the tooth structure between them. The remainder of the outline form is similar to the two-surface outline forms described previously in this chapter. Fig. 13.81 illustrates typical three-surface and four-surface restorations for this tooth.

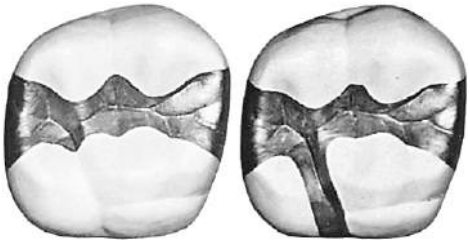


Figure 13.81 Typical three- and four-surface restorations for the maxillary first molar.

Maxillary Second Molar with a Caries Lesion on the Distal Portion of the Facial Surface

Close examination of the distal portion of the facial surface of the maxillary second molar may reveal decalcification, cavitation, or both. When enamel is only slightly demineralized (i.e. softened and rough), polishing with sandpaper disks may eliminate the superficial fault. Careful brushing technique, daily use of fluoride (i.e. rinses, toothpaste), and periodic applications of a fluoride varnish may prevent further mineral loss. When decalcification is as deep as the DEJ and a distal proximal caries lesion is also present; however, the entire distofacial cusp may need to be reduced and included in a mesioocclusodistofacial tooth preparation. The facial lesion may be restored separately if it is judged that the distofacial cusp would not be significantly weakened if left unrestored by amalgam. In that case, the mesioocclusodistal preparation would be restored first, followed by preparation and restoration of the facial lesion. When such sequential preparations are contraindicated, the preparation outline is extended gingivally to include the distofacial cusp (just beyond the caries lesion) and mesially to include the facial groove (Fig. 13.82). The No. 245 bur should be used to create a gingival floor (i.e. shoulder) perpendicular to the occlusal force when extending the distal gingival floor to include the affected facial surface. Inclusion of distofacial caries often indicates a gingival margin that follows the gingival tissue level. The width of the shoulder should be approximately 1 mm (or 0.5 mm inside the DEJ), whichever is greater. Some resistance form is provided by the shoulder. A retention groove should be placed in the axiofacial line angle of this distofacial extension, similar to the grooves placed in the proximal boxes. For additional retention, a slot may be placed, as discussed in Chapter 14.

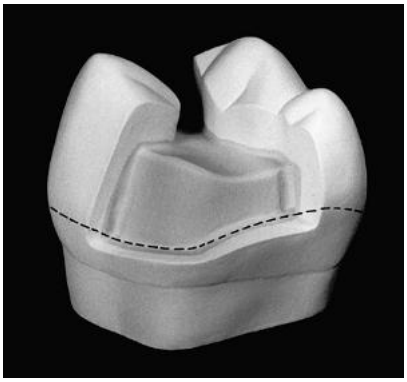


Figure 13.82 Mesioocclusodistofacial preparation of the maxillary second molar showing extension to include moderate to extensive caries lesion in the distal half of the facial surface. The outline includes the distofacial cusp and the facial groove. The *dotted line* represents the soft tissue level.

Mandibular First Molar

The distal cusp on the mandibular first molar may be weakened when positioning the distofacial wall and margin. Facial extension of the distofacial margin to clear the distal contact often places the occlusal outline in the centre of the cusp; this dictates relocation of the margin to provide a sound enamel wall and 90° amalgam that is not on a cuspal eminence. When the distal cusp is small or weakened or both, extension of the distal gingival floor and distofacial wall to include the distal cusp places the margin just mesial to the distofacial groove. Fig. 13.83 illustrates the ideal distofacial extension and a preparation design that includes the distal cusp.

Modifications in Tooth Preparation for Proximal Surfaces

Slot Preparation for Root Caries

Older patients who have gingival recession that exposes cementum (or dentin) may experience caries lesion development on the proximal root surface that is appreciably gingival to the proximal contact (Fig. 13.84A). Assuming that the contact does not need restoring, the tooth preparation usually is approached from the facial direction and has the form of a slot (Fig. 13.84B and E). A lingual approach is used when the caries is limited to the linguoproximal

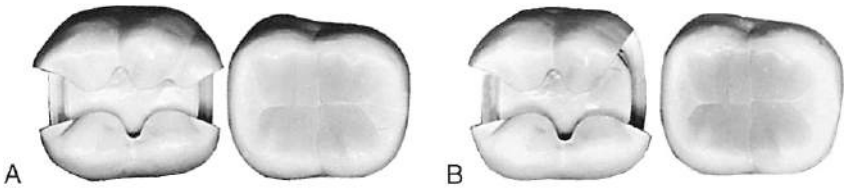


Figure 13.83 Mandibular first molar. (A) Ideal distofacial extension and (B) entire distal cusp included in preparation outline form.

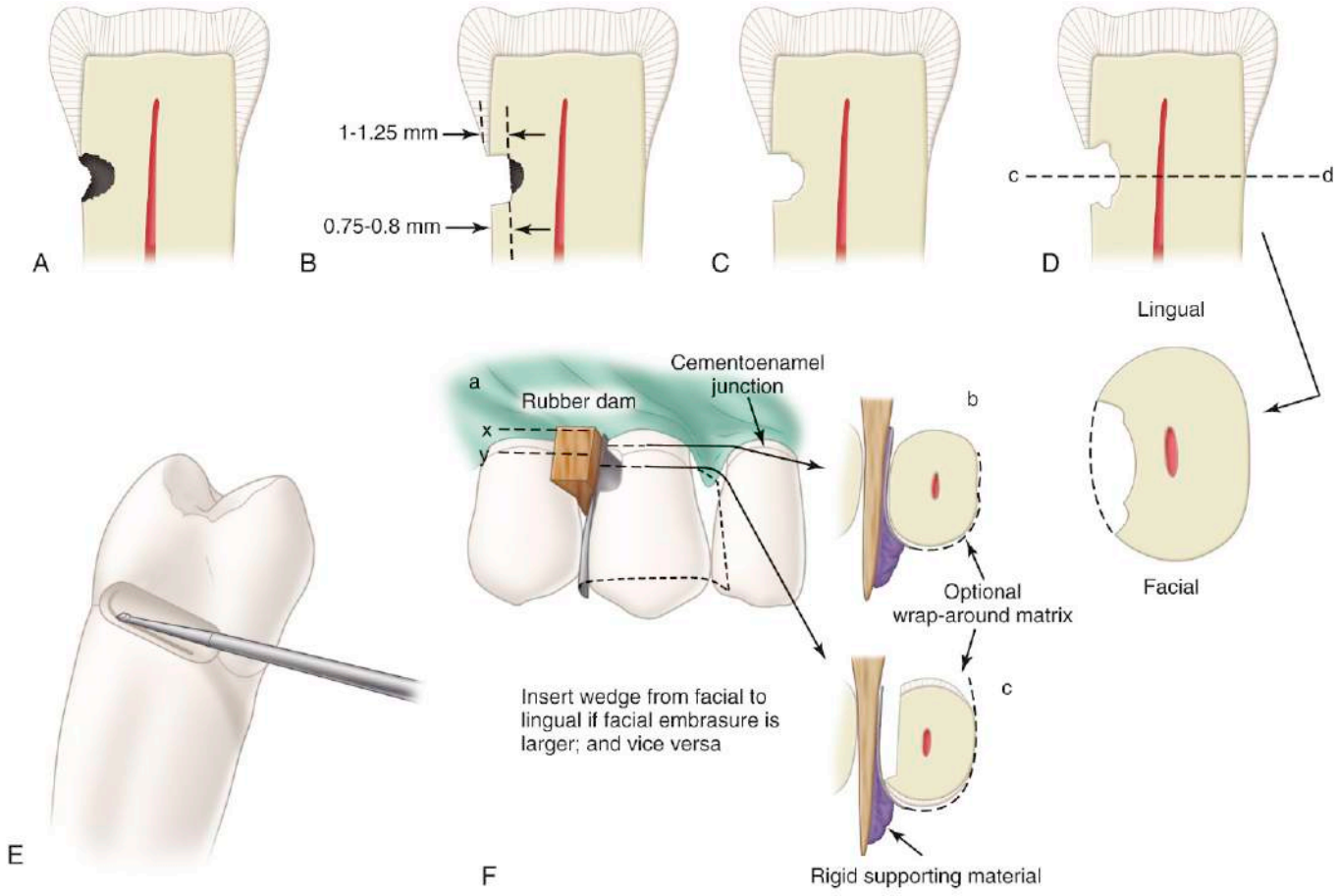


Figure 13.84 Slot preparation. (A) Mesiodistal longitudinal section illustrating a caries lesion. The proximal contact is not involved; (B) initial tooth preparation; (C) tooth preparation with soft carious dentin removed; (D) the retention grooves are shown in longitudinal section and the transverse section through plane cd illustrates the contour of the axial wall and the direction of the facial and lingual walls and (E) preparing the retention form to complete the tooth preparation; (F) matrix for slot preparation: *a*, facial view of wedged matrix; *b*, wedged matrix as viewed in transverse cross-section; *x*, gingival to gingival floor; *c*, wedged matrix as viewed in transverse cross-section and *y*, occlusal to gingival floor.

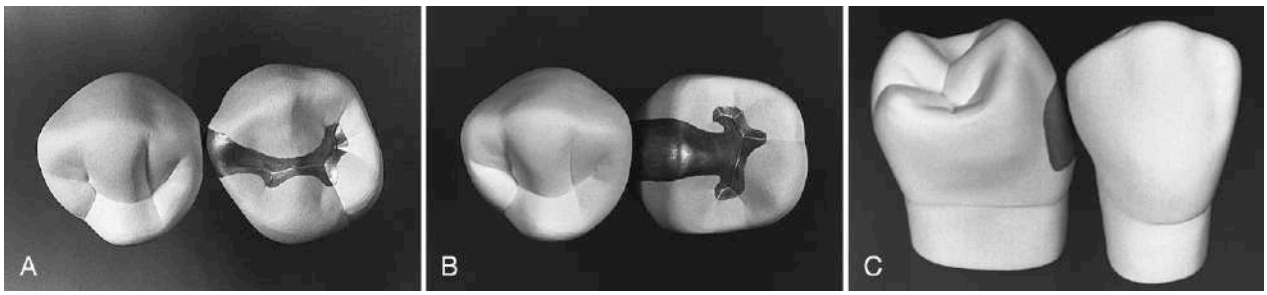


Figure 13.85 Restoration outlines for rotated teeth. (A) Mesioocclusal outline for the mandibular premolar with 45° rotation; (B) mesioocclusal outline for the mandibular premolar with 90° rotation; (C) slot preparation outline for the restoration of a small mesial lesion involving the proximal contact of the mandibular premolar with 90° rotation.

surface. Amalgam is particularly indicated for slot preparations if isolation is difficult.[5]

The initial outline form is prepared from a facial approach with a No. 2 or No. 4 bur using high speed and air-water spray. Outline form extension to sound tooth structure is at a limited depth axially (i.e. 0.75–1 mm at the gingival aspect [if no enamel is present], increasing to 1–1.25 mm at the occlusal wall [if the margin is in enamel]) (Fig. 13.84B). If the occlusal margin is in enamel, the axial depth should be 0.5 mm inside the DEJ. During this extension, the bur should not remove any soft dentin from the axial wall deeper than the outline form initial depth. The remaining soft dentin (if any) is removed during final tooth preparation (Fig. 13.84C). The external walls should form a 90° cavosurface angle. With a facial approach, the lingual wall should face facially as much as possible; this aids condensation of amalgam during insertion. The facial wall must be extended to provide access and visibility (convenience form) (Fig. 13.84D). In the final tooth preparation, the No. 2 or No. 4 bur should be used to remove any remaining soft dentin on the axial wall. A liner or base (or both) is applied, as indicated.

A No. 1/4 bur is used to create retention grooves in the occlusoaxial and gingivoaxial line angles, 0.2 mm inside the DEJ or 0.3–0.5 mm inside the cemental cavosurface margin (Fig. 13.84D and E). The depth of these grooves is ~0.25–0.5 mm and the bur is directed to bisect the angle formed by the junction of the occlusal (or gingival) and axial walls. Ideally, the direction of the occlusal groove is slightly more occlusal than axial, and the direction of the gingival groove would be slightly more gingival than axial. A dentin desensitizer may be placed before or after application of the matrix. The matrix for the slot preparation is illustrated in Fig. 13.84F.

Rotated Teeth

Tooth preparation for rotated teeth follows the same principles as for normally aligned teeth. The outline form for a mesioocclusal tooth preparation on the rotated mandibular second premolar (Fig. 13.85A) differs from normal in that its proximal box is displaced facially because the proximal caries lesion involves the mesiofacial line angle of the tooth crown. When the tooth is rotated 90° and the 'proximal' lesion is on the facial or lingual surface, and orthodontic correction is declined or ruled out, the preparation may require an isthmus that includes the cuspal eminence (Fig. 13.85B). If the lesion is small, consideration should be given to slot preparation. In this instance, the occlusal margin may be in the contact area or slightly occlusal to it (Fig. 13.85C).

Unusual Outline Forms

Outline forms should conform to the restoration requirements of the tooth and not to the classic example of a Class II tooth preparation. As mentioned earlier, a dovetail feature is not required in the occlusal step of a single proximal surface preparation unless a fissure emanating from the occlusal step is involved in the preparation. Another example is an occlusal fissure that is segmented by coalesced enamel (as illustrated previously for mandibular premolars and the maxillary first molars). This condition should be treated with individual amalgam restorations if the preparations are separated by approximately 0.5 mm or more of sound tooth structure (Fig. 13.86).[27,72]

Adjoining Restorations

It is permissible to repair or replace a defective portion of an existing amalgam restoration if the remaining portion of the original restoration has adequate resistance and retention forms. Adjoining restorations on the occlusal surface occur more often in molars because the dovetail of the



Figure 13.86 Restoration of the mesioocclusal tooth preparation, with the central fissure segmented by coalesced enamel.

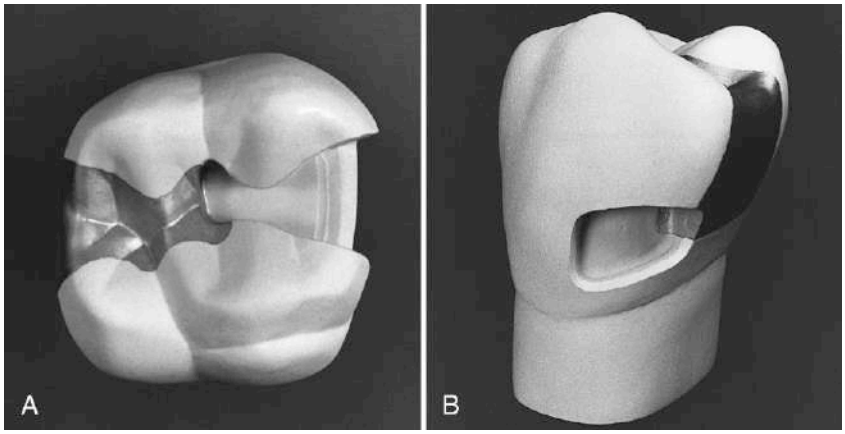


Figure 13.87 Adjoining restorations. (A) Adjoining mesioocclusal tooth preparation with distoocclusal restoration so that the new preparation does not weaken the amalgam margin of the existing restoration; (B) preparing and restoring a Class II lesion before preparing and restoring a Class V lesion contiguous with it eliminates condensation problems that occur when both lesions are prepared before either is restored.

new restoration usually may be prepared without eliminating the dovetail of the existing restoration. Where the two restorations adjoin, care should be taken to ensure that the outline of the second restoration does not weaken the amalgam margin of the first (Fig. 13.87A). The intersecting margins of the two restorations should be at a 90° angle as much as possible. The decision to join two restorations is based on the assumption that the first restoration, or a part of it, does not need to be replaced and that the procedure for the single proximal restoration (compared with a mesioocclusodistal restoration) is less complicated and conserves tooth structure.

Occasionally, preparing an amalgam restoration in two or more phases is indicated, such as for a Class II lesion that is contiguous with a Class V lesion. Preparing both lesions before placing amalgam introduces condensation problems that may be eliminated by preparing and restoring the Class II lesion before preparing and restoring the Class V lesion (Fig. 13.87B). It is better to condense amalgam against a carious wall of the first preparation than to attempt condensation where no wall exists.

Abutment Teeth for a Removable Partial Denture

When the tooth is an abutment for a removable partial denture (RPD), the occlusoproximal outline form adjacent to the edentulous region may need additional extension if a rest seat is planned. Additional extension must be sufficient facially, lingually and axially to allow for the rest seat to be in the proximoocclusal restoration without jeopardizing its strength.

If the rest seat is to be within the amalgam, it is recommended that a *minimum of 0.5 mm of amalgam be present between the rest seat and the margins* (Fig. 13.88A). The portion of the pulpal wall apical to the planned rest seat is deepened 0.5 mm so that the total depth of the axiopulpal line angle measured on the faciolingual wall is 2.5 mm (Fig. 13.88B). A rest seat used for a tissue-borne (i.e. distal extension) partial denture may involve amalgam and enamel (Fig. 13.88C). In this case, no modification of the outline form of the tooth preparation is indicated (Fig. 13.88C). Note Fig. 13.88D illustrates the relationship of the tissue-borne RPD with the abutment tooth (Fig. 13.88C).

All aspects of the Class II preparation, regardless of various levels of complexity, should be accomplished with precision. The preparation should then be cleaned, carefully inspected as indicated in the General Concepts Guiding Preparation for Amalgam Restorations section and any final modifications completed.

Restorative Technique for Class II Amalgam Preparations

Desensitizer Placement

A dentin desensitizer is placed on the prepared tooth structure per manufacturer instructions. Excess moisture is removed without desiccating the dentin.

Matrix Placement

The primary function of the matrix is to enable proper restoration of anatomic contours and contact areas. The qualities of a good matrix include (1) rigidity, (2) establishment of proper anatomic contour, (3) restoration of correct proximal contact relation, (4) prevention of gingival excess, (5) convenient application and (6) ease of removal. The following information presents the technique of placement for the universal (Tofflemire) and precontoured matrix systems.

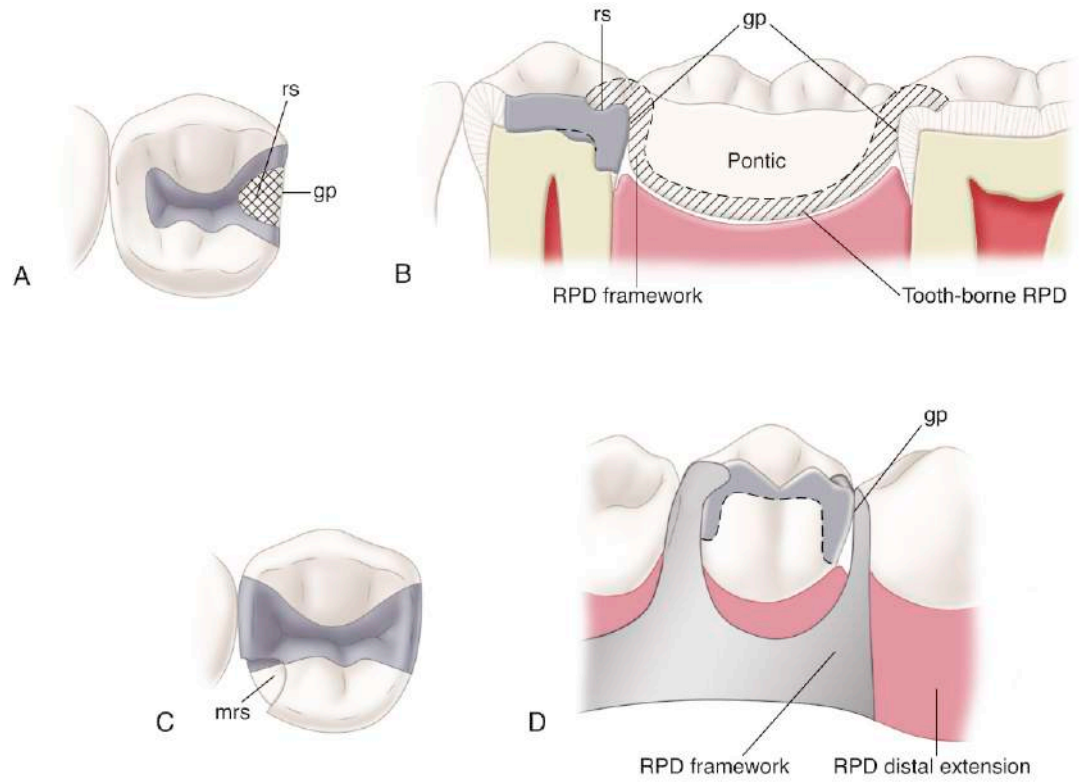


Figure 13.88 Abutment teeth with Class II restorations designed for a removable partial denture (*RPD*). (A) Occlusal view showing the location of the rest seat (*rs*) and the guide plane (*gp*) for a tooth-borne RPD. (B) Cross-sectional view illustrating deepened pulpal wall in the area of the rest seat (*rs*) to provide adequate thickness of amalgam. Note the relationship of the guide planes (*gp*) to the tooth-borne RPD. (C) Occlusal view showing the mesial rest seat (*mrs*) for the tissue-borne (i.e. distal extension) RPD prepared partly in a previous MOD amalgam restoration and partly in natural tooth structure. (D) Lingual view of a distal extension RPD showing the relationship of the RPD to the Class II restoration and the guide plan (*gp*). Note that the figure shows an M–D cross-section through the MOD amalgam and the occlusal portion of the facial aspect of the distal marginal ridge is visible.

Universal Matrix

The universal matrix system (designed by B.R. Tofflemire) is ideally indicated when three surfaces (i.e. mesial, occlusal, distal) of a posterior tooth have been prepared (Fig. 13.89). This system is also commonly used for the two-surface Class II restoration.

A definite advantage of the Tofflemire matrix retainer is that it may be positioned on the facial or lingual aspect of the tooth. However, lingual positioning requires the contra-angled design of the retainer (which may be used on the facial aspect as well) (Fig. 13.90; also Fig. 13.89). The retainer and the band are generally stable when in place. The retainer is separated easily from the band to expedite removal of the matrix system from the tooth. Matrix bands of various occlusogingival widths are available (Fig. 13.89). A small Tofflemire retainer is available for use with the primary dentition.

Types of bands

The universal matrix band itself does not meet all the requirements of an ideal matrix band.

(1) *Uncontoured bands*. The conventional, flat universal matrix band must be shaped (i.e. burnished) to reproduce natural anatomic contour and resultant proximal contact. Uncontoured bands are available in two thicknesses:

- 0.05 mm
- 0.038 mm

Burnishing the thinner band to create contour is more difficult, and the band is less likely to retain its contour when tightened around the tooth. The uncontoured band must be burnished before assembling the matrix band and the retainer. Burnishing must occur in the areas corresponding to the proximal surface to be restored after the band is positioned around the tooth.

(2) *Pre-contoured bands*. Pre-contoured bands for the universal retainer are commercially available and need little or no adjustment before being placed in the retainer or after being positioned around the tooth (Fig. 13.91). Although pre-contoured bands are more expensive, they generally are preferred over the uncontoured bands. The difference in cost may be justified as less chair time is required during the restorative phase. When cotton roll isolation is used, the Tofflemire retainer helps hold the cotton roll in place (Fig. 13.92).

Burnishing the Matrix Band

This means that the metal band is intentionally deformed occlusogingivally with a suitable hand instrument to produce a rounded or convex surface that (when in place around the tooth) produces a restoration that is symmetric



Figure 13.89 Straight (A) and contra-angled (B) universal (Tofflemire) retainers. Bands with varying occlusogingival widths are available.

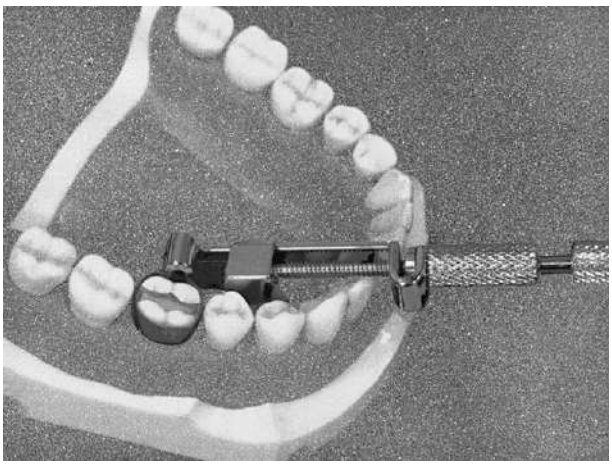


Figure 13.90 Lingual positioning requires a contra-angled universal retainer.

in contour with the adjacent proximal surface (Fig. 13.93). The Nos. 26–28 burnisher is generally recommended for burnishing the band. The band should be placed on a resilient paper pad because distortion of the metal matrix will not occur on a hard, non-resilient surface. The smaller round burnisher tip should first be used with firm pressure in back-and-forth, overlapping strokes along the length of the band until the band is deformed occlusogingivally in the appropriate areas. When the band is deformed, the larger egg-shaped end may be used to smooth the distorted band. If a convex surface is not obvious in the burnished areas when the band is removed from the pad, it has not been adequately burnished. The band may be burnished with the larger egg-shaped burnisher only, but more work is required to do so. It is not necessary to burnish the entire length of the band; focus effort only on distorting the matrix where the proximal contact areas are planned to occur.



Figure 13.91 Pre-contoured bands for a universal retainer.
(Pictured: Waterpik Getz Contour Matrix Bands; Courtesy: Water Pik Inc., Fort Collins, CO.)



Figure 13.92 Tofflemire retainer maintaining a cotton roll in the maxillary vestibule during condensation.

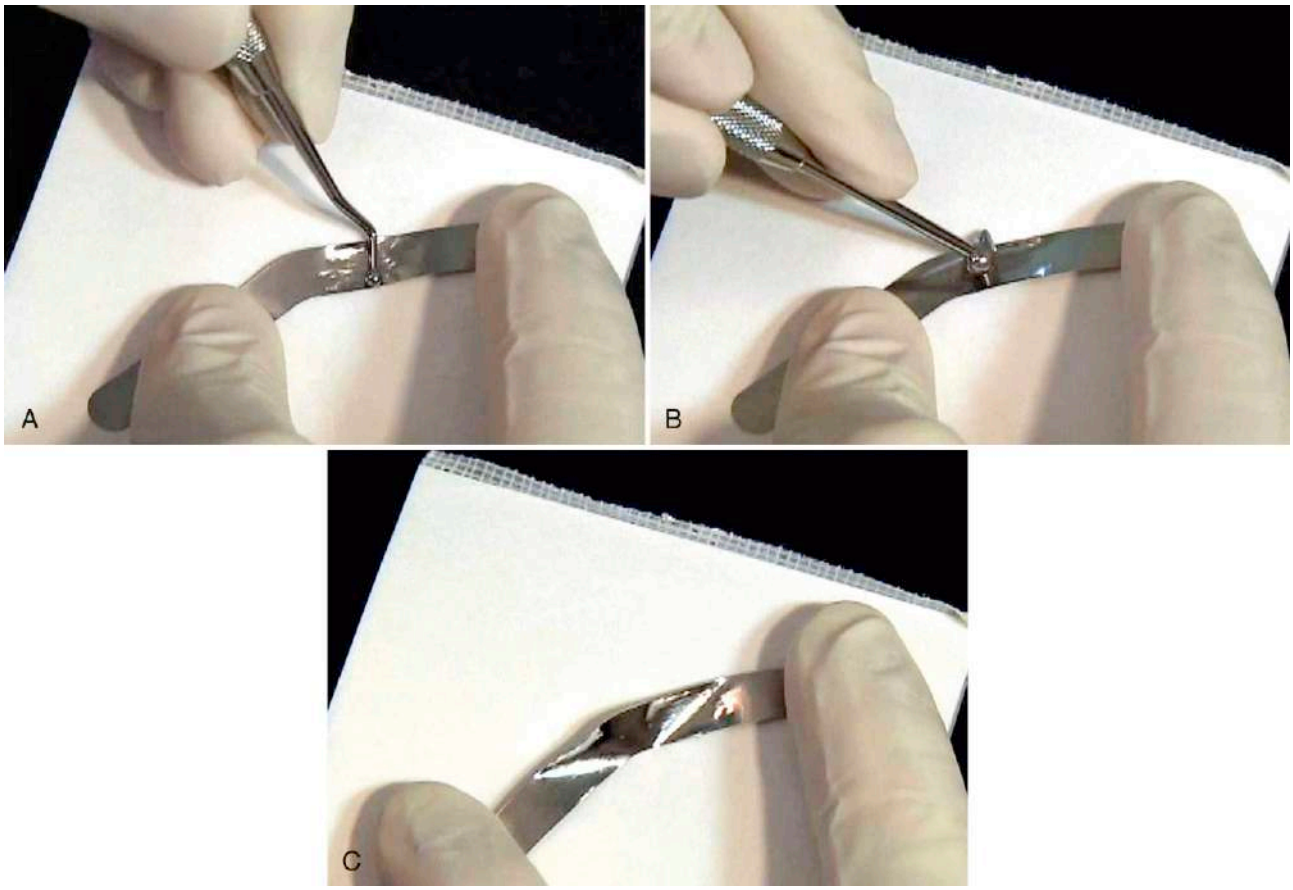


Figure 13.93 Burnishing the matrix band. (A) With the band on the pad, a small burnisher is used to deform the band; (B) a large burnisher used to smooth the band contour; (C) burnished matrix band for mesioocclusodistal tooth preparation.
(Courtesy: Aldridge D. Wilder, DDS.)

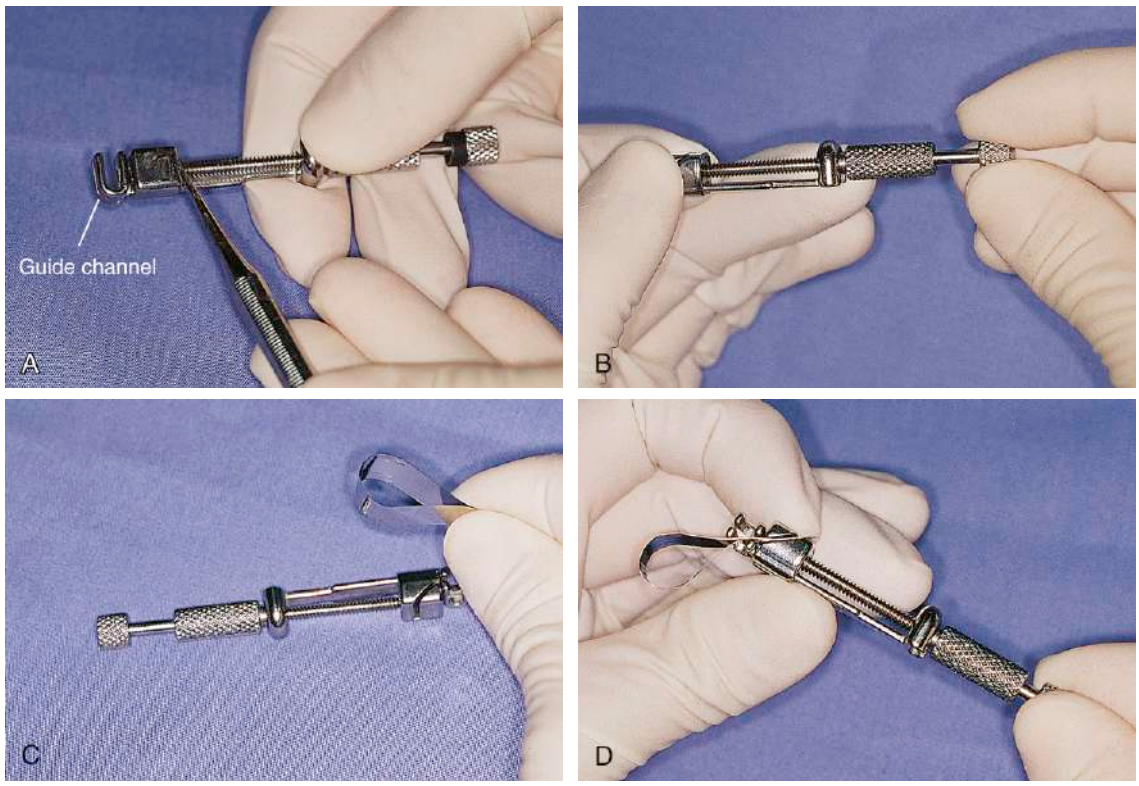


Figure 13.94 Positioning the band in a universal retainer. (A) Explorer pointing to the locking vice located immediately adjacent to the guide channel. The gingival aspect of the vice is shown in this view. (B) The pointed spindle is released from the locking vice when the small knurled nut is turned counter-clockwise. (C) The band is folded to form a loop and to be positioned in the retainer (occlusal edge of band first). (D) The spindle is tightened against the band in the locking vice.
(From Daniel SJ, Harfst SA, Wilder RS: *Mosby's dental hygiene: concepts, cases, and competencies*, ed 2, St. Louis, Mosby, 2008.)

Preparing the Retainer to Receive the Band

Step 1: The larger of the knurled nuts is turned counter-clockwise until the locking vice is positioned adjacent to the guide channel on the end of the retainer (Fig. 13.94A).

Step 2: Next, while holding the large nut, the dentist turns the smaller knurled nut counter-clockwise until the pointed spindle is free of the slot in the locking vice (Fig. 13.94B).

Step 3: The matrix band is folded end to end, forming a loop (Fig. 13.94C). When the band is folded, the gingival edge has a smaller circumference as compared with the occlusal edge so as to accommodate the difference in tooth circumferences at the gingival and, the more occlusal, contact levels.

Step 4: The band is positioned in the retainer so that the slotted side of the retainer is directed gingivally to permit easy subsequent separation (in an occlusal direction) of the retainer from the band after the restoration has been inserted. This is accomplished by placing the occlusal edge of the band in the correct guide channel (i.e. right, left, or parallel to the long axis of the retainer), depending on the location of the tooth.

Step 5: The two ends of the band are placed in the slot of the locking vice so that the ends are aligned with the edge of the vice, and the smaller of the knurled nuts is turned clockwise to tighten the pointed spindle against the band within the locking vice (Fig. 13.94D).

Placing the Band with Retainer on the Prepared Tooth

Step 1: If proximal wedges were used during tooth preparation, the wedges are removed at this point and the matrix band is fitted around the tooth (allowing the gingival edge of the band to be positioned at least 1 mm apical to the gingival margin). Damaging the gingival attachment with the sharp edge of the matrix band should be avoided. Care should be taken not to trap the rubber dam between the band and the gingival margin. If the dam material is trapped between the band and the tooth, the septum of the dam should be stretched and depressed gingivally to reposition the dam material.

Step 2: Once the matrix band is in place, the larger knurled nut is rotated clockwise to tighten the band slightly. The explorer should be used along the gingival margin to determine that the gingival edge of the band extends beyond the preparation margins. When the band is correctly positioned, the band is securely tightened around the tooth.



Figure 13.95 The band may be trimmed for the shallower gingival margin, permitting the matrix to extend farther gingivally for the deeper gingival margin on the other proximal surface.



Figure 13.96 Using a mirror from the facial or lingual position to evaluate the proximal contour of the matrix band.

When one of the proximal margins is deeper gingivally, the Tofflemire mesioocclusodistal band may be modified to prevent damage to the gingival tissue or attachment on the more shallow side. The band may be trimmed for the shallow gingival margin, permitting the matrix to extend farther gingivally for the deeper gingival margin (Fig. 13.95).

The mouth mirror is positioned lingually to observe the proximal contours of the matrix through the interproximal space (Fig. 13.96). The occlusogingival contour should be convex, with the height of contour at proper contact level such that the metal band is touching the adjacent proximal surface in the precise location where the contact should occur. The matrix is also observed from an occlusal aspect allowing evaluation of the position of the contact area in a faciolingual direction. If proper proximal contour and contact is not observed, it may be necessary to remove the retainer and reburnish the band. Minor alterations in contour and contact may be accomplished without removal from the tooth. The backside of the blade of the 15–8–14 spoon excavator (i.e. Black spoon) is excellent for improving contour in the area where the proximal contact is to occur. If a smaller burnishing instrument is used, the dentist should use care not to create a grooved or bumpy surface that would result in a restoration with an irregular proximal surface. Ideally the band should be positioned 1 mm apical to the gingival margin or deep enough to be engaged by the wedge (whichever is less) and 1–2 mm above the adjacent marginal ridge(s) to allow for adequate condensation of the amalgam in the marginal ridge areas.

Wedging Techniques

I. *Single wedge technique*

After the matrix contour and extension are evaluated, a wedge is placed in the gingival embrasure or embrasures using the following technique:

Step 1: Break off approximately 0.5 in (1.2 cm) of a round toothpick.
Step 2: Grasp the broken end of the wedge with the No. 110 pliers.
Step 3: Insert the pointed tip from the lingual or facial embrasure (whichever is larger), slightly gingival to the gingival margin.
Step 4: Wedge the band tightly against the tooth and margin (Fig. 13.97A).

The wedge may be lightly moistened with water or a water-soluble lubricant to facilitate its placement. If the wedge is placed occlusal to the gingival margin, the band is pressed into the preparation, which will result in an abnormal concavity in the proximal surface of the restoration (Fig. 13.97B). The wedge should not be so far apical to the gingival margin that the band will not be held tightly against the gingival margin. This improper wedge placement results

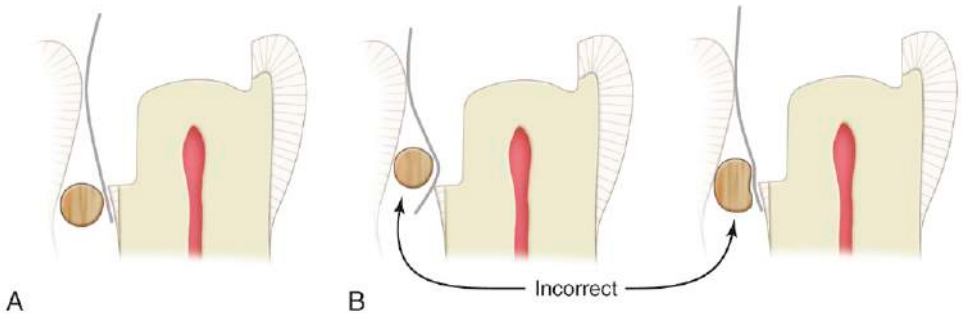


Figure 13.97 (A) Correct wedge position and (B) incorrect wedge positions.

in gingival excess of amalgam (i.e. amalgam 'overhang') caused by the band moving slightly away from the margin during condensation of the amalgam. Such an overhang often goes undetected and may result in irritation of the gingiva or an area of plaque accumulation, which may increase the risk of secondary caries.

To be effective, a wedge should be positioned as close to the gingival margin as possible without being occlusal to it.

II. *Piggy-back wedging technique*

If the wedge is significantly apical of the gingival margin, a second (usually smaller) wedge may be placed on top of the first to wedge adequately the matrix against the margin (Fig. 13.98).

This type of wedging is particularly useful for patients whose interproximal tissue level has receded.

III. *Double wedging technique*

The gingival wedge should be tight enough to prevent any possibility of an overhang of amalgam in at least the middle two-thirds of the gingival margin (Fig. 13.98A and B).

- *Double wedging* refers to using two wedges—one from the lingual embrasure and one from the facial embrasure (Fig. 13.98E and F).
- Double wedging is permitted (if access allows), securing the matrix when the proximal box is wide faciolingually. Two wedges ensure that the gingival corners of a wide proximal box can be properly condensed; they also help minimize gingival excess.
- Double wedging should be used only if the middle two-thirds of the proximal margins can be adequately wedged. However, because the facial and lingual corners are accessible to carving, proper wedging is important to prevent gingival excess of amalgam in the middle two-thirds of the proximal box (Fig. 13.98B).

IV. *Wedge wedging technique*

Occasionally, a concavity may be present on the proximal surface that is adjacent to the gingival margin. This may occur on a surface with a fluted root, such as the mesial surface of the maxillary first premolar (Fig. 13.98G1). A gingival margin located in this area may be concave (Fig. 13.98G2). To wedge a matrix band tightly against such a margin, a second pointed wedge may be inserted between the first wedge and the band (Fig. 13.98G3 and G4).

The wedging action between teeth should provide enough separation of adjacent teeth to compensate for the thickness of the matrix band. This ensures a positive contact relationship after the matrix is removed (after the condensation and initial carving of amalgam). If a Tofflemire retainer is used to restore a two-surface Class II preparation, the single wedge must provide enough separation to compensate for two thicknesses of band material. The tightness of the wedge is tested by pressing the tip of an explorer firmly at several points along the middle two-thirds of the gingival margin (against the matrix band) to verify that it cannot be moved away from the gingival margin (Fig. 13.99). As an additional test, the dentist attempts to remove the wedge (using the explorer with moderate pressure) after first having set the explorer tip into the wood near the broken end. Moderate pulling should not cause dislodgment. Often the rubber dam has a tendency to loosen the wedge through rebounding of the dam stretched during wedge placement. Lubrication of the wedge and stretching the interproximal dam septa, in the direction opposite to the direction of wedge placement, before and during wedge placement may prevent rebound dislodgment. The stretched dam is released after the wedge is inserted.

Round versus Triangular Wedge

Some situations may require a triangular wedge that is modified (by knife or scalpel blade) to conform to the approximating tooth contours (Fig. 13.100).

The *round toothpick wedge is usually the wedge of choice with conservative proximal boxes*, however, because its wedging action is more occlusal (i.e. nearer the gingival margin) than with the triangular wedge (Fig. 13.101A and B).

The *triangular (i.e. anatomic) wedge is recommended for a preparation with a deep gingival margin*. The triangular wedge is positioned similarly to the round wedge, and the goal is the same. When the gingival margin is deep (cervically), the base of the triangular wedge more readily engages the tooth gingival to the margin without causing excessive soft tissue displacement. The anatomic wedge is preferred for deeply extended gingival margins because its greatest cross-sectional dimension is at its base (Fig. 13.101C and D).

To maintain gingival isolation attained by an anatomic wedge placed before the preparation of a deeply extended gingival margin, it may be appropriate to withdraw the wedge a small distance to allow passage of the band between the loosened wedge and the gingival margin. Tilting (i.e. canting) the matrix into place helps the gingival edge of the band slide between the loosened wedge and the gingival margin. The band is tightened, and the same wedge is firmly reinserted.

Supporting the matrix material with the blade of a Hollenback carver during the insertion of the wedge for the difficult deep gingival restoration may be helpful.[28] The tip of the blade is placed between the matrix and the gingival margin, and then the 'heel' of the blade is leaned against the matrix and adjacent tooth (Fig. 13.102). In this position, the blade supports the matrix to help in positioning the wedge sufficiently gingivally and preventing the wedge from pushing the matrix into the preparation. After the wedge is properly inserted, the blade is gently removed.

All aspects of the band and wedge are assessed and any desired final corrections are made. The matrix band and wedge must provide appropriate proximal contact and

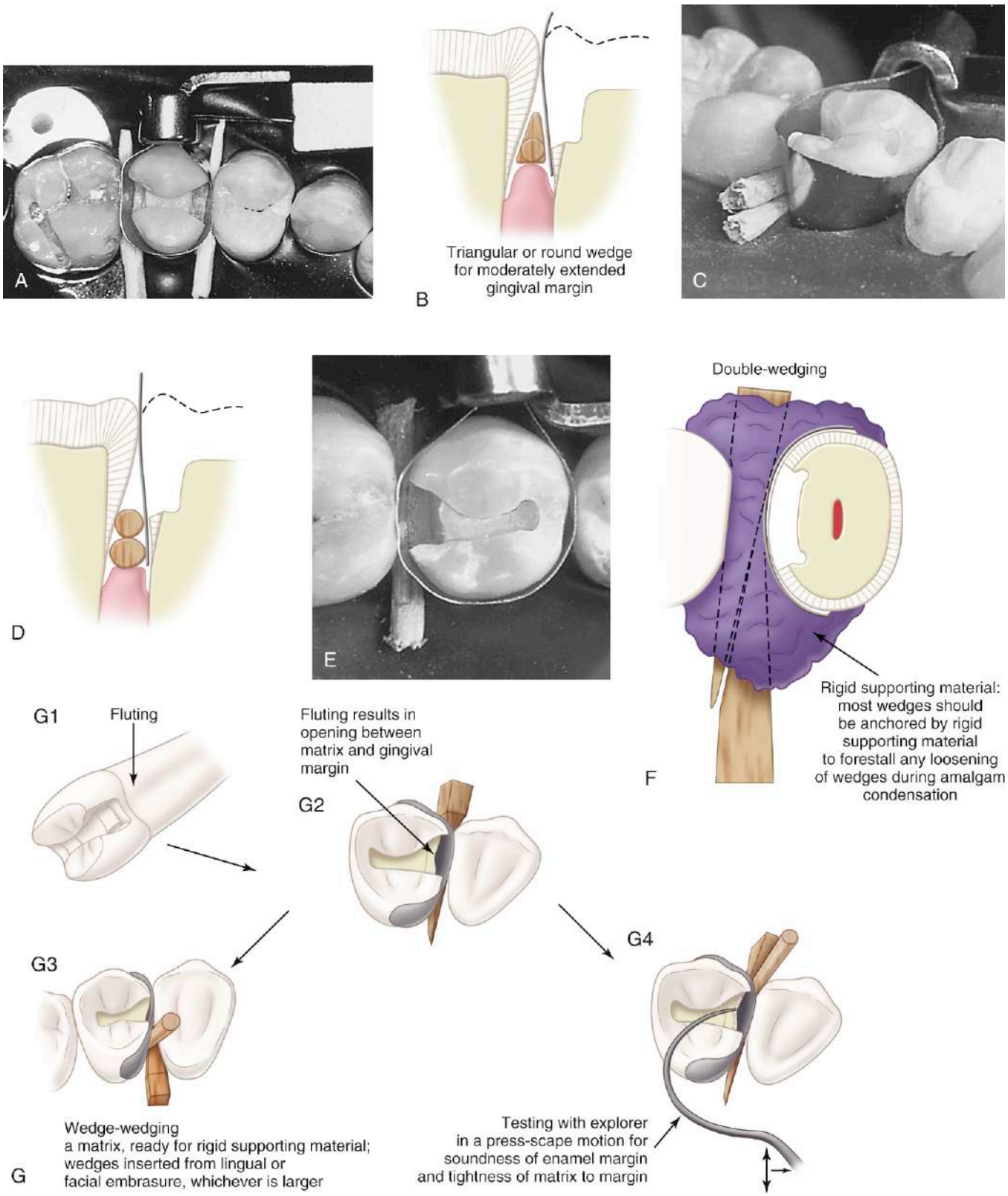


Figure 13.98 Various double-wedging techniques. (A and B) Proper wedging for the matrix for a typical mesioocclusodistal preparation; (C and D) technique to allow wedging near the gingival margin of the preparation when the proximal box is shallow gingivally, the interproximal tissue level has receded, or both; (E and F) double wedging may be used with faciolingually wide proximal boxes to provide maximal closure of the band along the gingival margin and (G) another technique may be used on the mesial aspect of the maxillary first premolars to adapt the matrix to the fluted (i.e. concave) area of the gingival margin (G1, G2); a second wedge inserted from the lingual embrasure (G3); testing the adaptation of the band after insertion of the wedges from the facial aspect (G4).

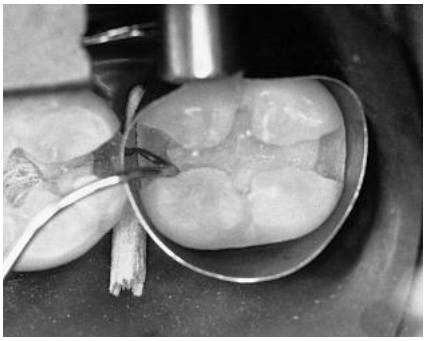


Figure 13.99 Use of the explorer tip (with pressure) to ensure proper adaptation of the band to the gingival margin.

contour (Fig. 13.96). After wedging, relaxing the tension of the band by turning the larger knurled nut of the Tofflemire retainer a quarter turn counter-clockwise will help to ensure proximal contact. If loosening the Tofflemire band still does not allow for contact with an adjacent tooth, a custom-made band with a smaller angle may be used. Reducing the angle of the band increases the difference in length (i.e. circumferences) of the gingival and occlusal edges. To reduce the angle of the band, the operator folds it as shown in Fig. 13.103, burnishes for appropriate occlusogingival contour (in the contact areas), and inserts the band into the Tofflemire retainer.

Occasionally, the interdental spacing is so large that a suitably trimmed tongue blade is required to wedge the matrix (Fig. 13.104). Occasionally, it is impossible to use a wedge to secure the matrix band. In this case, the band must be sufficiently tight to minimize the gingival amalgam excess. Because the band is not wedged, special care must be exercised to place small amounts of amalgam in the gingival floor and condense the first 1 mm of amalgam lightly but thoroughly in a gingival and lateral direction. The condensation of additional increments is then carefully continued in a gingival direction using a larger condenser with firm pressure. Condensation against an unwedged matrix may cause a gross extrusion of amalgam beyond the gingival margin. The excess amalgam must be removed with a suitable carver immediately after matrix removal.

Rigid-Material Supported Sectional Matrix

An alternative to the universal matrix band is the use of a properly contoured sectional matrix that is wedged and supported by a material that is rigid enough to resist

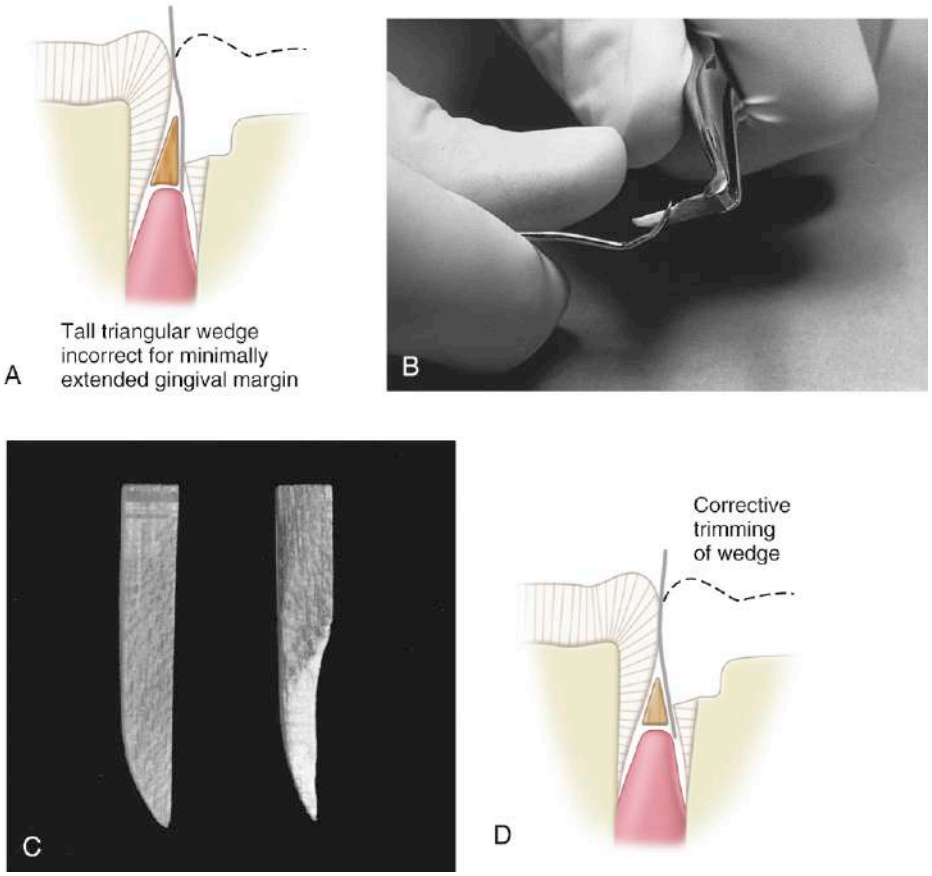


Figure 13.100 Modified triangular (i.e. anatomic) wedge. (A) Depending on the proximal convexity, a triangular wedge may distort the matrix contour; (B) a sharp-bladed instrument may be used to modify the triangular steepness of the wedge; (C) modified and unmodified wedges are compared and (D) properly modified triangular wedge prevents distortion of the matrix contour.

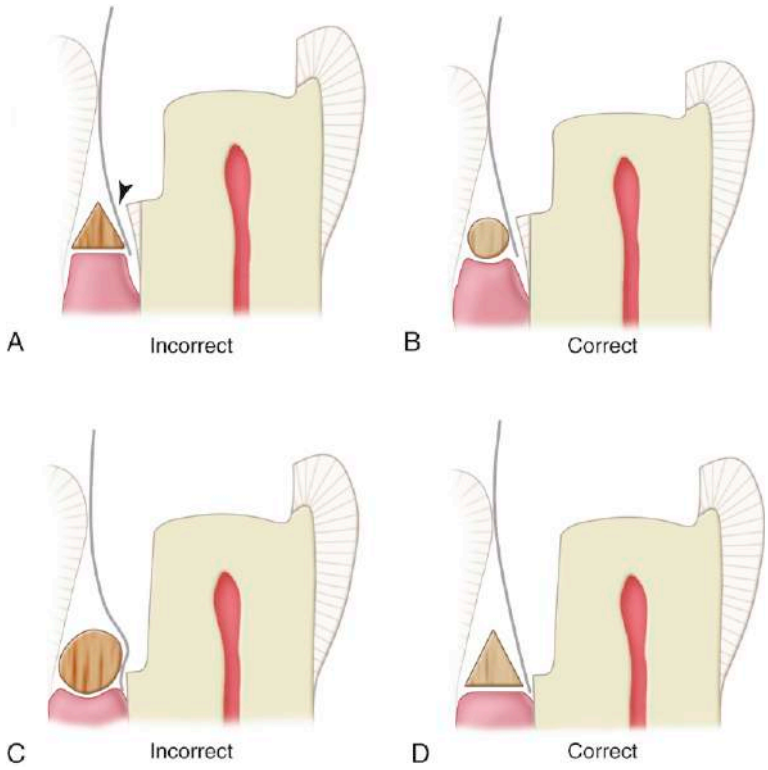


Figure 13.101 Indications for the use of a round toothpick wedge versus a triangular (i.e. anatomic) wedge. (A) As a rule, the triangular wedge does not firmly support the matrix band against the gingival margin in conservative Class II preparations (*arrowhead*); (B) the round toothpick wedge is preferred for these preparations because its wedging action is nearer the gingival margin; (C) in Class II preparations with deep gingival margins, the round toothpick wedge crimps the matrix band contour if it is placed occlusal to the gingival margin and (D) the triangular wedge is preferred with these preparations because its greatest width is at its base.

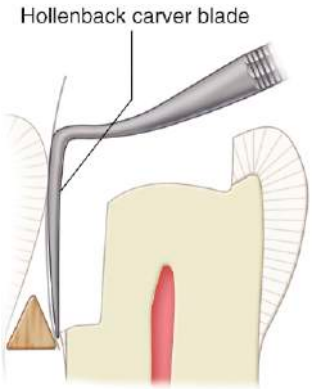


Figure 13.102 Supporting the matrix with the blade of a Hollenback carver during wedge insertion.

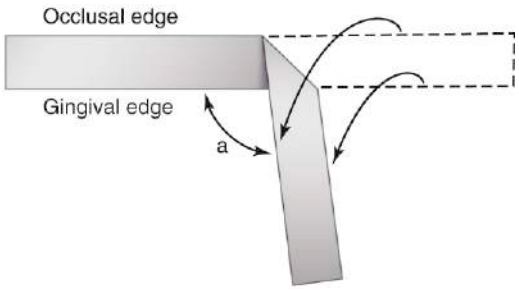


Figure 13.103 The custom-made matrix strip is folded, as indicated by arrows. The smaller angle (*a*) compared with the angle of the commercial strip increases the difference between the lengths of the gingival and occlusal edges.

condensation pressure. The supporting material selected must be easy to place and to remove. Examples include light-cured, thermoplastic and *quick-setting rigid PVS materials* (Fig. 13.105). The gingival wedge is positioned interproximally to secure the band tightly at the gingival margin to prevent any excess of amalgam (i.e. overhang). The wedge also separates teeth slightly to compensate for the thickness of the section of matrix material. The wedge is placed before the application of rigid supporting material.

The proximal surface contour of the matrix should allow the normal slight convexity between the occlusal and middle-thirds of the proximal surface when viewed from the lingual (and facial) aspect. Proximal surface restorations often display an occlusogingival proximal contour that is too straight, thereby causing the contact relationship to be located too far occlusally (with little or no occlusal embrasure). This condition allows food impaction between teeth, with resultant injury to the interproximal gingiva and supporting tissues, and invites conditions leading to demineralization of the adjacent proximal surfaces. The



Figure 13.104 A custom-made tongue blade wedge may be used when excessive space exists between adjacent teeth.

proximal surface contour of the matrix should also provide the correct form to the proximal line angles, both facially and lingually. If these contours are not present, the facial and lingual embrasures of the restoration are too open, inviting food impaction and injury to underlying supporting tissues. Correct and incorrect contours and matrix correction steps are illustrated in Figs. 13.106 and 13.107.

The matrix should be tight against the facial and lingual margins on the proximal surface so that the amalgam may be well condensed at the preparation margins. In addition, when the matrix is tight against the tooth, minimal carving is necessary on the proximal margins after the matrix is removed.

Pre-contoured Sectional Matrix Strips

Commercially available sectional metal strips (e.g. Palodent System; DENTSPLY Caulk, Milford, DE) are pre-contoured and ready for application to the tooth (Fig. 13.108). These strips have limited application when used for amalgam because of their rounded contour. They usually are most suitable for mandibular first premolars and the distal surface of maxillary canines. The contact area of the adjacent tooth occasionally is too close to allow placement of the contoured Palodent strip without causing a dent in the strip's contact area, making it unusable.

Condensation and Carving of the Amalgam

Proper condensation of amalgam results in close material adaptation to the preparation walls (limiting marginal voids) and minimal inclusion of internal voids. In addition, careful overstepping condensation technique

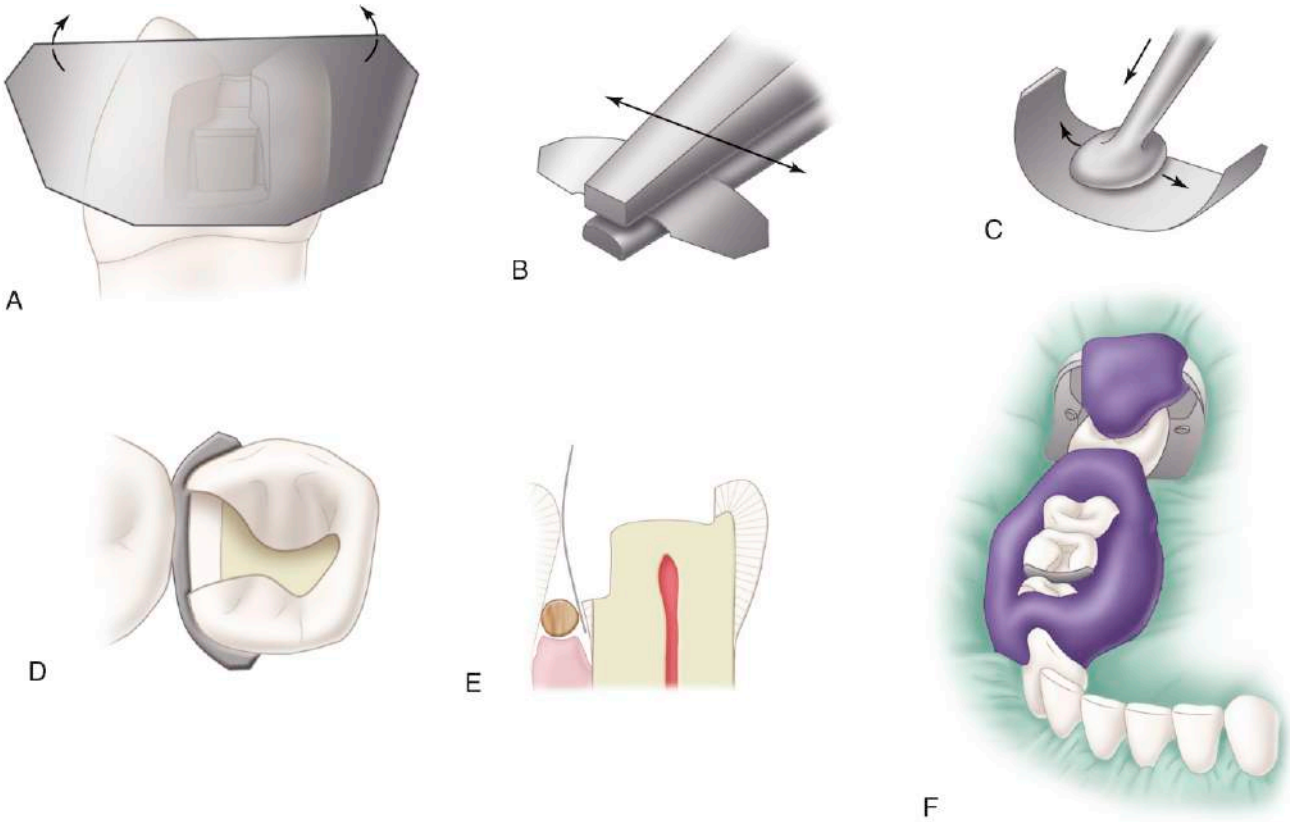


Figure 13.105 Rigid material-supported sectional matrix. (A) The shape of the stainless steel strip after trimming; (B) the strip contoured to the circumferential contour of the tooth (fingers can be used); (C) burnishing the strip to produce occlusogingival contact contour (*left* and *right arrows* indicate the short, back-and-forth motion of the burnisher); (D) contoured strip in position; (E) matrix strip properly wedged and (F) completed rigid material-supported sectional matrix.

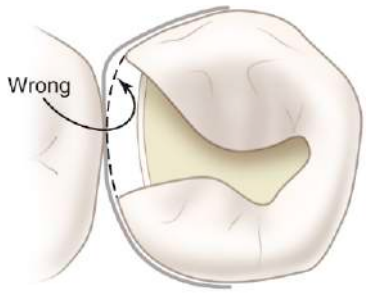
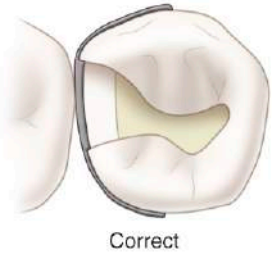


Figure 13.106 Alteration of the sectional matrix contour to provide the correct form to the proximofacial line angle region.



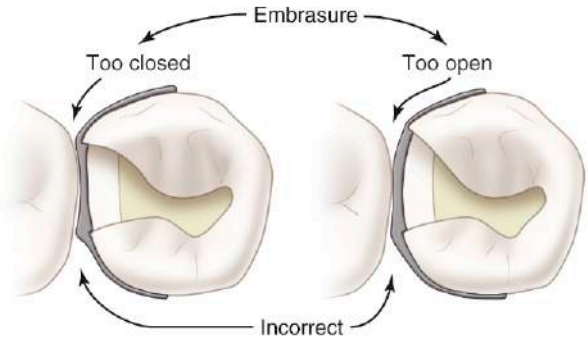


Figure 13.107 Correct or incorrect facial and lingual embrasure form is determined by the shape of the sectional matrix strip.



Figure 13.108 Pre-contoured metal strips.
(The Palodent System; courtesy: DENTSPLY Caulk, Milford, DE.)

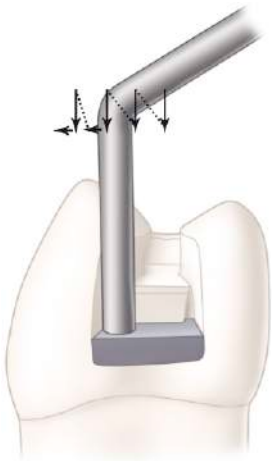


Figure 13.109 Lateral and occlusogingival force is necessary to condense amalgam properly into the proximal grooves and into the angles at the junction of the matrix band and the margins of the preparation.

eliminates excess mercury and results in a stronger restoration that is less prone to corrosion. Care should be taken to choose condensers that are best suited for use in each part of the tooth preparation and that may be used without binding in the preparation or between the preparation and the matrix band.

Step 1: Condensation should occur in the proximal box(es) first by transfer of enough amalgam with the amalgam carrier to *fill the gingival portion by about 1 mm*. This first increment is condensed in a *gingival and lateral direction* with sufficient force to adapt the amalgam to the gingival floor.

Step 2: Additional amalgam is carefully condensed against the *proximal margins of the preparation* and into the proximal retention grooves (if present). Firm, facially and lingually directed condensation (i.e. lateral condensation) with the side of the condenser should be accomplished along with the application of gingivally directed force (Fig. 13.109).[87] Condensation strokes in a gingival and lateral direction help to ensure that no voids occur internally or at the faciogingival or linguogingival line angles. Mesial (or distal) condensation of the amalgam in the proximal box is accomplished to ensure proper contour and proximal contact with the adjacent tooth.

Step 3: The procedure of adding and condensing continues until *amalgam reaches the level of the pulpal wall*. The size of the condenser is changed (usually to a larger one), if indicated, and amalgam is condensed in the remaining proximal portion of the preparation concurrently with the occlusal portion. It may be necessary to return to a smaller condenser when condensing in a narrow extension of the preparation or near the proximal margins. A smaller condenser face is more effective at condensing, provided it does not significantly penetrate the amalgam already in place.

Step 4: The *occlusal margins* are covered and overpacked by at least 1 mm using a large condenser, ensuring that the



Figure 13.110 Pre-carve burnishing with a large burnisher. (Courtesy: Aldridge D. Wilder, DDS.)



Figure 13.111 Using No. 110 pliers, the matrix band should be removed in a linguoocclusal (arrow) or facioocclusal direction (not just in an occlusal direction).

margins are well condensed, especially in the area of the marginal ridge.[88]

Proper condensation of the amalgam in the marginal ridge area(s) reduces the risk of marginal ridge fracture during matrix removal. The plasticity and wetness of the amalgam mass should be monitored during condensation. Proper condensation requires that the mix should be neither wet (i.e. mercury rich) nor dry and crumbly (i.e. mercury poor). Amalgam that is beginning to set should be discarded and a new mix obtained to complete the condensation. Before carving procedures are initiated, pre-carve burnishing of the occlusal portion with a large egg-shaped or ball burnisher should be accomplished (Fig. 13.110).[52] As mentioned previously, pre-carve burnishing is a form of condensation.

Carving the Occlusal Portion

Carving of the occlusal portion (including the occlusal embrasure or embrasures) may be accomplished with the matrix assembly in place while the amalgam is reaching its initial set. Early removal of the band increases the risk of marginal ridge fracture. Careful carving of the occlusal portion should begin immediately after condensation and burnishing. *Sharp discoid instruments* of suitable size as well as the *Hollenback carver* may be effective at re-establishing the occlusal anatomy. The marginal ridge is carved confluent with the tooth's anatomy such that it duplicates the height and shape of the adjacent marginal ridge (Fig. 13.18). An explorer or small Hollenback carver may be used to carefully define the occlusal embrasure. Remembering the pattern of occlusal contacts (that were evaluated before tooth preparation), observing the height of the adjacent marginal ridge and knowing where the preparation cavosurface margins are located all aid in completing the carving of the occlusal surface, including the marginal ridge and occlusal embrasure. If the restoration has extensive proximal involvement of the tooth, the occlusal carving should be accomplished quickly.

Removal of the Matrix Band and Completion of Carving

Step 1: The retainer is removed from the band after turning the small knurled nut counter-clockwise to retract the pointed spindle. The matrix is removed after ensuring hardening of the amalgam to avoid fracture of the marginal ridge during band removal.
Step 2: The end of the index finger may be placed on the occlusal surface of the tooth to stabilize the band as the retainer is removed.
Step 3: The No. 110 pliers are used to tease the band free from one contact area at a time by pushing or pulling the band in a *linguo-occlusal* (or *facio-occlusal*) direction and, if possible, in the direction of wedge insertion (Fig. 13.111). A straight occlusal direction should be avoided during matrix removal to prevent breaking of the marginal ridges.
Step 4: The wedge may be left in place to provide separation of teeth while the matrix band is removed, and then finally the wedge is removed. By maintaining slight interdental separation, the wedge reduces the risk of fracturing of amalgam.

Clinical Notes

Band removal in a straight occlusal direction should be avoided so as to limit the risk of marginal ridge fracture.

Matrix band removal allows access to carve the proximal portions of the restoration. Efficient use of time is required so as to gain access to the proximal areas while the amalgam remains carvable. Proper matrix form should result in a proximal surface that is nearly completed, with proper contact evident and minimal carving required except to remove a possible small amount of excess amalgam at the proximal facial and lingual margins, at the faciogingival and linguogingival

corners, and along the gingival margin. Amalgam knives (scalers, Nos. 34 and 35) are ideal for removing gingival amalgam excess (Figs. 13.21 and 13.22). They also are ideal for refining the embrasure form around the proximal contacts (Fig. 13.19). The secondary (or 'back') edges on the blades of amalgam knives are occasionally helpful, using a push stroke. The Hollenback carver No. 3 and (occasionally) the side of the explorer may also be suitable instruments for carving these areas. However, the explorer may not refine the margins and contour as effectively as the amalgam knife if the set of the amalgam is close to completion. When the proximal carving is completed, the occlusal surface contouring is completed. Occasionally, occlusal contouring may require the use of an abrasive stone or finishing bur if the setting of the amalgam is nearing completion.

When carving the margins of the restoration, the cutting surface of the carving instrument is held in a perpendicular orientation. Carving should be parallel to the margins, however, with the adjacent tooth surface being used to guide the carver. The existence of the proximal contact is verified visually by using the mouth mirror. When carving is completed, the rubber dam is removed and the occlusion is assessed and adjusted, as needed.

Before the patient is dismissed, *thin unwaxed dental floss* may be passed through the proximal contact(s) once to remove any amalgam shavings on the proximal surface of the restoration and to assess the gingival margin. Passing the floss multiple times through the contact area of an amalgam that is still early in its set may result in removal of some surface amalgam and loss of contact with the adjacent tooth. When positioned in the gingival embrasure, the floss is wrapped around the restored tooth and positioned apical to the gingival margin of the restoration. The floss is moved in an occlusogingival direction. The floss not only removes amalgam shavings but also smooths the proximal amalgam and detects any gingival overhang of amalgam. If an overhang is detected, further use of an amalgam knife is necessary.[89] Alternatively, a finishing strip may be used to remove the amalgam excess and finish the proximal surface gingival to the contact area. Floss also may be used to verify that the intensity of the contact is similar to that of neighbouring teeth. Final rinsing and evacuation of the oral cavity is then accomplished. The patient is advised to avoid chewing with the restored tooth for 24 h.

Finishing and Polishing of the Amalgam

Finishing and polishing of the occlusal surface of the Class II amalgam should follow the approach presented in the General Concepts Guiding Restoration with Amalgam section. Finishing and polishing of the proximal surface may be indicated where the proximal amalgam is accessible. This area usually includes the facial and lingual margins and areas occlusal to the contact. The remainder of the proximal surface is often inaccessible; however, the matrix band should have imparted sufficient smoothness to it.

If amalgam along the facial and lingual proximal margins was slightly over-carved, the enamel margin may be felt as the explorer tip passes from amalgam across the margin onto the external enamel surface. Finishing burs or sandpaper disks, rotating at slow speed, may be used to smooth the enamel–amalgam margin. Sandpaper disks also may be used to smooth and contour the marginal ridge. Inappropriate use of sandpaper disks may 'ledge' the restoration around the contact, however, resulting in inappropriate proximal contours.

In small preparations, the facial and lingual proximal margins are generally inaccessible for finishing and polishing. Fine abrasive disks or the tip of a sharpened rubber polishing point or edge of a sharped abrasive rubber polishing cup may be used to polish the proximal portion that is accessible. When proximal margins are inaccessible to finishing and polishing with disks or rubber polishing points, and some excess amalgam remains (e.g. at the gingival corners and margins), amalgam knives occasionally may be used to trim amalgam back to the margin and to improve the contour, though a flame-shaped finishing bur may ultimately be required. Fig. 13.112 provides an

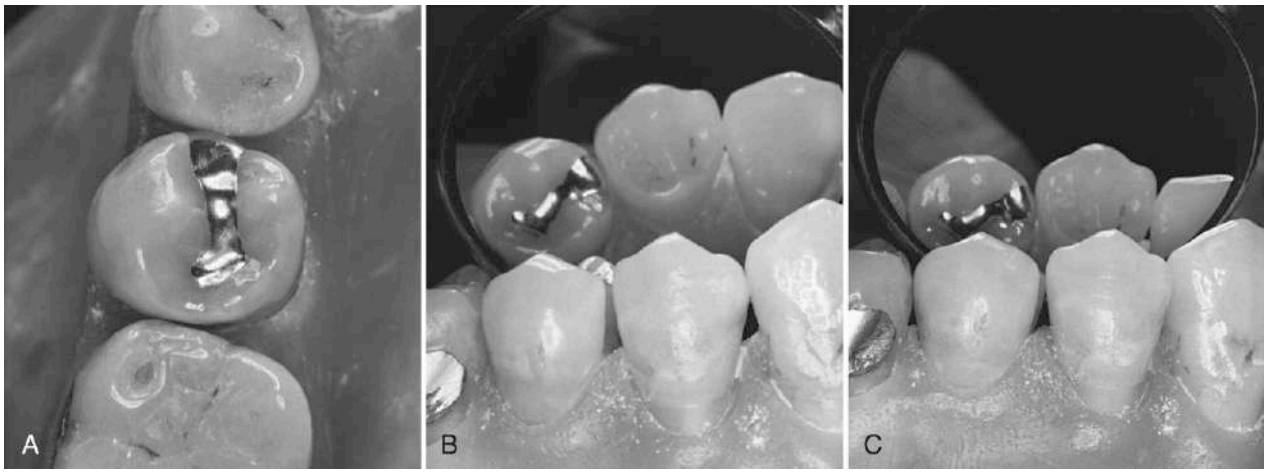


Figure 13.112 Polished mesioocclusal amalgam restoration. Note the conservative extension. (A) Occlusal view; (B) mesiofacial and occlusal views of the mesiofacial margin; (C) facial and occlusal views of the proximal surface contour and the location of the proximal contact.

example of a finished and polished Class II amalgam restoration.

Quadrant Dentistry

When several teeth in a quadrant have caries lesions, the experienced dentist usually treats them all at the same appointment rather than individually. Quadrant dentistry increases efficiency and reduces chair time for the patient. The use of the rubber dam is particularly important in quadrant dentistry. For maximal efficiency, when a quadrant of amalgam tooth preparations is planned, each rotary or hand instrument should be used on every tooth where it is needed before being exchanged.

When restoring a quadrant of Class II amalgam tooth preparations, it is permissible to apply matrix bands on alternate preparations in the quadrant and restore teeth two at a time. Banding adjacent preparations requires excessive wedging to compensate for a double thickness of band material and makes the control of proximal contours and interproximal contacts difficult. Extensive tooth preparations may need to be restored one at a time. If proximal boxes differ in size, teeth with smaller boxes should be restored first because often the proximal margins are inaccessible to carving if the larger adjacent box is restored first. In addition, smaller boxes may be restored more quickly and accurately because more tooth structure remains to guide the carver. If the larger proximal box is restored first, the gingival contour of the restoration may be damaged when the wedge is inserted to secure the matrix band for the second, smaller restoration. If the adjacent proximal boxes are similar in size, the banding of alternate preparations should be started with the most posterior preparation because this allows the patient to close slightly as subsequent restorations are inserted (Fig. 13.113).

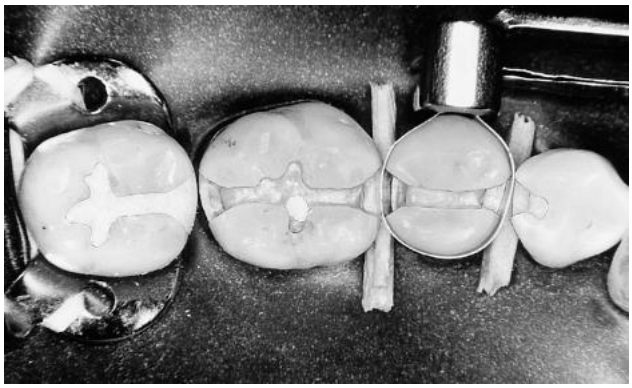


Figure 13.113 Quadrant dentistry. Unless otherwise indicated, a quadrant of Class II preparations with similarly sized proximal boxes may be restored using two bands simultaneously if they are placed on every other prepared tooth. It is recommended that the most distal tooth be restored first.

Before restoring the second of two adjacent teeth, the proximal contour of the first restoration should be carefully established. Its anatomy serves as the guide to establish proper proximal contours leading to correct embrasure forms and contact position/size. If necessary, a finishing strip may be used to refine the contour of the first proximal amalgam (Fig. 13.114). The finishing strip is indicated, however, only where the proximal clearance is present. Using a finishing strip between contacting amalgam restorations may compromise or eliminate the proximal contact. Properly placed amalgam restorations have the potential to last for decades (Fig. 13.115).

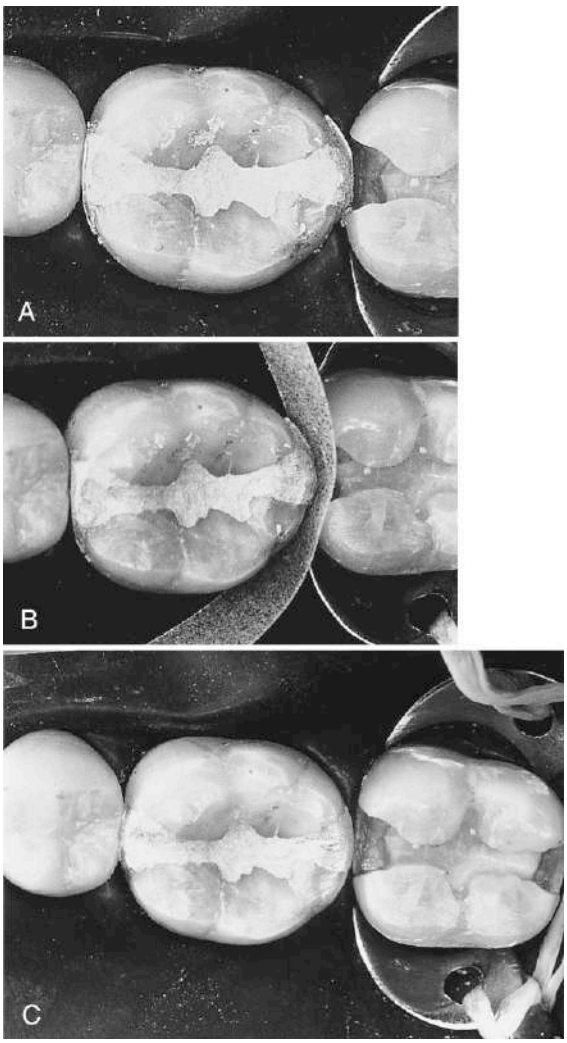


Figure 13.114 When the first of two adjacent Class II preparations is restored, proper contour may be established by using a finishing strip before the second is restored. (A) Before using the strip; (B) removing the over-contour; (C) verification of proper contour may be accomplished by viewing the restoration with a mirror from the occlusal, facial and lingual positions. The proximal contour also may be evaluated after matrix placement by noting the symmetry between the restored surface and the burnished matrix band.

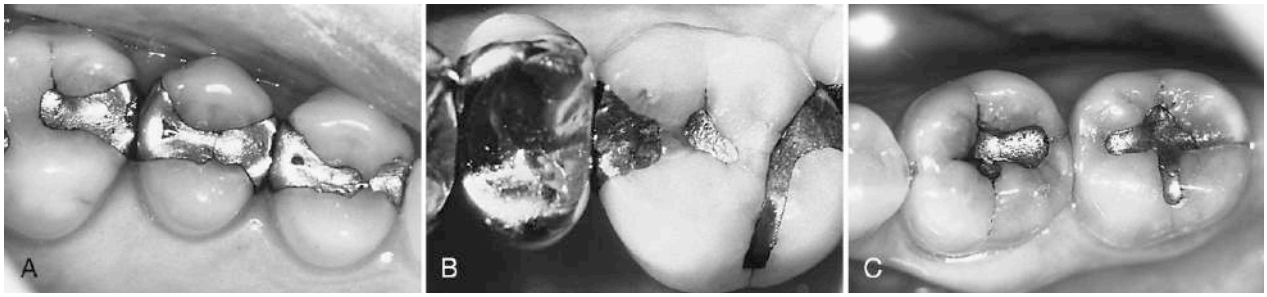


Figure 13.115 Clinical examples of long-term amalgam restorations: (A) 44-year-old amalgams; (B) 58-year-old amalgams in the first molar; (C) 65-year-old amalgams in molars.
(Parts A and B, Courtesy: Drs. John Osborne and James Summitt.)

Summary

The use amalgam remains a predictable, cost-effective and safe means for the restoration of posterior (and some anterior) teeth that are missing various amounts of tooth structure. The design of a successful tooth preparation must be based on the material properties of dental amalgam and the successful use of primary and secondary means of restoration retention. Restoration of normal anatomic contours and therefore normal clinical function may be readily accomplished with dental amalgam.

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